# Design, Manufacture and Testing of a Liquid Rocket Engine Injector Test Rig

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# Abstract

The combustion efficiency and stability of a liquid rocket engine are influenced by the distribution and atomization of propellants within the combustion chamber. High combustion efficiency and stable combustion require finely atomized propellants distributed evenly across the chamber cross-section, in the correct oxidiser to fuel (O/F) ratio. The injector is the primary component that controls the distribution and atomization of propellants within the combustion chamber. Therefore, the injector fuel and oxidiser domains require in-depth analyses to ensure the correct propellant conditions are present during combustion.

The South African First Integrated Rocket Engine (SAFFIRE) is a liquid oxygen/kerosene engine. With a thrust of 27.5 kN, it has been designed to provide propulsion to a small satellite launch vehicle. The injector was designed using computational fluid dynamic (CFD) software to ensure the propellants were evenly distributed across the chamber cross-section. The design concluded that the propellants were evenly distributed with a maximum distribution deviation of ±2 %. Although extensive CFD analyses were conducted during the design phase of the SAFFIRE injector, a comprehensive test program was required to establish the actual performance. This study presents the experimental procedure and results used to validate the modelled performance of the SAFFIRE injector. Cold flow tests, with water as an analogue fluid, were performed using a custom-developed test rig to determine the required injector pressure drop to drive the fuel and oxidiser at mass flow rates of 2.28 kg/s and 5.66 kg/s respectively. Propellant distribution tests were also conducted to ensure the correct O/F ratio was evenly distributed across the chamber cross-section. This study employed a two-test method to mitigate the effects of the transient periods experienced during start-up and shutdown. For comparative data sets, the fluid properties for the simulations were updated to match the analogue fluid properties used during testing.

The fuel and oxidiser simulations were validated using the experimental data gathered from the cold flow tests. It was concluded that the most appropriate turbulence model and wall treatment for the fuel domain, containing internal passages and small diameter holes, was a K-Epsilon turbulence model with a low $y^+$ wall treatment. The simulated pressure drop was found to deviate by 2.48 % from the experimental value, with the average distribution deviating by 5.46 %. For the oxidiser domain, containing mainly open channel axial flow, the K-Epsilon turbulence model with a high $y^+$ wall treatment was concluded to be the most appropriate model. The simulated pressure drop was found to deviate by 14.59 % from the experimental value, with the average distribution deviating by 8.18 %.

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# Nomenclature

| Symbol | Description | Units |
|---|---|---|
| $C_\mathrm{d}$ | Coefficient of discharge | [-] |
| $C_\mathrm{v}$ | Flow coefficient | [-] |
| $d$ | Diameter | [m] |
| $D_\mathrm{orifice}$ | Propellant deviation | [%] |
| $F_\mathrm{s}$ | Safety factor | [-] |
| $F1$ | Fuel ring 1 | [-] |
| $F2$ | Fuel ring 2 | [-] |
| $F3$ | Fuel ring 3 | [-] |
| $F4$ | Fuel ring 4 | [-] |
| $FF$ | Fuel film cooling ring | [-] |
| $K$ | Coverage factor | [-] |
| $K$ | Head loss coefficient | [-] |
| $K_\mathrm{v}$ | Flow factor | [$m^3$/h.$bar^{0.5}$] |
| $\dot{m}$ | Mass flow rate | [kg/s] |
| $m_\mathrm{d}$ | Sample measurement | [-] |
| $m_\mathrm{orifice}$ | Orifice sample mass | [g] |
| $n$ | Number of observations | [-] |
| $N$ | Number of orifices | [-] |
| $Oh$ | Ohnesorge number | [-] |
| $p$ | Order of convergence | [-] |
| $P$ | Pressure | [Pa] |
| $\Delta P$ | Change in pressure | [Pa] |
| $P_\mathrm{zero}$ | Zero grid spacing continuum value | [-] |
| $Q$ | Flow rate | [$m^3$/s] |
| $r$ | Refinement ratio | [-] |
| $Re$ | Reynolds number | [-] |
| $s$ | Standard deviation | [-] |
| $SG$ | Specific gravity | [-] |
| $u$ | Uncertainty | [-] |
| $U$ | Expanded uncertainty | [-] |
| $v$ | Mean jet velocity | [m/s] |
| $We$ | Weber number | [-] |

| Greek symbols | Description | Units |
|---|---|---|
| $\gamma$ | Surface tension | [N/m] |
| $\delta$ | Device resolution | [-] |
| $\mu$ | Dynamic viscosity | [Pa. s] |
| $\rho$ | Density | [kg/m$^3$] |
| $\sigma$ | Standard deviation | [-] |

| Abbreviation | Description |
|---|---|
| AIAA | American Institute of Aeronautics and Astronautics |
| ASME | American Society of Mechanical Engineers |
| ASReG | Aerospace Systems Research Group |
| BURPG | Boston University Rocket Propulsion Group |
| CAD | Computer-aided Design |
| CFD | Computational Fluid Dynamics |
| CHPC | Centre for High-Performance Computing |
| DNS | Direct Numerical Simulation |
| EDM | Electrical Discharge Machining |
| EMRTC | Energetic Materials Research and Testing Center |
| FEA | Finite Element Analysis |
| GCI | Grid Convergence Index |
| L/D | Length to Diameter ratio |
| LES | Large Eddy Simulation |
| LOX | Liquid Oxygen |
| LPL | Liquid Propulsion Laboratory |
| NASA | National Aeronautics and Space Administration |
| NI | National Instruments |
| O-F-O | Oxidiser – Fuel – Oxidiser |
| O/F | Oxidiser to Fuel Ratio |
| P&ID | Piping and Instrumentation Diagram |
| PDPA | Phase Doppler Particle Analyser |
| PIV | Particle Image Velocimetry |
| RANS | Reynolds-average Navier-Stokes |
| REM | Richards Extrapolation Method |
| SAFFIRE | South African First Integrated Rocket Engine |
| SEM | Standard Error of the Mean |
| SMD | Sauter Mean Diameter |

| | |
|---|---|
| UI | User Interface |
| UKZN | University of KwaZulu-Natal |
| USC | University of Southern California |
| WEDM | Wire Electrical Discharge Machining |

# 1. Introduction

## 1.1 Liquid Rocket Propulsion

A liquid rocket engine injector must supply propellants to the combustion chamber in the correct proportions and pressures to yield an efficient and stable combustion process (Huzel & Huang, 1992). These parameters are based on the O/F ratio and the chamber pressure of an engine. The injector is also responsible for sealing the top of the combustion chamber where it experiences high pressures and temperatures during combustion. Therefore, both the flow and structural design are critical to the overall performance of a liquid rocket engine. Computational fluid dynamic software has become a useful tool within the design phase for evaluating the performance of a liquid rocket engine injector, but the modelled results must be validated through experimental methods. Within the propulsion industry, the process of testing a system or component with non-reactive analogue fluids is known as cold flow testing (Gill & Nurick, 1976). Cold flow testing methods are used to test the performance outputs of a system or component in a safe and controlled environment without the complexities introduced by combustion.

The injector design influences the combustion stability and designing for stability can reduce combustion efficiency. To achieve high levels of combustion efficiency, uniform distribution of the desired mixture ratio and fine atomization is required (Elverum & Morey, 1959). Typically, a 15-20% pressure drop across the injector system is targeted (Huzel & Huang, 1992). This pressure drop allows for slight pressure fluctuation to occur during operation without causing reverse flow within the injector flow domains. According to the National Aeronautics and Space Administration (NASA), pressure fluctuations of ±5 % of the engine chamber pressure are considered acceptable for stable combustion (Huzel & Huang, 1992). When designing for stability or performance, parameters such as the element type and layout must be considered. The element layout affects the mixing interaction between elements, mass distribution and effective mixture-ratio distribution. These factors are known to influence the stability of a liquid rocket engine (Huzel & Huang, 1992; Sutton & Biblarz, 2001).

## 1.2 SAFFIRE Programme Overview

The SAFFIRE program comprises the paper design of a liquid rocket engine with a predicted thrust of 27.5 kN. The combustion chamber is thermally insulted using an ablative liner. Cooling is assisted by a fuel-rich region near the chamber wall, known as film cooling. The injector design consists of a concentric ring injector that has been designed to supply liquid oxygen (LOX) and kerosene to the combustion chamber. The injector is designed to produce an even mass flow distribution, such that the mass flow rate through each orifice is within ±2 % of the nominal mass flow rate. This nominal mass flow rate is based on even distribution for the correct O/F ratio. To achieve maximum engine thrust the

SAFFIRE injector must perform as designed. The current performance of the SAFFIRE injector is based solely on CFD analyses. Therefore, a comprehensive cold flow testing campaign is required to validate the actual performance of the SAFFIRE injector. The SAFFIRE engine was designed to provide propulsion for a small satellite launcher. Therefore, the actual injector performance is required to ensure the launcher can successfully deliver a payload to orbit.

## 1.3 Injector Cold Flow Testing

Various methods of cold flow testing have been developed to determine specific performance outputs of liquid rocket engines. Cold flow testing with an analogue fluid is used to test the pressure drop and propellant distribution of an injector (Elverum & Morey, 1959). Extensive research data is available for different element types/shapes and the resulting propellant distribution (Elverum & Morey, 1959; McHale, 1974; Riebling, 1967; Rupe, 1973). Based on these studies and others, an appropriate element configuration and angle were selected for the SAFFIRE injector: a triplet element configuration at $30^{\circ}$ (Martin, 2019). This element type relies on two outer LOX streams impinging on a central fuel stream to atomise and distribute the propellants. The major factor that determines the impingement performance is the momentum of each stream. An unbalanced momentum will result in a skewed spray fan leading to hot spots on the chamber walls (Elverum & Morey, 1959). The pressure drop across the injector is the main factor that determines the flow rate of each propellant. It also decouples the combustion chamber from the feed system and prevents any back propagation of propellants caused by pressure spikes in the combustion chamber (Huzel & Huang, 1992). Therefore, the pressure drop and propellant distribution are performance parameters that must be evaluated in an injector test programme as they directly influence the impingement and the overall performance of the engine.

## 1.4 CFD Error and Uncertainty

Verification and validation are processes that ensure a simulated problem is being solved correctly. CFD verification is the process of ensuring the partial differential equations selected to describe the computational model are being solved correctly to yield an accurate solution. This is purely a numerical analysis. Validation is the process of ensuring these partial differential equations accurately describe the physical model being solved based on experimental data (AIAA, 1998). Before a model can be validated, it must first be verified to avoid a false validation (Pelletier, 2010). Verification involves investigating the errors within a CFD solution, namely insufficient spatial and temporal discretization convergence, insufficient convergence of an iterative procedure, computer roundoff and computer programming errors (Oberkampf & Trucano, 2002).

## 1.5 Problem Statement and Research Objectives

The SAFFIRE injector design includes the following operating performance parameters: 1) the required pressure drops for the design mass flow rates and 2) the propellant distributions for the fuel and oxidiser domains. This experimental study aims to validate the performance parameters through cold flow testing methods.

The study comprises the following research objectives.

1. Develop an experimental testing methodology to determine the pressure drop across separate oxidiser and fuel injector domains, for nominal mass flow rates, and propellant distribution for the SAFFIRE injector.
2. Develop and commission a SAFFIRE injector test stand incorporating an appropriate control system, flow control valves and data acquisition hardware to obtain the required experimental data.
3. Obtain comparative data sets for the simulation and experimental results by conducting cold flow tests for the oxidiser and fuel domains, using water as an analogue fluid, and simulating the performance of the injector domains using the analogue fluid properties.
4. Determine the accuracy of the simulation results based on the deviation from the experimental data and validate the final performance of the injector.

The results of this study will aid future developments of injector design and optimization by determining the accuracy of the selected simulation model.

## 1.6 Dissertation Outline

A brief introduction to liquid rocket engine injectors, CFD verification and CFD validation is given in Chapter 1. An overview of the SAFFIRE programme is presented followed by the research problem statement and objectives.

Chapter 2 covers literature on liquid rocket engine testing, with specific reference to cold flow testing. Numerous cold flow testing methods and the corresponding outputs are investigated. Furthermore, CFD validation techniques and methods are covered, including evaluating experimental error and statistical analysis of experimental data.

Chapter 3 summarizes the test stand design, including the plumbing setup, valve types and actuation, control system, pressurization system and the data acquisition hardware. The test facility zones and sub-systems are shown and described in detail.

Chapter 4 describes the calibration process of the test stand and the pressurization system. This ensured a relatively constant tank pressure is maintained during operating conditions. Preliminary tests were also conducted to determine and decrease the valve actuation delay.

Chapter 5 covers the methodology followed for the CFD validation. This includes the test article designs, geometric changes, simulation updates, steady state testing methods and the respective outputs. The propellant collection method for the distribution tests is shown and discussed.

Chapter 6 covers the process of determining the uncertainty of each experimental data point obtained through physical testing. A justification of the two-test method used to obtain the mass flow rate measurement is provided. Lastly, the updated simulation results modelled using the analogue test fluid are verified.

Chapter 7 summarises the comparison of the experimental data and the CFD simulation results. The simulations included all design changes made due to machining limitations and practical considerations. The differences in pressure drop and propellant distribution with respect to the experimental data are investigated. The accuracy of the selected turbulence models and wall treatments are determined based on the experimental data gathered.

A final review of the experimental data and CFD simulation results is presented in Chapter 8. To conclude, guidelines are discussed on the best suited CFD model parameters, namely turbulence model and wall treatment, to aid future injector designs. Recommendations for future work and problems encountered are also discussed.

# 2. Literature Review

## 2.1 Introduction

Computational fluid dynamic software has become a useful and effective tool for modelling the performance of liquid rocket engine injectors, including modelling the pressure drop, propellant distribution (Martin, 2019), evaluating the performance of different element configurations (Thirupathi et al., 2015), and modelling the combustion process from a resulting injector spray (Cheng & Farmer, 2002). Employing CFD methods to optimise the performance of liquid rocket engines is by no means a novel concept as NASA have been using this approach since the 1960s. One early use was the development of codes to compute combustion stability in liquid hydrogen/oxygen rocket chambers (Priem & Guentert, 1962). Recent developments in computational hardware and commercially available CFD packages have increased the availability of complex solvers, leading to an increase in accuracy and streamlined the required setup for CFD simulations. In theory, these models can produce accurate results depending on the physical problem being modelled. A problem arises, though, when determining the accuracy of these CFD models as experimental data pertaining to the specific geometry and flow conditions do not exist (Oberkampf & Trucano, 2002).

## 2.2 Cold Flow Testing Methods

Cold flow testing involves substituting explosive propellants with non-reactive analogue fluids to determine the performance of a system or component. The method of cold flow testing a liquid rocket engine injector is dependent on the propellant phases and the performance parameters being investigated, for example, pressure drop, propellant atomization and mixing.

Basic cold flow calibration tests are used to calibrate flow resistance and visually assess the quality of the injected propellant streams (Huzel & Huang, 1992). This flow resistance value is compared to computational techniques and indicates the required inlet pressures to achieve the correct flow rates with the operating chamber pressure. This type of testing was conducted on the SAFFIRE injector to determine the required inlet pressures to drive the propellants at the design mass flow rates. An example of a cold flow test is shown in Figure 2-1 for a hybrid rocket. Here the pressure drop across the injector is being determined using nitrous oxide without any combustion taking place.

Figure 2-1: Hybrid rocket engine cold flow test (Broughton, 2018)

Cold flow testing an engine with a gaseous fuel and liquid oxidiser propellant combination involves using water as a substitute for the oxidiser and high-pressure air for the fuel. For mixing tests these substitute propellants are directed through the injector simultaneously and are expelled to atmosphere. The resulting spray is collected in a grid located a set distance away from the injector faceplate. The propellant mixture ratio can then be determined for specific grid locations (Fang, 1983; Huzel & Huang, 1992).

Cold flow mixing tests for a bi-liquid propellant combination are conducted using two non-soluble liquids and the resulting spray is collected in a grid system after impingement has occurred. The oxidiser to fuel ratio for each grid location can be measured to determine the propellant distribution. This ensures the entire cross-section of the combustion chamber receives the correct O/F ratio. These tests are also conducted using a single element setup to determine the effects of the impingement angle, orifice length or orifice shape (Riebling, 1967; Rupe, 1973). Figure 2-2 shows the distribution of a single triplet element with varying stream momentums (Gill & Nurick, 1976). Alternatively, the mixing and propellant distribution for an entire injector setup can be evaluated (McHale, 1974). The propellant distribution of the fuel and oxidiser domains for the SAFFIRE injector were conducted with a multiple element setup. However, the distribution was determined prior to impingement.

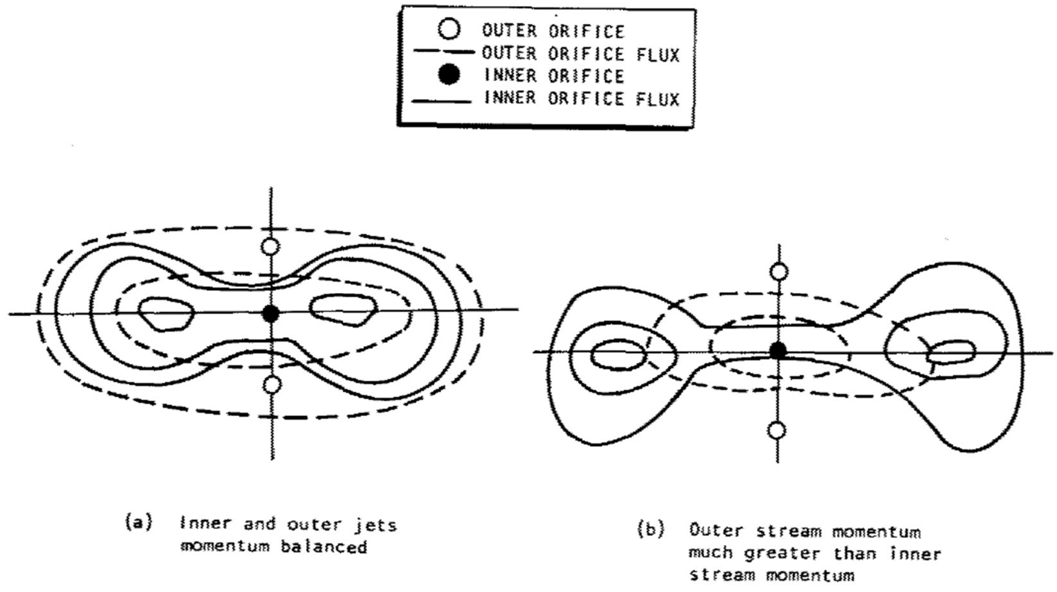


Figure 2-2: Spray fan distributions of an unlike triplet element (Gill & Nurick, 1976)

Supercritical conditions can influence the atomization and mixing within a combustion chamber. These conditions exist when the chamber pressure is at or near the critical pressure of the propellants. This is desirable due to the decrease in surface tension at critical pressures as surface tension is known to have a dominant role in the atomization process (Lefebvre, 1998). This theory was confirmed through visual observations of liquid nitrogen at its critical pressure for a coaxial injector by Chehroudi and Talley (2010). Further developments for impinging jets were investigated by Chehroudi (2012) and it was concluded that the supercritical jets display gas-like jet properties.

The atomization process determines the combustion efficiency and relative sensitivity to destabilizing influences (Huzel & Huang, 1992). The SAFFIRE injector design considered numerous factors that aid atomization such as element type, impingement momentum and impingement angle (Martin, 2019). Historically, atomisation characteristics were assessed through qualitative methods conducted during cold flow testing by visually inspecting the spray of an injector. Quantitative methods were also developed based on the collection of small wax droplets or low-melting metals solidifying after impingement. More advanced tests were developed that used laser and holographic measurement methods. However, development slowed when larger engines were found to be mixing-limited rather than vaporisation-limited. This is due to larger engines employing supercritical injection which is less sensitive to hydraulic-atomization characteristics (Huzel & Huang, 1992).

An effective tool for evaluating the performance of a model or prototype is to consider dynamic similarity, provided that the flows are both incompressible and no cavitation exists. Dynamic similarity exists when the flow characteristics of two steady state flows result in force distributions, where identical types of forces are parallel and vary in magnitude by a constant factor. Geometric similarity is also required for the two flow domains. Using the Buckingham Pi theorem, the governing dimensionless groups for a flow phenomenon can be found. Each independent dimensionless group must be equal for both the model and prototype to ensure dynamic similarity exists (Fox, McDonald & Pritchard, 2012). For an incompressible flow with no free surface, the Reynolds number must be equal for identical forces and pressure coefficients (Fox *et al.*, 2012). The Reynolds number for a flow within a pipe of diameter $d$, mass flow rate $\dot{m}$ and dynamic viscosity $\mu$ can be calculated using Equation 2-1:

$$Re = \frac{4\dot{m}}{\pi d \mu} \tag{2-1}$$

For two different fluids within the same geometry, the Reynolds numbers for each flow can be made equal by varying the flow rate based on the ratio of the viscosities as it is the only remaining independent variable. Therefore, the required mass flow rate of fluid 1 ($\dot{m}_1$with dynamic viscosity $\mu_1$) that yields the same Reynolds number as fluid 2 (with mass flow rate $\dot{m}_2$ and viscosity $\mu_2$) is given by Equation 2-2:

$$\dot{m}_1 = \dot{m}_2 \frac{\mu_1}{\mu_2} \tag{2-2}$$

By extension, liquid rocket engines can be tested with analogue fluids at different flow rates to model the force and pressure coefficients of the actual propellants at the design flow rates. However, certain phenomena, such as atomization, which is not solely dependent on the density and viscosity of the propellant, will not be accurately modelled at this dynamically similar flow rate. When using water as an analogue fluid for LOX the required mass flow rate must increase by a factor of roughly seven to meet the requirements of dynamic similarity. This increased mass flow rate often increases the cost and complexity of a test stand (Moruzzi, Fessl, Prochnicki, *et al.*, 2018).

## 2.3 Pressurization Systems

Liquid rocket engines are often designed to operate at high chamber pressures due to the increase in combustion efficiency. The propellants must be delivered to the combustion chamber at a high pressure to ensure there is no reverse flow during operation. There are currently three main methods of increasing the pressure of the propellants to overcome the line losses and injector pressure drop, namely turbopumps, electrically driven pumps and blowdown systems (Sutton & Biblarz, 2001). Blowdown systems can be separated into two configurations. Both systems make use of a high-pressure ullage to drive the propellants to the combustion chamber. The more stable approach uses a regulator to ensure a constant ullage pressure throughout the burn. The simplified approach does not include a regulator and the injector inlet pressure and chamber pressure decrease during the duration of the burn (Sutton & Biblarz, 2001). Although the SAFFIRE engine was designed for an electric turbopump setup, a regulated blowdown system was used for the test stand because of the simplicity of the configuration. A schematic of a regulated blowdown feed system is shown in Figure 2-3.

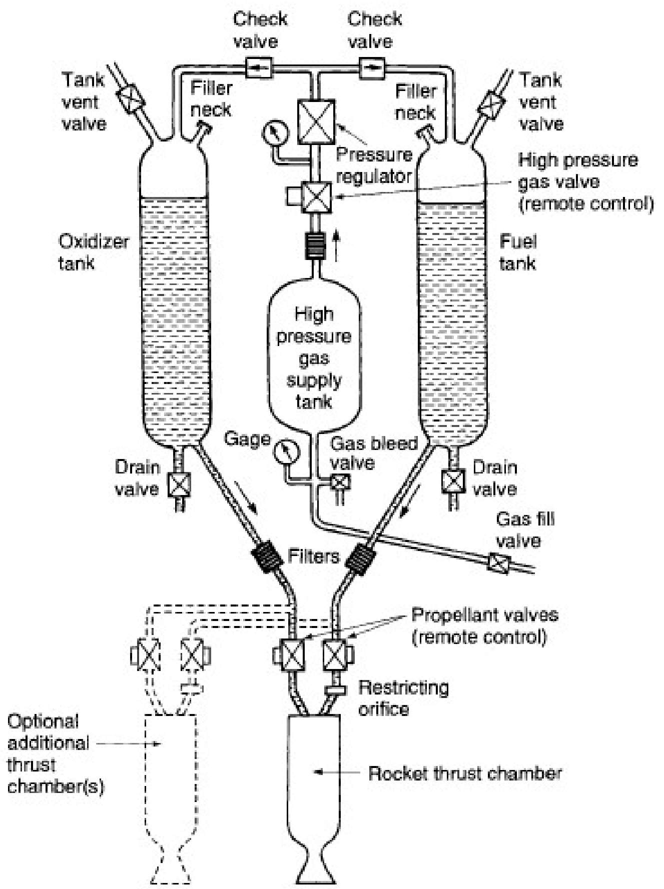


Figure 2-3: Regulated blowdown feed system (Sutton & Biblarz, 2001)

## 2.4 Propellant Distribution

One of the many functions of an injector is to distribute and mix the propellants in the combustion chamber while ensuring that the correctly proportioned mixture of fuel and oxidiser has an even mass flow rate and composition across the combustion chamber cross-section (Sutton & Biblarz, 2001). One factor affecting the distribution of propellants is the injector orifice pattern which is closely related to the internal manifolds and feed passages. A method of increasing the likelihood of even distribution is to employ a large complex geometry which results in low passage velocities. This increases the internal volume of the injector which results in a larger "dribble" flow, leading to poor combustion performance during engine shut down. However, this only represents a problem with vehicles that require an accurate terminal velocity (Sutton & Biblarz, 2001).

Propellant distribution is also a function of the type of element configuration used. Different types of injector elements are shown in Figure 2-4. The SAFFIRE injector contains triplet elements with two oxidiser streams impinging on a single centre fuel stream.

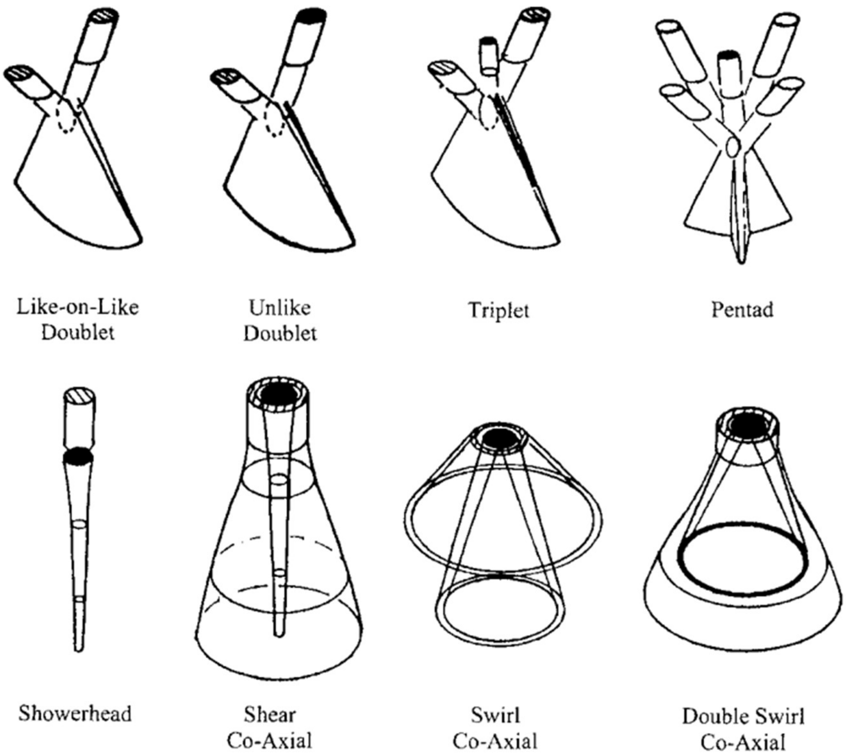


Figure 2-4: Injector element types (Ito, 1995)

Experimental methods were used to evaluate the performance of numerous element types and configurations using a single element setup, but these experiments do not consider inter-element mixing and rather focus on the element type and distribution (Elverum & Morey, 1959; Riebling, 1967; Rupe, 1973). An example of inter-element mixing is shown in Figure 2-5 for two like-doublet elements positioned next to each other.

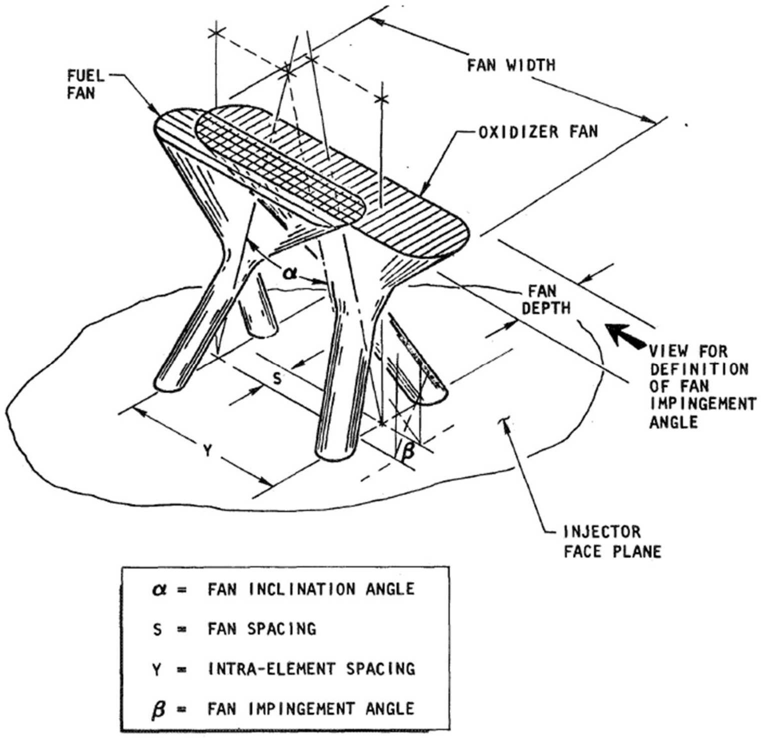


Figure 2-5: Geometric factors affecting inter-element mixing of like doublet elements (Falk & Burick, 1973)

An oxidiser – fuel – oxidiser (O-F-O) triplet element configuration has two oxidiser streams impinging on a single central fuel stream. This type of configuration is known for its increased performance due to high levels of mixing. However, triplet elements tend to have combustion stability issues (Huzel & Huang, 1992). Testing of propellant mixing for a liquid-liquid propellant combination is performed by collecting the resulting spray of two non-soluble liquids in a grid system during cold flow tests. An example of a test system designed to evaluate propellant distribution, developed by NASA in 1974, is shown in Figure 2-6. This setup was used to evaluate the distribution of non-circular elements (McHale, 1974).

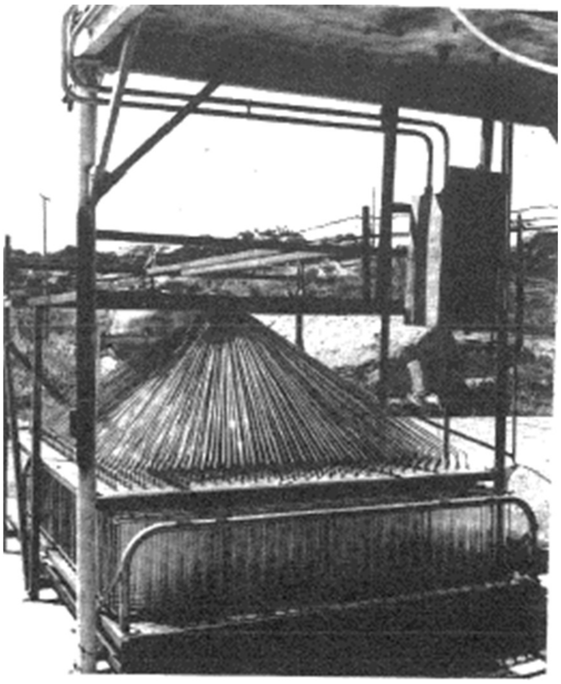

Figure 2-6: Injector spray collection system (McHale, 1974)

## 2.5 Atomization

Atomization is a process that takes place over several steps, with general agreement that there are at least two stages. Firstly, primary atomization is the initial disintegration of a jet into ligaments and large droplets. Secondary atomization is the breakup of the ligaments and large droplets into smaller droplets (Kenny, Moser, Hulka, *et al.*, 2006). The Weber number ($We$) is a dimensionless number used to describe a fluid's inertia compared to its surface tension. The Weber number is given by Equation 2-3:

$$We = \frac{\rho v^2 d_{\text{jet}}}{\gamma} \tag{2-3}$$

Here, $\rho$ is the density of the fluid, $v$ is the mean jet velocity, $d_{\text{jet}}$ is the undisturbed jet diameter and $\gamma$ is the surface tension of the fluid. Ohnesorge conducted a dimensionless analysis based on photographic experiments and classified liquid jet break up into three regimes based on a dimensionless parameter *Oh* (Ohnesorge number) (Wang, 2016). The *Oh* number is given by Equation 2-4:

$$Oh = \frac{\sqrt{We}}{Re} \tag{2-4}$$

Here, *Re* is the Reynolds number given by Equation 2-1. Based on the Ohnesorge number, a fluid's properties affect the atomization process, namely surface tension, density and viscosity. Therefore, when investigating atomization using an analogue fluid, these properties must be representative of the actual propellant. A fluid with a high surface tension and high density will resist breakup into smaller droplets. Similarly, a fluid's viscosity also prevents breakup into smaller droplet sizes. Therefore, fluids with a high viscosity have a larger average droplet size. This also increases the time taken for the fluid to atomize, as shown in Figure 2-7.

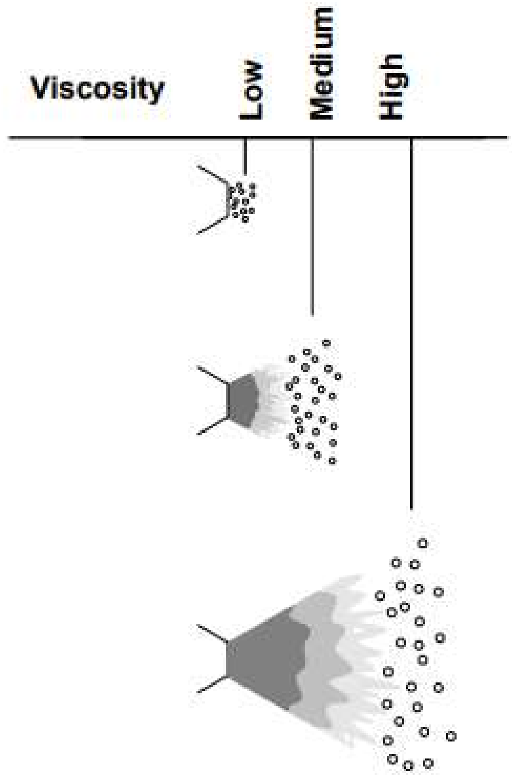


Figure 2-7: Viscosity, droplet size and time taken to atomize (Graco, 1995)

The atomization process in a liquid rocket engine is a pressure atomization, also known as airless atomization. This occurs when high pressures force a fluid through a small nozzle and the fluid emerges as a solid stream or sheet at high speeds. There is no high-pressure air supply directly assisting with the initial atomization. It is the friction between the fluid and the air/combustion gases that breaks up the stream into ligaments and eventually droplets. This process becomes more complex when multiple streams impinge on each other and the interaction also depends on the conditions present at the nozzle outlet. Factors such as the pressure and temperature can affect the atomization. The relative velocity between the fluid and the gas affects the droplet size, such that the higher the pressure before the nozzle the smaller the resulting droplet size due to the increased velocity (Rupe, 1973).

Modern methods of evaluating the droplet size produced by injectors include the use of laser scattering devices. These use a helium-neon laser beam and can measure the Sauter mean diameter (SMD) of droplets between 0.1 µm and 1750 µm (Bouziane, de Morais Bertoldi, Lee, *et al.*, 2017).

## 2.6 Existing Test Stands

Liquid rocket engine test stands are designed to test for specific performance parameters, namely thrust, combustion efficiency, combustion stability, pressure drop, propellant mixing and supercritical injection. Usually, test stands are designed with hot fire capabilities. However, they may also be constructed specifically for cold flow testing to quantify the performance of an injector safely. Both hot fire and cold flow test stands usually include a similar propellant feed system and control system. Several universities currently have hot fire and cold flow testing capabilities. A review of the published work is discussed below. Hybrid test stands are also reviewed for additional information due to the similar nature of the test stand requirements. An example of a university-based hot fire test stand is shown in Figure 2-8.

Figure 2-8: BURPG vertical hot fire test stand (Boston University, 2018)

The liquid test stand developed by the Boston University Rocket Propulsion Group (BURPG) has both cold flow and hot fire test capabilities. The fluid systems used include high-pressure nitrogen, liquid nitrous oxide and liquid isopropyl alcohol. The main valves operate using a high-pressure pneumatic system operating at 1000 psi (69 bar). Solenoid valves allow for reliable and quick operation of the main propellant valves. The system includes an engine purge capability that operates at 150 psi (10.3 bar) and is controlled by a servo-operated ball valve. A water suppression system is also included for hot fire tests and is remotely or automatically activated based on abort conditions (Boston University, 2018).

The Liquid Propulsion Laboratory (LPL) at the University of Southern California (USC) has a dedicated cold flow test stand for evaluating the flow coefficients of newly developed injectors, Figure 2-9. The flow coefficient, $C_{\mathrm{v}}$, is a measure of the flow rate through a device based on the pressure drop and the specific gravity of the fluid ($SG$). The flow coefficient is an imperial term, with the metric equivalent flow function, $K_{\mathrm{v}}$, given by Equation 2-5:

$$K_{\mathrm{v}} = Q\sqrt{\frac{SG}{\Delta P}} \tag{2-5}$$

Here, $Q$ is the mass flow rate, $\Delta P$ is the change in pressure across the flow device and $SG$ is the specific gravity of the fluid.

Figure 2-9: LPL cold flow test stand (Moruzzi *et al.*, 2018)

The LPL cold flow test stand uses a regulated blow down pressurization system with an electro-pneumatic regulator to control the tank ullage pressure. The mass flow rate was initially determined by monitoring the force produced by the spray of the injector. However, turbulence in the collector decreased the accuracy of the mass flow rate measurement. Therefore, the total mass collected over the test period was used to calculate the mass flow rate. Using this method, the LPL is able to evaluate the flow coefficient with an error of less than 1 % (Moruzzi *et al.*, 2018).

The University of California in San Diego designed a mobile test stand to hot fire test liquid rocket engines. The test stand is known as Colossus, Figure 2-10. The propellants, namely LOX and kerosene, are stored on the trailer in separate tanks made from seamless stainless-steel tubes with welded end flanges. A regulated blowdown system with a dome loaded regulator is used to drive the propellants to the combustion chamber. This ensures a constant ullage pressure is maintained for the full duration of the burn. The control system employs National Instruments hardware with a LabVIEW software control system. For cold flow testing, isopropyl alcohol and cryogenic liquid nitrogen were used as analogue fluids. Isopropyl alcohol was selected as it cleans the internal plumbing system while also acting as an analogue fluid (Seshadri, 2016).

Figure 2-10: Colossus test stand (Seshadri, 2016)

The University of Alabama in Huntsville has facilities for cold flow and hot fire testing. The injector spray facility is used to cold flow test liquid and gaseous rocket injectors with inert fluids. The cold flow testing outputs are the flow performance, spray patterns and droplet size. These outputs can be observed at atmospheric pressure or at high chamber pressures to simulate supercritical injection. Previous studies conducted using this facility include impingement and the self-pulsation of swirl coaxial bipropellant injectors (Kenny, Hulka, Moser, *et al.*, 2009). The facility uses high-speed video cameras, a 2-D phase Doppler particle analyser (PDPA) and a 2-D particle image velocimetry (PIV) system to capture atomization and velocity profile characteristics.

The design and commissioning of a bi-propellant liquid rocket engine test facility were conducted by Youngblood (2015) at the Energetic Materials Research and Testing Center (EMRTC) in New Mexico.

The facility was used to test an ethanol/nitrous oxide engine. The results were used to validate an in-house combustion model predicting engine performance based on the propellants flow rates and engine dimensions. The facility is separated into three sections, namely the test stand, control console and fuel galley. The SAFFIRE cold flow test facility will have a similar layout due to the safety benefits. The facility used a regulated blowdown system with nitrogen as the pressurant gas and single stage pressure regulators to control the tank ullage pressures. The chamber pressure decreased during the burn duration due to a decrease in tank ullage pressure. This was caused by the large pressure drop across the regulator and nitrogen tank valve. The facility controls the valves and data acquisition with National Instruments hardware, coupled with a LabVIEW control system.

## 2.7 Ground Support Equipment

When performing cold flow tests ground equipment is required to control the various steps within each experiment. LabVIEW software uses a graphical programming approach such that control, data acquisition and debugging are made easier through a visual approach. LabVIEW was selected as the control system software for the SAFFIRE test stand because of the experience and hardware held within the Aerospace Systems Research Group (ASReG) at the University of KwaZulu-Natal (UKZN). Furthermore, LabVIEW software has been used extensively for previous ground support equipment as it provides a user interface that facilitates the required operations (Miklaszewski, Dadson, Fox, *et al.*, 2013; Moruzzi *et al.*, 2018; Youngblood, 2015).

Test stands are composed of numerous sub-systems including the propellant feed system, pressurization system and data acquisition system. These components require a control system to operate as a complete system. Valves are required to control pressurization and flow within the system and provide the means to start, stop and throttle the propellants. Sensors are needed to measure temperature, pressure, valve positions and thrust in the case of hot fire tests. Often redundant regulators and relief valves are added to increase the reliability of the system. Regulated pressurization systems are relatively simple when compared to turbo-pump engine cycles because they have fewer moving parts. On flight weight vehicles helium is used for pressurization because of its lighter weight. However, nitrogen is used on static ground test equipment due to its availability and low cost (Ito, 1995).

## 2.8 CFD Verification and Validation

The American Institute of Aeronautics and Astronautics (AIAA) (1998) defines verification as the process of determining the accuracy of a computational model and the conceptual model it represents. Validation is the process of determining whether a computational model accurately represents the real-world physics. When validating a computational simulation, the accuracy is based on experimental data.

Verification involves determining the accuracy of a CFD analysis based on benchmark solutions of simplified model problems. The objective of verification is to identify and quantify errors in a computational model. The errors arise from four sources, namely spatial discretization, temporal discretization, lack of iterative convergence and programming errors. Spatial discretization error can be identified by performing a mesh independence study to evaluate how computational simulation results vary with a decrease in mesh size.

Validating a computational simulation according to the AIAA's recommendations is a lengthy process and involves a building block approach. Complex geometries must be separated into sub-systems which are tested individually. Often even unit problem geometries are manufactured to test for specific flow phenomena. A suggested validation assessment method is provided, in which the complete system is broken down to single-unit problems (Figure 2-11). The complete system is the actual hardware or system being validated. It includes all flow features and physical flow effects occur simultaneously. Complete system analyses often result in limited measurement data collected during operating conditions. Therefore, the data collected has a higher degree of uncertainty when compared to unit problems. The recommended phases are referred to as guidelines as practical limitations and the required degree of accuracy will dictate which phases are necessary (AIAA, 1998).

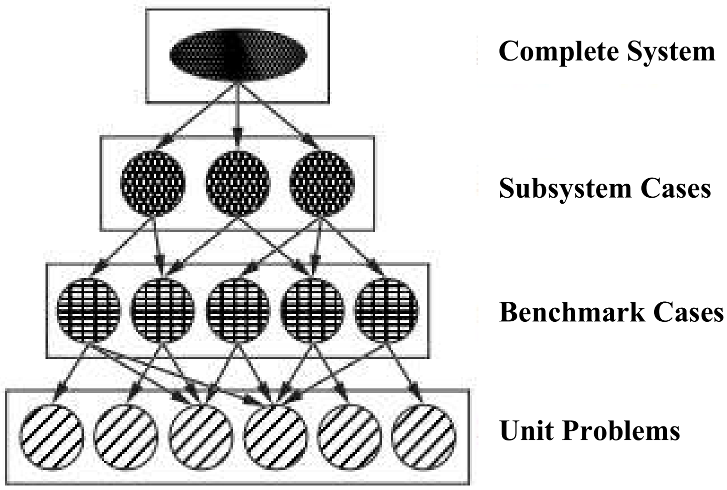


Figure 2-11: Validation phases (AIAA, 1998)

A general guideline suggested by Aeschliman and Oberkampf (1997) is to develop an experimental validation setup with the input of the CFD code developer. This process helps determine the necessary measurement tools to capture all flow physics and boundary conditions assumed by the flow. The development stages require this interaction. However, the process of collecting experimental and simulation data should be conducted independently to ensure there is no influence on either data set. A better understanding of both the simulation results and experimental data will result from comparing the final independent data from both sets. Oberkampf and Trucano (2002) also recommend supplying detailed information regarding the experimental setup and results, including accurate measurements instead of nominal or average data, surface roughness conditions, potential mismatches in experimental and simulation components and detailed dimensions of all model and instrumentation mounting locations.

Model geometry uncertainties arise from two sources, namely model geometry deviations and model imperfections. Geometry deviations are defined as differences between the fabricated model and the conceptual model arising from various sources, for example, poor tolerances, accidental surface damage or warpage from heating. Geometry imperfections are defined as model features that are not part of the original geometric model but which influence the flow and resulting measurements, for example, poor fabrication resulting in burrs on small diameter holes or a pressure leak between a pressure tap and transducer (Aeschliman & Oberkampf, 1998). Geometry deviations are a concern for the SAFFIRE injector as a small increase of an orifice diameter will result in a significant increase of the flow rate through the orifice.

Verification of a CFD simulation must be conducted before a CFD validation study to ensure a false validation does not occur (Pelletier, 2010). To verify a CFD simulation, a grid and time-step convergence analysis must be conducted. This involves evaluating the discretization error by refining the grid size and time step, ensuring the solution is independent of the mesh and time step size. When the grid size and time step approach zero the discretization error should asymptotically approach zero, excluding computer round-off errors (AIAA, 1998). Once in this asymptotic region, a Richards Extrapolation Method (REM) can be used to estimate the 'zero-grid' spacing and time step (Celik & Zhang, 1995). Another method of analysing the grid convergence is a Grid Convergence Index (GCI) (Roache, 1998). This is based on a generalised Richards extrapolation using two discrete solutions at two different grid spacings. This method provides an error band between the solution and asymptotic value. The GCI is given by Equation 2-6:

$$GCI_{1,2} = \frac{F_s \left| \frac{f_2 - f_1}{f_1} \right|}{r^p - 1} \tag{2-6}$$

Here, $F_s$ is a factor of safety equal to 1.25 for the comparison of three grids or more, $f$ is the monitored solution for the simulation and subscripts 1 and 2 describe the mesh size of the fine and coarse meshes respectively, $r$ is the refinement ratio and $p$ is the order of mesh convergence given by Equation 2-7:

$$p = \frac{\ln\left(\frac{f_3 - f_2}{f_2 - f_1}\right)}{\ln(r)} \tag{2-7}$$

Here, $f$ is the monitored solution for the simulation and subscripts 1, 2 and 3 describe the mesh size of the fine, medium and coarse meshes respectively. The GCI for the fine and medium meshes is given by $GCI_{1,2}$ and the medium and coarse meshes by $GCI_{2,3}$. A check can be conducted to determine if the grid convergence is within the asymptotic range. If the relationship given below in Equation 2-8 is true, it can be concluded that the grid convergence is within the asymptotic range.

$$\frac{GCI_{2,3}}{r^p \, GCI_{1,2}} \approx 1 \tag{2-8}$$

Lastly, by applying the REM the continuum value, $P_{\text{zero}}$, for a zero-grid spacing can be calculated from Equation 2-9:

$$P_{\text{zero}} = f_1 + \frac{f_1 - f_2}{r^p - 1} \tag{2-9}$$

Martin (2019) conducted CFD verifications on the SAFFIRE injector simulations and concluding that the simulation results were independent of the mesh size for the fuel and oxidiser domains. Since the simulation fluid properties were updated to the analogue fluid (water) and the volumes were remeshed, new simulation verifications were required for each domain.

Two practical issues must be considered during a validation assessment. First, the number of validation test cases and the accuracy required by each test case are highly application-dependant. Secondly, the validation process must be realistically achievable. To summarise, the validation process must allow for varying levels of accuracy with possible incremental improvements as time and funding permit (AIAA, 1998).

## 2.9 Turbulence Models and Wall Functions

To model a flow accurately within any CFD package, the physics models selected must accurately represent the conditions in which the flow exists. One physics model parameter that has a large impact on the results of a CFD simulation is the turbulence model. When turbulence is present, it is often the dominant flow phenomenon. Therefore, successfully modelling the turbulence will increase the accuracy of the results produced by the simulation (Sodja, 2007). Currently, two predominant methods exist for modelling turbulence, namely Direct Numerical Simulations (DNS) and Reynolds Averaged Navier-Stokes (RANS). Using a DNS involves modelling the large-scale turbulent fluctuations using time-dependent Navier-Stokes equations. This process requires immense computational resources because of the small-time step requirements. A simplification of this approach is the Large Eddy Simulation (LES) where only the large-scale turbulence is simulated directly with the smaller turbulence scales being modelled. Alternately, Reynolds Averaged Navier-Stokes (RANS) models are used to determine the averaged turbulence within a flow. Since the RANS equations are not being solved directly, models are required to calculate the Reynolds stresses. These models consist of simple eddy viscosity models or more complex models that solve the transport equations directly for the Reynolds stresses (Hirsch, 2007). A RANS model was selected to model the performance of the SAFFIRE injector.

The selected wall treatment of a CFD simulation dictates how the boundary layer is resolved. A low wall $y^+$ model solves the viscous sublayer near the wall whereas a high $y^+$ solver will approximate the boundary layer with a wall function. An all-wall $y^+$ model is a hybrid approach that solves the boundary layer using the low $y^+$ wall treatment for fine meshes and the high $y^+$ wall treatment for coarse meshes. A blending function is used for the transition layer (Siemens, 2019). Utilising a low $y^+$ wall treatment usually leads to an increase in computational resource requirements due to the addition of multiple small cells near the wall required to capture the viscous sublayer.

## 2.10 Uncertainty and Error

Measurement uncertainty is described as the parameter that is associated with a measured value. It characterizes the variation from the measurand, which is also referred to as the 'true value'. This uncertainty serves to represent the validity of a measured value, such that the measured value can be compared to those of other studies or measurements. Uncertainties can be evaluated by two different methods. A type A evaluation is derived statistically and a type B evaluation is determined through means other than statistical analyses. For example, calibration certificates, device specifications and published information are type B uncertainties. Type A uncertainties are determined through a probability distribution and the resulting uncertainties are expressed as a variance or standard deviation. Type B uncertainties are expressed as an estimated variance or standard deviation based on an assumed probability density function. The SAFFIRE injector test stand will contain type A and type B uncertainties.

Errors are classified as either random or systematic and arise during the measurement process. Random errors cannot be quantified, but they can be reduced by increasing the number of observations. These random effects are caused by unpredictable spatial and temporal variations of influencing quantities (JCGM, 2008). Systematic errors are unavoidable but can be reduced. If a reoccurring effect is recognised and quantified, it can be accounted for using a correction factor. However, this is only deemed appropriate if the quantity is significant when compared to the required accuracy of the measurement. Once a correction factor has been applied, the identified systematic error is assumed to be zero. Unidentified systematic errors still exist within the system. A calibration testing campaign was conducted with the SAFFIRE injector test rig to identify and mitigate systematic errors.

Repeatability is defined as the closeness of agreement between numerous measurements of the same measurand carried out under similar conditions (JCGM, 2008). Variables that must be kept constant include the observer and the measurement procedure. Additionally, the location must be the same and the period must be short. Reproducibility is described as the agreement of measured results of the same measurand that are carried out under different conditions (JCGM, 2008). The different conditions include the method of measurement, the observer, the measuring instrument, location and time. Reproducibility can be expressed quantitatively in terms of the dispersion characteristic of the result.

Experimental standard deviation $s(q_{\mathrm{j}})$ characterises the dispersion of the quantities $(q_{\mathrm{j}})$ of a repeated number of measurements ($n$) from the mean of the data set ($\bar{q}$) and is given by Equation 2-8:

$$s(q_{\mathrm{j}}) = \sqrt{\frac{\sum_{\mathrm{j}=1}^{n}(q_{\mathrm{j}} - \bar{q})^2}{n-1}} \qquad (2\text{-}8)$$

The standard error of the mean (SEM) describes the standard deviation of numerous mean quantities $\bar{q}_{\mathrm{k}}$, obtained from multiple repeated data sets ($n$). The SEM indicates the repeatability of numerous experiments as it shows how the repeated mean measurements vary. The SEM is a standard deviation given by Equation 2-9:

$$SEM = \frac{s(\bar{q}_{\mathrm{k}})}{\sqrt{n}} \qquad (2\text{-}9)$$

Here, $s(\bar{q}_{\mathrm{k}})$ is the standard deviation of the mean quantities $\bar{q}_{\mathrm{k}}$.

A measurement device's resolution also contributes to the uncertainty of its measurement. Since a range of input signals to the measurement instrument may result in the same output there exists uncertainty as to the true measurement of the device. This uncertainty can be described by a rectangular probability distribution using Equation 2-10:

$$\sigma^2 = \frac{(\delta)^2}{12} \tag{2-10}$$

Here $\delta$ is the resolution of the device and $\sigma$ is the standard deviation.

Measurements obtained by a calibrated device will have a specific uncertainty based on the device's specifications. This variance can then be assumed for the measurement obtained using the device. Furthermore, if a device's uncertainty is not provided, an assumption of the measurement variance can be made based on the nature of the measuring apparatus.

Often the process of determining a measurement will involve numerous uncertainties. These can be combined and are defined as combined standard uncertainties. Combined standard uncertainties are expressed as a standard deviation. When adding uncertainties together the combined standard uncertainty can be calculated by equating the positive square root of the variances (JCGM, 2008). The type A and type B uncertainties for the SAFFIRE injector test rig will be combined to determine the combined standard uncertainty.

An expanded uncertainty *U* is determined by multiplying the combined standard uncertainty with a coverage factor *k*. This term *U* expresses a large range that has a high probability of encompassing the measurand. The *k* value usually ranges from two to three and is determined by the coverage probability or level of confidence required for the interval (JCGM, 2008). The coverage factor can be reduced by increasing the number of observations. The number of observations for the SAFFIRE injector tests will depend on the required accuracy and selected confidence level.

The calculation of the coverage factor and confidence intervals is dependent on the sample size and probability distribution. For example, consider a set of three pressure transducer readings obtained from three repeated experiments that are normally distributed. For a 99% confidence interval, containing three observations, the required coverage factor is 9.925. If the number of observations is increased to six, then the required coverage factor for a 99% confidence interval is 4.032. This example demonstrates how even a small increase in observations can decrease the uncertainty of a measurement. The coverage factor is dependent on the sample distribution of the collected data. The central limit theorem states that the distribution of a sum of random variables will tend more towards a normal distribution as the number of random variables increase. Experimental error comprises the summation of numerous random errors. Therefore, it can be concluded by the central limit theorem that the experimental measurements will follow a normal distribution (Peters, 2001).

Furthermore, statistical analysis is usually conducted on an average of experimental data that contains numerous random errors. This is the further summation of values that are likely to be normally distributed. Based on central limit theorem, it can be assumed that measurement data is normally distributed irrespective of the sample size (Peters, 2001).

A confidence interval for a normally distributed population with a known population mean $\mu$ and variance $\sigma^2$ can be calculated using equation 2-11:

$$\bar{x} - \frac{z_{\alpha/2}\,\sigma}{\sqrt{n}} \leq \mu \geq \bar{x} + \frac{z_{\alpha/2}\,\sigma}{\sqrt{n}} \qquad (2\text{-}11)$$

Here, $n$ is the number of samples, $\bar{x}$ is the sample mean and $Z_{\alpha/2}$ is the upper percentage point of the standard normal distribution. For a population where the mean $\mu$ and variance $\sigma^2$ is unknown a common solution is to replace the population standard deviation $\sigma$ with the sample standard deviation *S*. The variable $Z$ determined from the normal distribution table is now given by *t* obtained from a t-distribution table. The confidence interval can then be calculated using Equation 2-12. The t-distribution is symmetrical like a normal distribution. However, the t-distribution variance is dependent on the degrees of freedom which is defined as *n*-1 where *n* is the sample size (Peters, 2001).

$$\bar{x} - \frac{t_{\frac{\alpha}{2},n-1}\,S}{\sqrt{n}} \leq \mu \geq \bar{x} + \frac{t_{\frac{\alpha}{2},n-1}\,S}{\sqrt{n}} \qquad (2\text{-}12)$$

## 2.11 Orifice Pressure Drop and Tolerances

The pressure drop through an orifice for a given mass flow rate can be calculated using Equation 2-13. Equation 2-13 is based on an empirical model for orifices with a length to diameter ratio (L/D) of 3 or greater since the coefficient of discharge ($C_d$) approaches 0.82 asymptotically when L/D $\geq$ 3.

$$\Delta P = \frac{0.811 K \dot{m}^2}{\rho d^4 N_{\text{orifices}}^2} \qquad (2\text{-}13)$$

Here, $\Delta P$ is the change in pressure across the orifice, *K* is the head loss coefficient equal to 1.7 for sharp edge orifices, $\dot{m}$ is the mass flow rate, $\rho$ is the density of the fluid, *d* is the diameter of the orifices and *N* is the number of orifices. This equation can be used to estimate the change in flow rate through an orifice resulting from machining tolerances. Orifices with small diameters become sensitive to slight diameter variations due to machining tolerances. Often, a $\Delta P$ tolerance is provided based on the orifice diameter or in locations where an unpredictable flow will cause engine failure. The J-2 injector employed such precautions. Smaller $\Delta P$ tolerances were specified on the elements near the combustion chamber wall. For low $\Delta P$ tolerances, a machining tolerance of ±0.005 mm is recommended. For a larger $\Delta P$ tolerance, machining tolerances of +0.008 mm are acceptable (Gill & Nurick, 1976).

## 2.12 SAFFIRE Injector

In this study, the performance of the SAFFIRE injector was validated using the SAFFIRE injector test rig. The test rig design and testing methodology were developed considering the literature reviewed in this chapter. A cross-section of the SAFFIRE injector designed by Martin (2019) is shown in Figure 2-12. It consists of two separate flow domains, namely the fuel and oxidiser domains. These domains have separate inlets and outlets. Therefore, the propellants are kept separate until they mix, atomize and combust within the combustion chamber. The validation of the SAFFIRE injector included determining the required pressure drops for the oxidiser and fuel domains, for the design mass flow rates, and the propellant distribution for each domain. Separate fuel and oxidiser test articles were manufactured and tested independently on the SAFFIRE test stand.

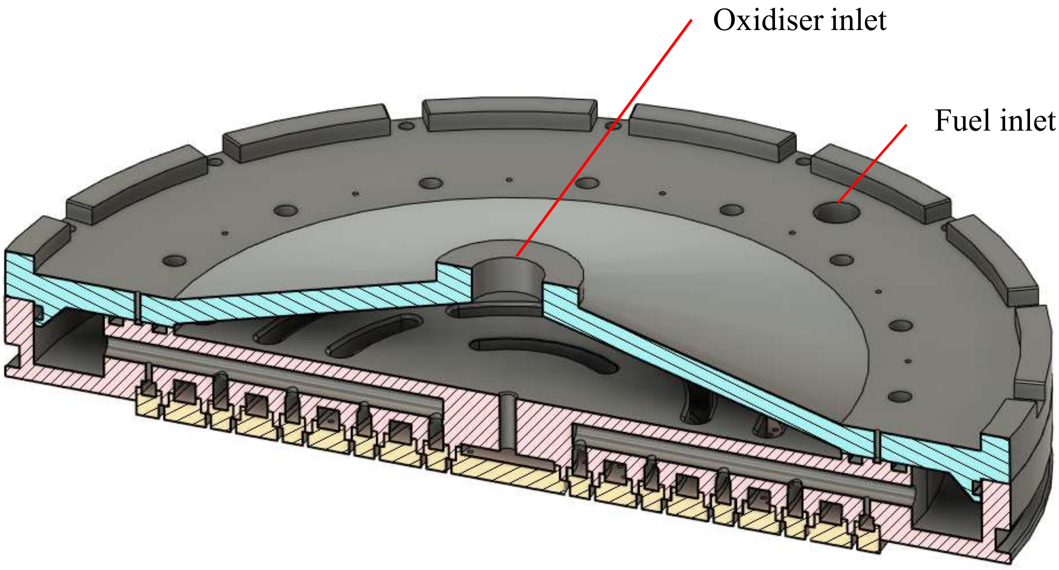


Figure 2-12: SAFFIRE injector cross-section (Martin, 2019)

# 3. Test Facility

## 3.1 Introduction

The test facility was designed to determine the performance of a liquid rocket engine injector through cold flow testing techniques. In this study, the test facility was developed to validate the modelled performance of the SAFFIRE injector conducted by Martin (2019). The required performance parameters were the fuel and oxidiser inlet pressures for a mass flow rate of 2.28 kg/s and 5.66 kg/s respectively. Secondly, the propellant distribution for each domain was determined. The facility used a regulated blowdown system to drive the analogue fluid to the injector inlet at a constant pressure. Therefore, the cold flow testing results were collected during a relatively steady-state period. The test facility layout ensured all test articles and high-pressure zones were isolated during testing.

## 3.2 Test Facility

The test facility contains three main areas, namely the control room, tank storage and blast-proof area. The tank storage and blast-proof area are considered hazardous and are closed off when testing is conducted. The control room is a safe area where test personnel conduct and monitor test proceedings. The test facility schematic and zone designations are shown in Figure 3-1.

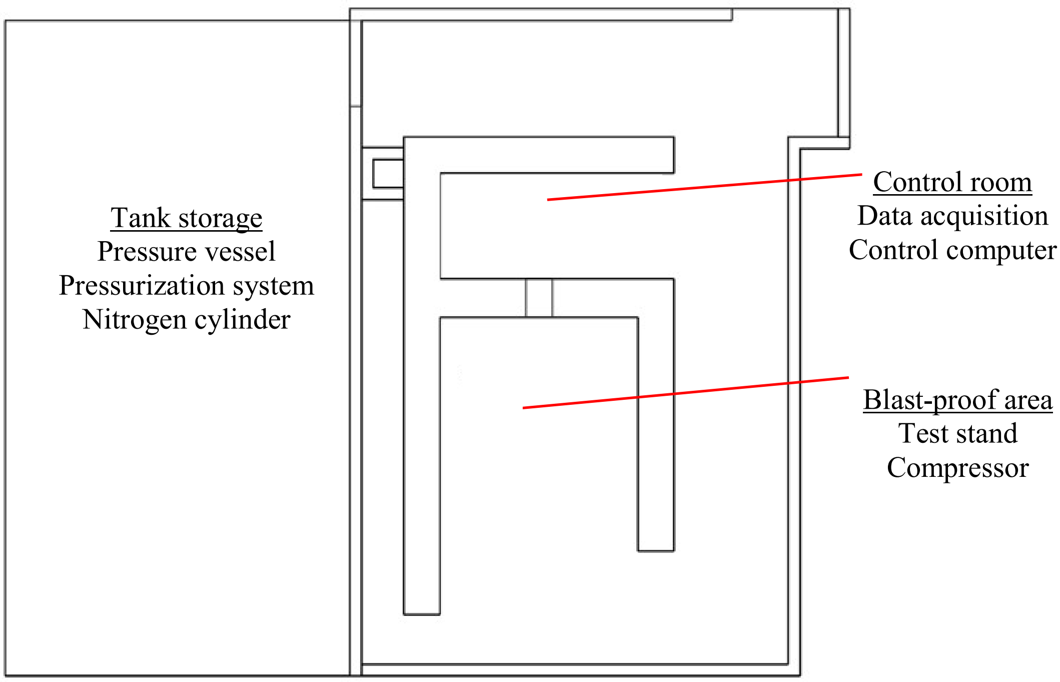


Figure 3-1: Test Facility Layout

### 3.2.1. Propellant Storage & Pressurization

The analogue fluid was driven to the injector inlet by a high-pressure ullage in the propellant tank. The ullage pressure was controlled using a single-stage regulator and a high-pressure nitrogen supply. The pressure vessel and pressurization system are shown in Figure 3-2. The nitrogen was supplied to the tank through a flexible hose with a quick connect fitting. The pressurisation inlet was located at the top of the tank.

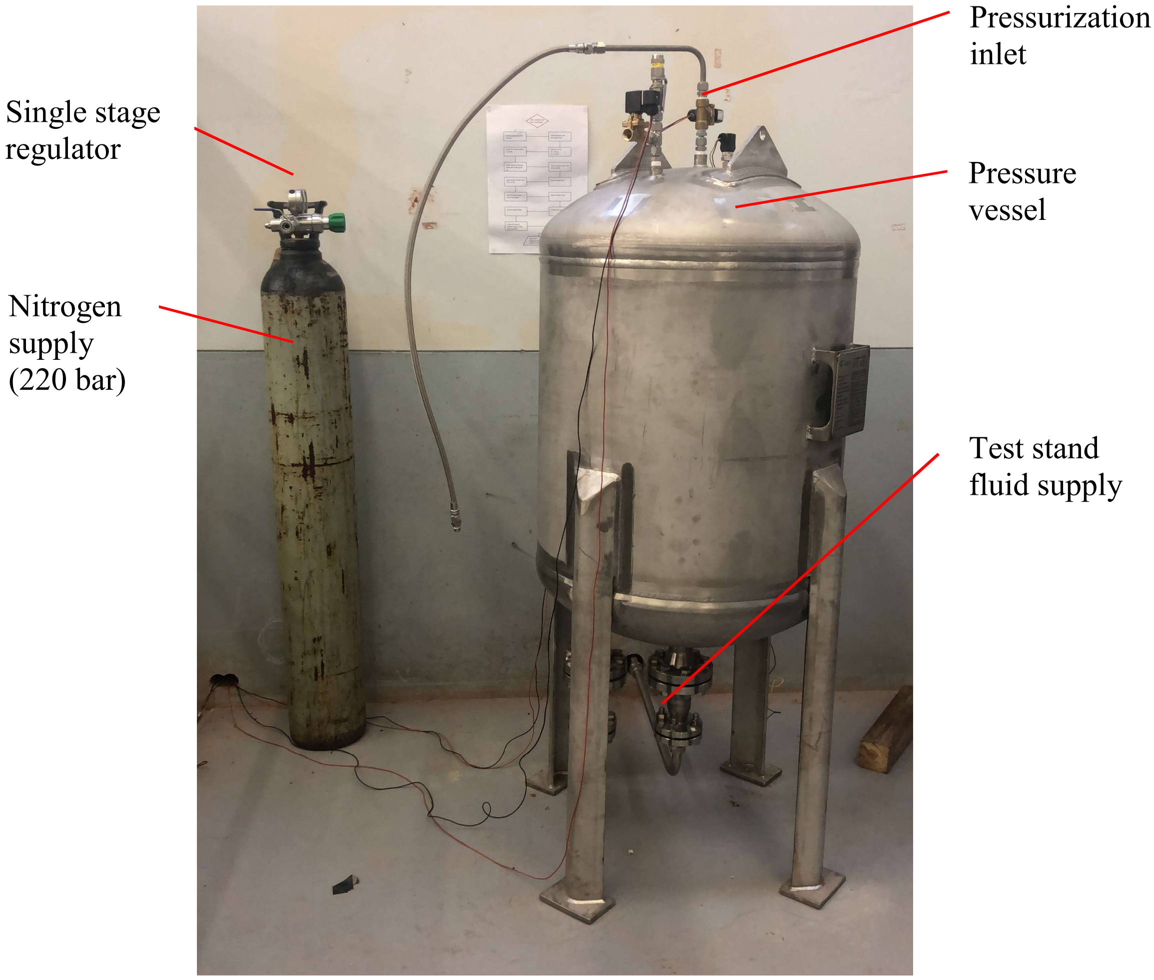


Figure 3-2: Pressure vessel and pressurization system

The pressurization system components were in the tank storage area and included the pressure vessel and a high-pressure nitrogen cylinder. A discharge line ran through the wall to supply the test stand with the analogue fluid.  The pressure vessel was designed to the American Society of Mechanical Engineers (ASME) standards with a working pressure of 30 bar. Seven ports on the tank were used to pressurize, vent, fill/drain, inspect and monitor the tank level. The pressure vessel was designed for category four hazardous fluids, namely kerosene and helium. For the cold flow tests the pressure vessel contained water and nitrogen. The water supply was filtered through a y strainer to remove any small particle contamination. This filter reduced the probability of blocking an orifice or passage within the injector. The tank was filled through the fill port located at the bottom of the tank. The fill port was positioned next to the main discharge port running to the test stand. The tank fluid level was inspected before pressurizing the tank using a manometer located next to the tank.

The pressurization system included two solenoid valves, one pressure relief valve, two pressure transducers, a high flow single stage pressure regulator and a high-pressure cylinder of nitrogen. These components are shown in Figure 3-3. The pressure transducer located on the tank monitored the ullage pressure during operation. The pressure transducer on the pressurant inlet line measured the set pressure of the regulator before a test was conducted. This was required as the analogue gauge on the outlet side of the pressure regulator lacked the required resolution for setting the outlet pressure. A solenoid valve was positioned between the regulator and the tank pressurant inlet to control the inlet of pressurant.

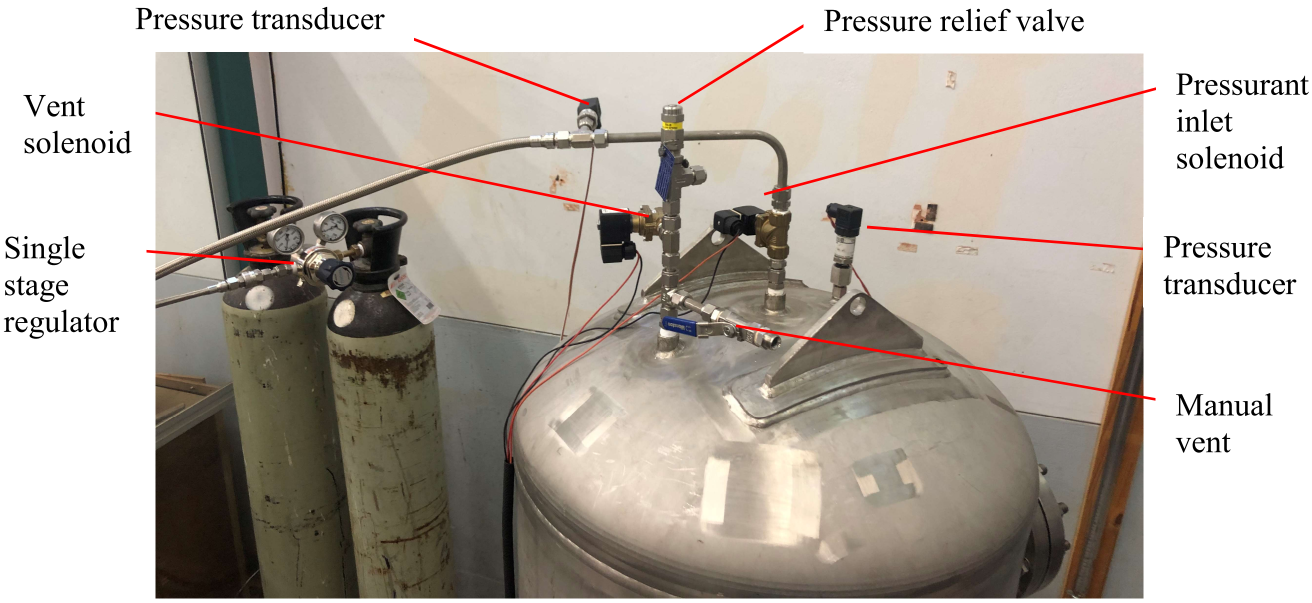


Figure 3-3: Pressurization system

A piping and instrumentation diagram (P&ID) is shown in Figure 3-4. It shows all the major components used by the test stand. The components functions are given in Table 3-1.

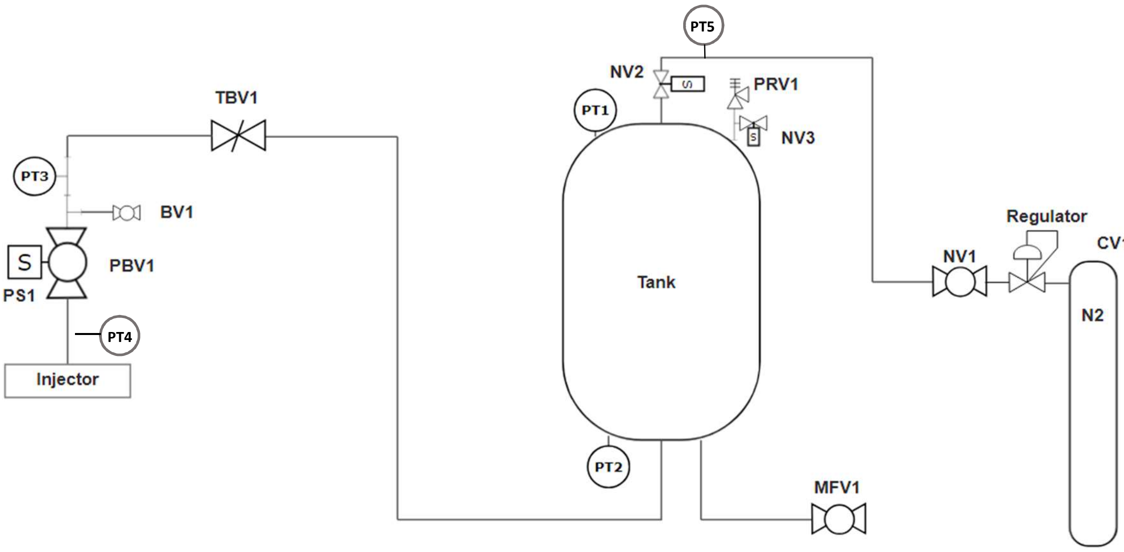


Figure 3-4: P&ID

Table 3-1: P&ID description

| Code | Description | Function |
|---|---|---|
| CV1 | Manual cylinder valve 1 | Isolates the nitrogen cylinder |
| N2 | Nitrogen cylinder | High-pressure nitrogen supply |
| Regulator | Single-stage high flow regulator | Regulates the nitrogen supply |
| NV1 | Solenoid nitrogen valve 1 | Isolates the pressurant system |
| NV2 | Solenoid nitrogen valve 2 | Pressurant inlet valve |
| PRV1 | Pressure relief valve 1 | Passive pressure relief (30 bar) |
| NV3 | Solenoid nitrogen valve 3 | Vent valve |
| PT1 | Pressure transducer | Ullage pressure |
| PT2 | Pressure transducer port | Manometer port |
| TBV1 | Throttle butterfly valve | Throttling capabilities |
| PT3 | Pressure transducer | Main valve inlet pressure |
| BV1 | Manual purge ball valve | Purge line |
| PBV1 | Pneumatic ball valve 1 | Main isolation valve |
| PT4 | Pressure transducer | Injector inlet pressure |
| PS1 | Pilot solenoid valve | Main ball valve actuation |
| MFV1 | Manual fill valve | Fill and drain |
| PT5 | Pressure transducer | Pressurant line pressure |
| Injector | Fuel or LOX test articles | Injector mounting position |

### 3.2.2. Test Stand

The test stand shown in Figure 3-5 was positioned in the blast-proof area. It includes the supply line, a throttle valve and the main pneumatic valve. The compressed air supply for the main pneumatic valve was supplied from a compressor also stored in this area. Figure 3-5 displays the test stand during the assembly process without an injector test article attached. The throttle valve was left fully open during testing as throttling was not investigated. The supply line is shown at the bottom of the tank coming through the wall from the pressure vessel. The supply line runs up the side of the test stand where it is isolated by the main ball valve.


Figure 3-5: Test stand during assembly in a blast-proof area

A detailed view of the test stand with the fuel test article attached is shown in Figure 3-6. The main additions from Figure 3-5 are the test article, injector mounting bracket and two pressure transducers.

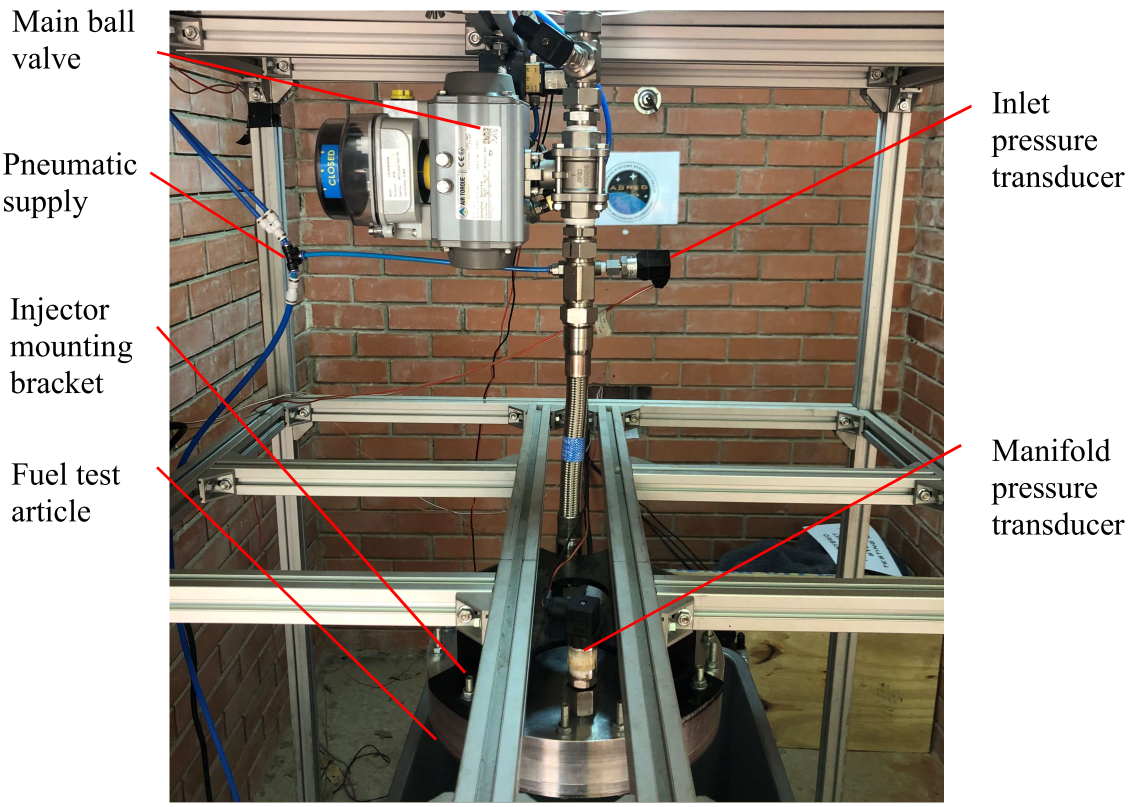


Figure 3-6:Test stand with the fuel test article

### 3.2.3. Control Room and System

The control room contained the control computer and the data acquisition hardware. The control system included a National Instruments (NI) C-DAQ chassis and input modules with a LabVIEW software user interface to control the cold flow test stand. The hardware for the data acquisition and control system was wired into the control box located in the control room. This included the NI chassis and modules, a relay for hardwired safety switches and a 24 V power supply as shown in Figure 3-7.

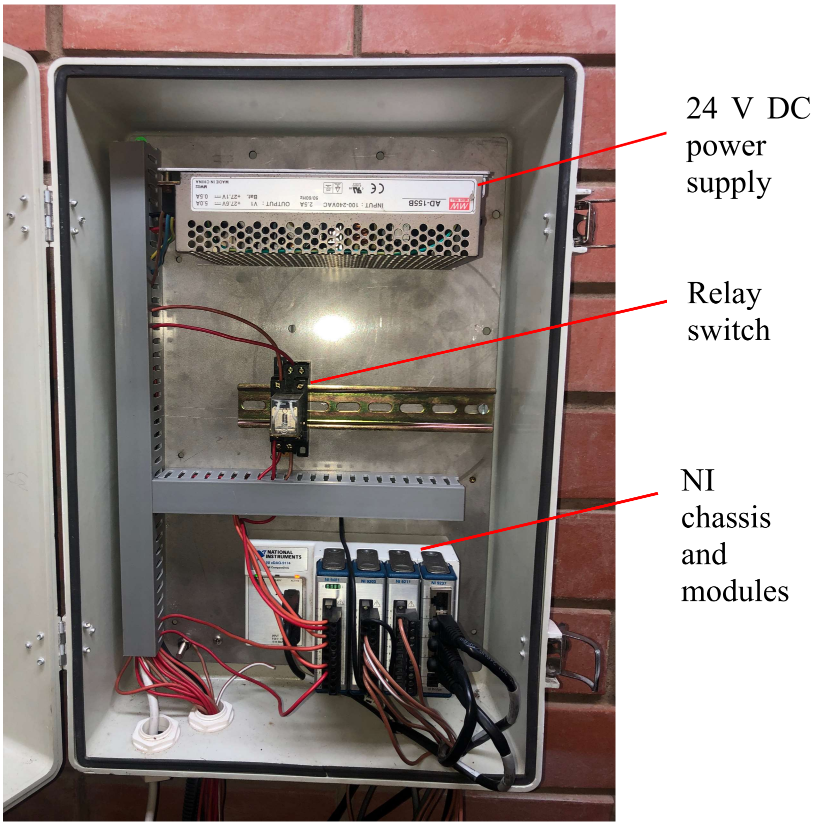


Figure 3-7: Main control box

The designed LabVIEW user interface is shown in Figure 3-8. The required inputs are listed in Table 3-2 with the default values. The control system automatically sequenced the timing of the valves after the 'start auto sequence' button was activated.

Table 3-2: LabVIEW input parameters

| Parameter | Description | Default input |
|---|---|---|
| Auto vent set pressure | Pressure at which safety valve opens | 15 bars |
| Test duration | Total test time | 12 seconds |

The LabVIEW user interface (UI), Figure 3-8, allowed the user to operate the pneumatic valve, pressurisation inlet valve and vent valve without starting the auto sequence. This control setup allowed the user to prime the injector before a test. Furthermore, the user could increase or decrease the tank ullage pressure using the inlet and vent solenoids. The UI displayed the real-time pressure measurements taken during a test. The safety panel allowed the user to set the auto vent pressure and displayed the tank ullage pressure.


Figure 3-8: LabVIEW cold flow testing user interface

Before a test was conducted the following procedure was followed.

1. The pressure vessel was filled with water to 90 % of its 378 L capacity. The level measurement of the tank was obtained using a manometer installed next to the tank.
2. The pressurant supply line was connected to the tank with a flexible hose and quick connect fitting.
3. The regulator outlet pressure was set to the required ullage pressure plus the regulator pressure ‘droop’ for the mass flow rate of nitrogen through the regulator.
4. The final test stand safety and setup procedure shown in Figure 3-9 was conducted. This involved ensuring the tank and test stand were set up correctly before the auto sequence was started.

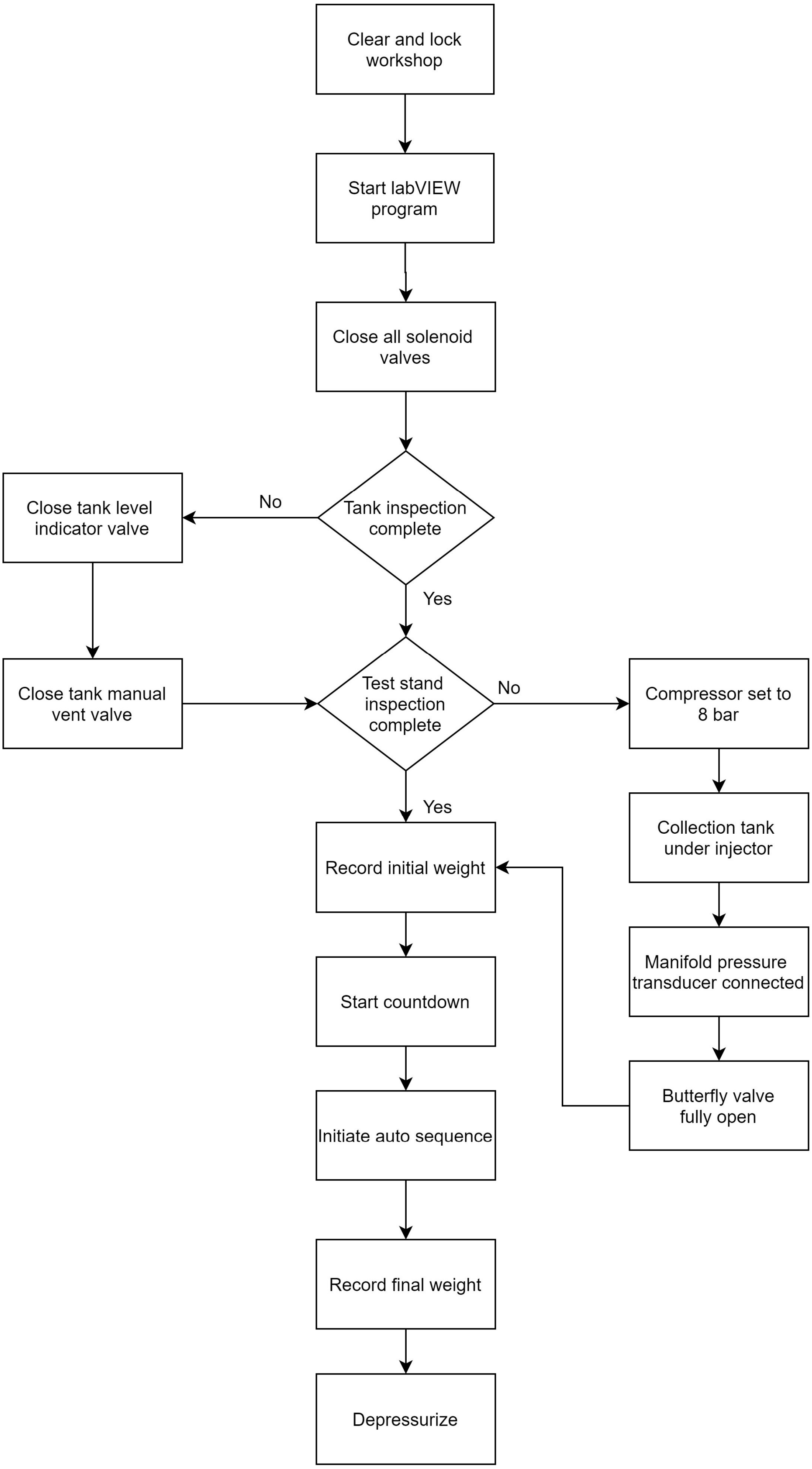


Figure 3-9: Testing safety and setup procedure

The control system comprised two loops, namely the main loop and the auto sequence loop. The main loop controlled the data acquisition and valve relays. The safety relief code ran within this loop. If the tank ullage pressure exceeded the user-defined vent pressure, the vent solenoid valve would open. Furthermore, if the control system were to malfunction a permanently installed pressure relief valve with an opening pressure of 30 bar acted as a fail-safe. The main loop control code is shown in Appendix D.

The data acquisition module in this loop reads all the pressure transducer inputs at a rate of 500 Hz. The received outputs from the pressure transducers are 4-20 mA current signals. These signals are scaled depending on the pressure transducer range. The pressure transducer type, measurement range and accuracy are listed in Table 3-3.

Table 3-3: Pressure transducer locations and accuracies

| Pressure transducer location | Type | Measurement range (bar) | Accuracy (%) |
|---|---|---|---|
| Pressure vessel ullage | S10 | 0-25 | 0.25 |
| Pressurant line inlet | A10 | 0-100 | 0.5 |
| Main ball valve (upstream) | A10 | 0-100 | 0.5 |
| Main ball valve (downstream) | A10 | 0-100 | 0.5 |
| Manifold pressure tap | S10 | 0-25 | 0.25 |

Once the start auto-sequence button was activated the following actions were initiated. The program was instructed to save all incoming data to file. The main pneumatic valve and the pressurant inlet solenoid valve were instructed to open simultaneously. After the user-defined duration 'test duration' had passed the main pneumatic and inlet solenoid valves were instructed to close. Writing the data to file was halted after an additional five seconds to ensure all data collected had been recorded. The output file contained pressure readings obtained during the test duration. The weight of the water collected was measured and recorded with the data file for post-processing. The LabVIEW auto sequence loop is given in Appendix D.

The auto sequence controlled the timing of the main pneumatic ball valve and the pressurant solenoid valves. A LabVIEW relay module was used to operate the tank solenoid valves and the pilot solenoid valve that supplied the main pneumatic ball valve with 8 bar air pressure. These solenoids were powered by the 24 V DC power supply. The solenoids positions and functions are listed in Table 3-4. Each tank solenoid valve was hardwired with a manual override switch. In the event of a loss of power, the system was designed such that the solenoid valves would return to the normally closed position.

Table 3-4: Solenoid valve locations and function

| Valve | Function | Activation |
|---|---|---|
| Vent solenoid | Depressurizes the tank | Automatic |
| Pressurization inlet solenoid | Controls the inlet of pressurant gas | Automatic with manual override switch |
| Pilot solenoid | Controls the actuation of the main pneumatic ball valve | Automatic with manual override switch |

## 3.3 Summary

The test facility was designed to cold flow test injectors in a safe and controlled environment. The control system used NI hardware and LabVIEW software to control valves and monitor pressures at various points on the test stand. Software with an in-built ‘auto sequencer’ managed the valve timing, pressurisation system and data acquisition during the duration of a test. The test duration was a user-defined input. The system also included hardwired safety switches to safely shut down the system in the event of a control system malfunction.

# 4. Test Stand Calibration

## 4.1 Introduction

Initial tests aimed to determine and quantify the timing and mechanical delays produced by the control system and physical hardware. Furthermore, experiments were conducted to determine the line losses through the piping system. These losses were added to the estimated injector pressure drop to set the tank ullage pressure, such that the required fuel and LOX mass flow rates were achieved. Sources of error and uncertainty were identified and mitigated during the calibration process where possible.

## 4.2 Tank Pressure and Manifold Pressure Relationship

A regulated blowdown system was used to pressurise the analogue fluid during testing. This included a high-pressure nitrogen cylinder and a single stage regulator. The regulator reduced the cylinder pressure from 220 bar to a user-defined outlet pressure. The regulator was responsible for maintaining a constant tank pressure during the full duration of a test. Initial testing concluded the regulator was unable to maintain a constant tank pressure because of the high mass flow rate of nitrogen producing a pressure drop across the regulator. This is known as pressure ‘droop’ and is illustrated in Figure 4-1.

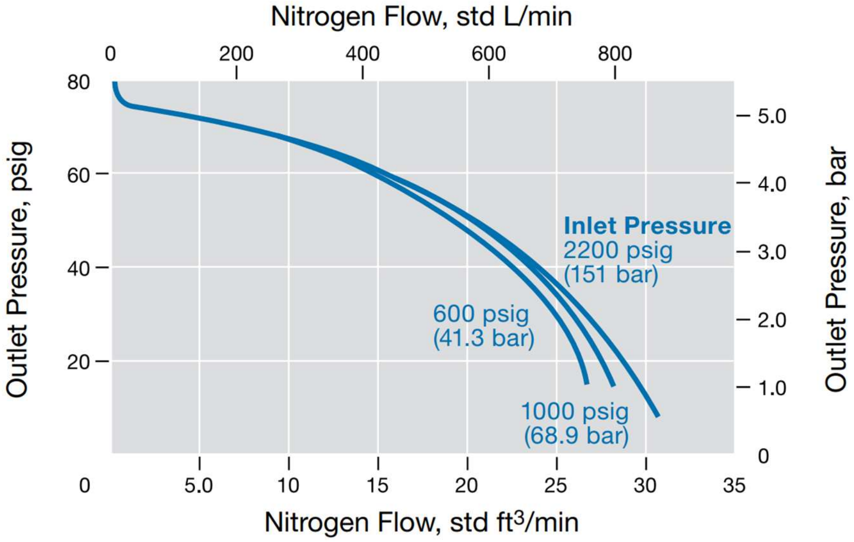


Figure 4-1: Swagelok regulator pressure curve

The regulator droop resulted in a decrease of the manifold pressure over the test duration, shown in Figure 4-2. This graph resembles the injector pressure of a simplified blowdown system without a regulator. This is the result of an insufficient amount of nitrogen entering the tank.

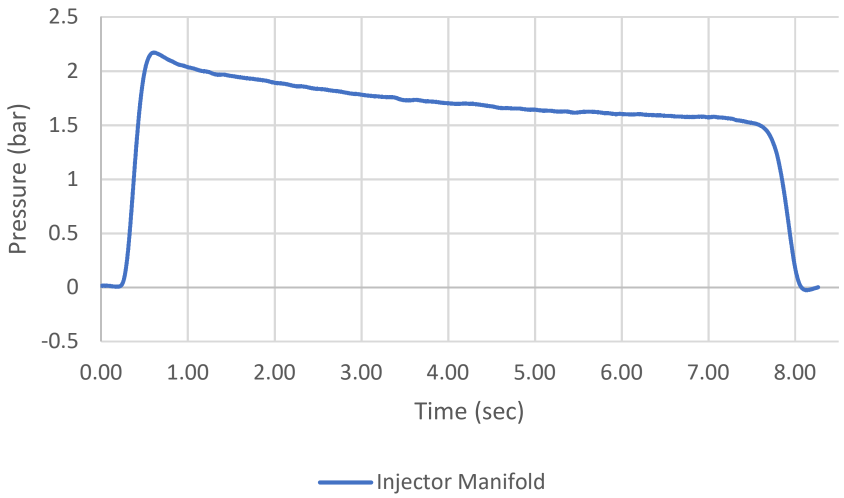


Figure 4-2: Decreasing manifold pressure

A solenoid valve was installed between the regulator and the tank pressure inlet port. This solenoid was referred to as the pressurant inlet solenoid described in Chapter 3. The solenoid decoupled the tank and the regulator pressures. It allowed for an initial increase of the regulator pressure to account for the regulator pressure 'droop'. This allowed the tank pressure to be set to the required ullage pressure to deliver the correct mass flow rate while ensuring a constant tank pressure was maintained during the duration of a test. The increased regulator pressure would drop to the tank ullage pressure due to the regulator pressure 'droop', resulting in a constant tank ullage pressure.

An auto sequence is a list of commands that are carried out sequentially once the initial command has been activated. The auto sequence described in Chapter 3 ensured the inlet solenoid valve opened simultaneously with the main ball valve. The tank ullage pressure fluctuations were reduced to 0.1 bar using this method. The manifold pressure using this method is shown in Figure 4-3.

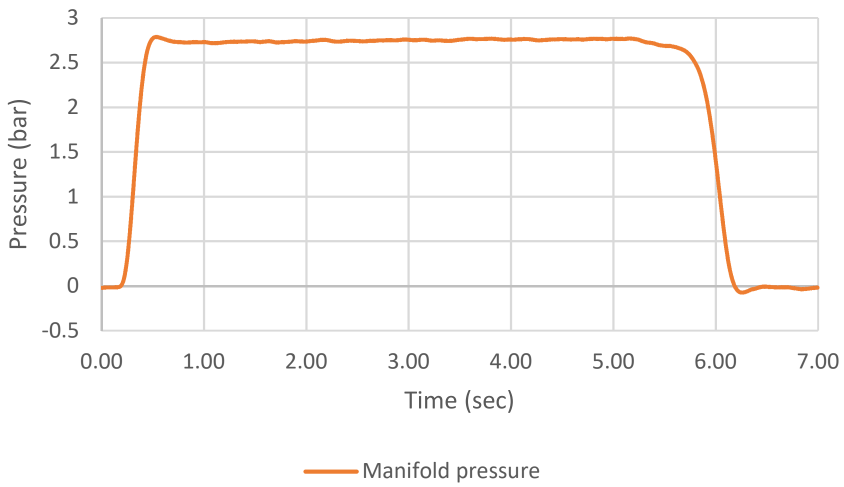


Figure 4-3: Relatively constant manifold pressure

## 4.3 Line Loss Calibration

Calibration tests were conducted to determine the accuracy of the analytical pressure drop through the piping system. The analytical values were based on the major and minor losses through the lines and fittings for the design mass flow rate. The tank ullage pressure was set to overcome the estimated line losses for the resulting mass flow rates of 2.28 kg/s and 5.66 kg/s. The mass flow rates were determined and the initial ullage pressure was altered accordingly until the correct mass flow rates were achieved. The line pressure drops were 1.8 bar and 5.3 bar for the fuel and LOX mass flow rates respectively. These pressure drops were used to set the starting tank pressure for the final injector tests. The actual line pressure drop varied from the analytical calculations by roughly 10%. This was due to the incorrect estimation of the flow coefficient for the butterfly valve.

## 4.4 Main Valve Calibration

A positioner located on the main pneumatic ball valve contained two relays and a visual display to indicate an open or closed position. The opening and closing times of the main valve were determined using the two relays. The mechanical switches on the relays were activated by two cams on the main valve stem. An example of a positioner with the opening cam (green) and closing cam (red) are shown in Figure 4-4. Calibration tests were conducted to compare the opening and closing valve times to the total test duration. A longer start-up and shutdown time would increase the transient periods of a test and reduce the accuracy of the mass flow rate calculation. The relay outputs were connected to the LabVIEW current input module. The input signals were monitored during operation and the response times were determined. Based on the relay specifications, a maximum internal armature delay of 40 ms could be expected.

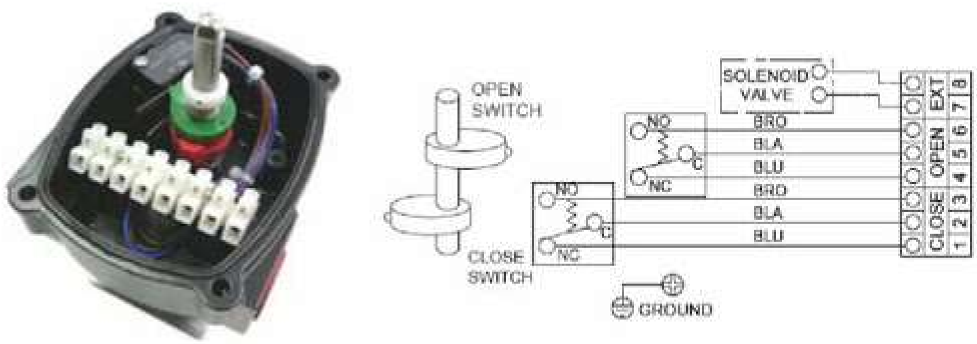


Figure 4-4: Main valve positioner (APL, 2006)

During testing, immediately after the main value was actuated, the pressure in the injector manifold increased to the setpoint based on the tank ullage pressure as shown in Figure 4-3. The injector manifold pressure resulted in a specific mass flow rate. The initial start-up ramp was found to be 600 ms. This was with a single-acting pneumatically actuated ball valve with an air supply pressure of 6 bar. The main valve was converted to a double-acting valve by removing the springs and adding a second pilot solenoid valve. The pneumatic pressure was also increased to 8 bar. The start-up ramp was decreased to 300 ms. Lastly, two dump valves were positioned after the solenoid valves to reduce the start-up and shut-down ramp times to approximately 250 ms. The dump valves reduced the actuation time since the air pressure resisting the motion of the actuator piston was expelled from the pneumatic cylinder faster. The decreased start-up ramp was deemed acceptable as the start-up and shutdown durations equated to 4.2 % of the total test duration for a twelve-second test. The final construction of the pneumatically actuated ball valve is shown in Figure 4-5.

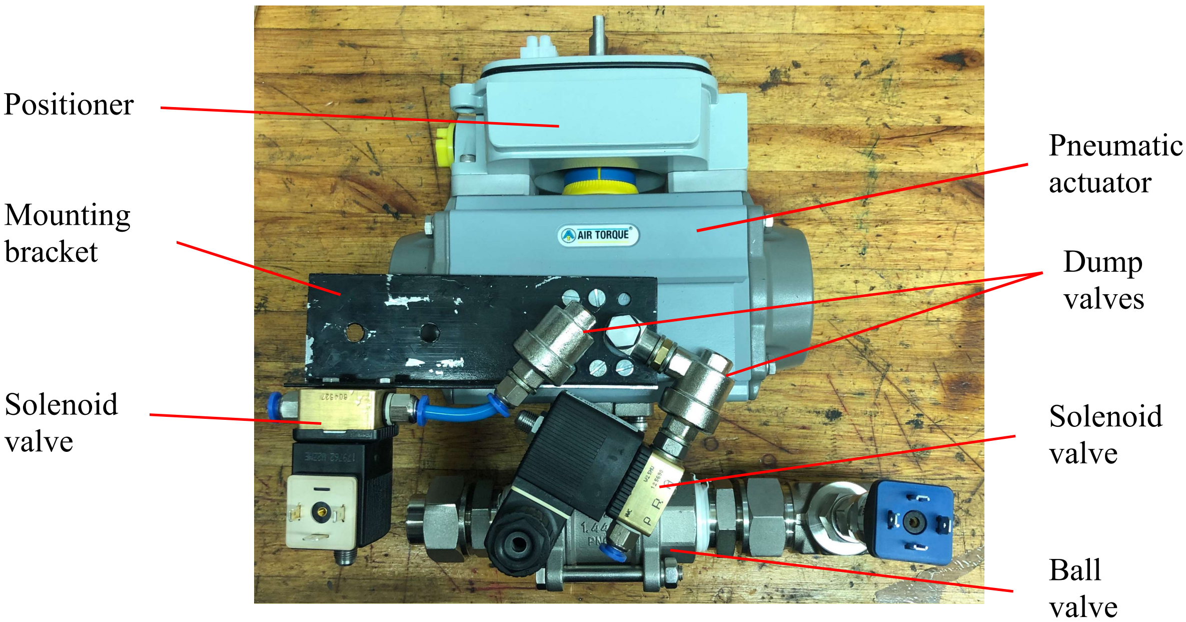


Figure 4-5: Double acting pneumatic ball valve

## 4.5 Summary

Calibrating the test stand determined the required tank pressure for the design mass flow rates for both the fuel and oxidiser domains. The issue of maintaining a constant tank pressure using a single-stage regulator was resolved by installing a solenoid valve to decouple the tank and regulator initial set pressures. The mass flow rate measurement accuracy was increased by decreasing the main valve actuation time.

# 5. Validation Methodology

## 5.1 Introduction

The CFD simulations conducted by Martin (2019) modelled the required pressure drop across the SAFFIRE injector for each flow domain to drive the LOX and kerosene at mass flow rates of 5.66 kg/s and 2.28 kg/s respectively. Another performance output was the propellant distribution. As part of this study the accuracy of the modelled pressure drop and propellant distribution were compared to the experimental data collected using the SAFFIRE cold flow test stand. The test stand operated with water, as an analogue fluid, pressurized by a regulated blow down system. The mass flow rate and propellant distribution were calculated using the mass collected over a steady state period. The manifold pressure was monitored during injector tests and combined with the flow rate to calculate the flow function for each domain.

## 5.2 Validation Methodology

Martin (2019) conducted two CFD simulations for the SAFFIRE injector. The oxidiser and fuel simulations comprised the oxidiser domain, with LOX as the simulation fluid, and the fuel domain, with kerosene as the simulation fluid. Validating the simulations involved machining two separate test articles since it was not practical to test both domains at the same time, nor could they be tested independently with a single test article. The fuel and oxidiser test articles were therefore separate and contained the fuel and oxidiser domains respectively. The test articles were inspected to ensure no geometric imperfections, such as burrs, were present. Secondly, each test article was analysed to identify any significant geometry deviations. The computer-aided design (CAD) models used by the simulations were updated to include any geometric deviations. The measurement and model update process ensured the CAD models being simulated and the physical articles being tested contained the same geometries. Therefore, a deviation between the simulation and experimental results would be the result of a simulation or experimental error. Simulation errors are categorised as a deviation from the experimental data. The experimental error was represented by a combined uncertainty.

The output parameters of the simulations were the mass flow rates for each orifice. These flow rates were used to calculate the total mass flow rate and the propellant distribution for each injector domain. To validate the performance of each domain, the simulations fluid properties were updated to water, such that the simulation fluid and analogue fluid used for testing were the same. The oxidiser and fuel simulations required remeshing to capture the flow accurately. This was caused by the change in density and viscosity of the fluid. Updated performance results for the pressure drop and propellant distribution were obtained using the updated simulations. The accuracy of each simulation was determined based on the deviation from the experimental results. The pressure measurement results from the simulation and cold flow tests were measured relative to atmospheric pressure. The validation process followed for the injector fuel and oxidiser domains is shown in Figure 5-1.

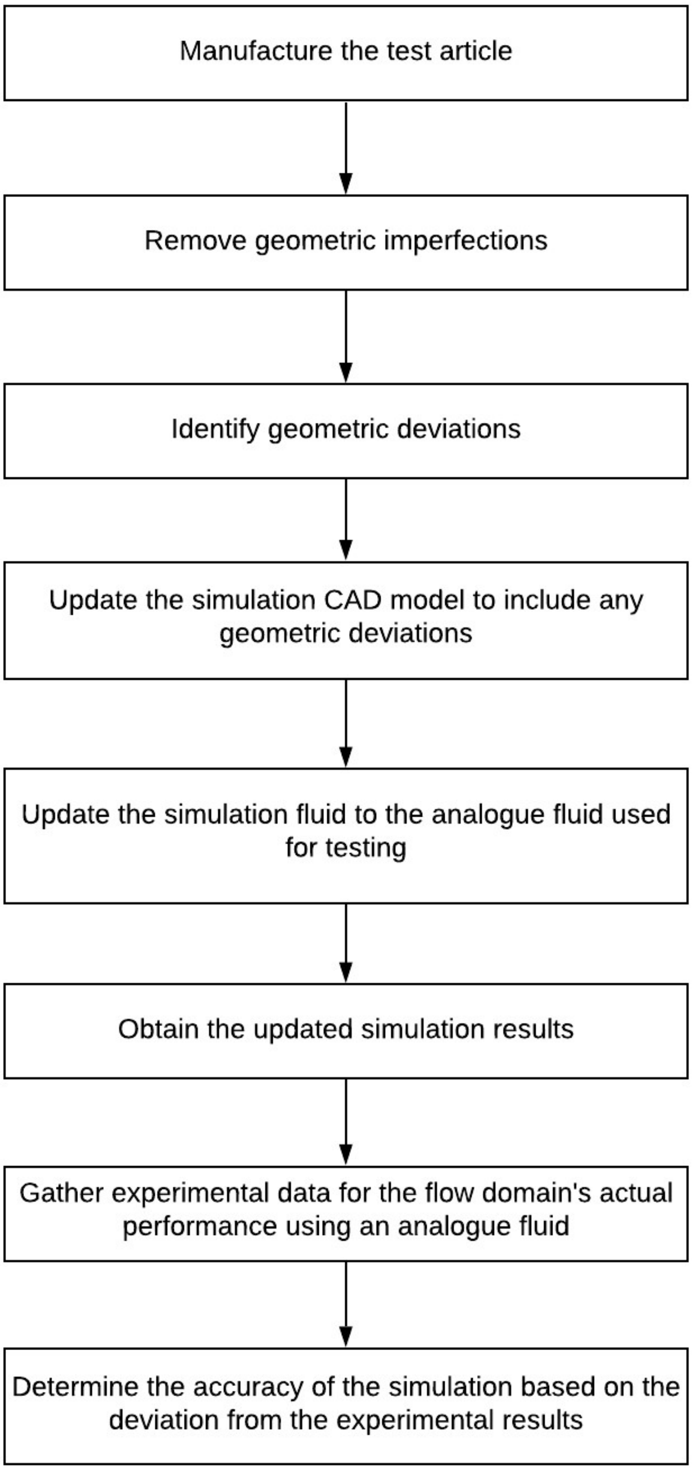


Figure 5-1: Validation process and outcomes

## 5.3 Steady State Comparison

The simulation results were based on a steady state model as a Reynolds Averaged Navier Stokes turbulence model was selected to reduce computational resource requirements. The experimental data points were collected during a steady state period and were averaged over the test duration. Therefore, the results are comparable as both the computational model and experimental data contain averaged results.

The transience flow rate experienced during start-up and shutdown was removed by employing a two-test method. A single full duration test of twelve-seconds was followed by a separate transient test of two-seconds. Subtracting the mass collected during the transient test from the total mass of the full duration test yielded a weight corresponding to a steady manifold pressure. Dividing the steady state weight by the difference in test durations resulted in the mass flow rate for the steady state period. The steady state and transient regions are shown in Figure 5-2. The full duration test data are labelled as “Full” and the transient valve test data are labelled as “Valve”. The portion labelled “Start-up transience” is the pressure in the manifold during the start-up sequence of a test. This is when water begins to flow through the injector. The “Steady state” pressure is when the flow is steady through the injector. Finally, the “Shut down transience” is when the main ball valve is closed and the flow through the injector decreases then stops. To obtain a mass flow rate two tests are required. These two tests result in a single data point for the mass flow rate. Numerous sets of two tests were conducted to obtain repeated measurements for the mass flow rate.

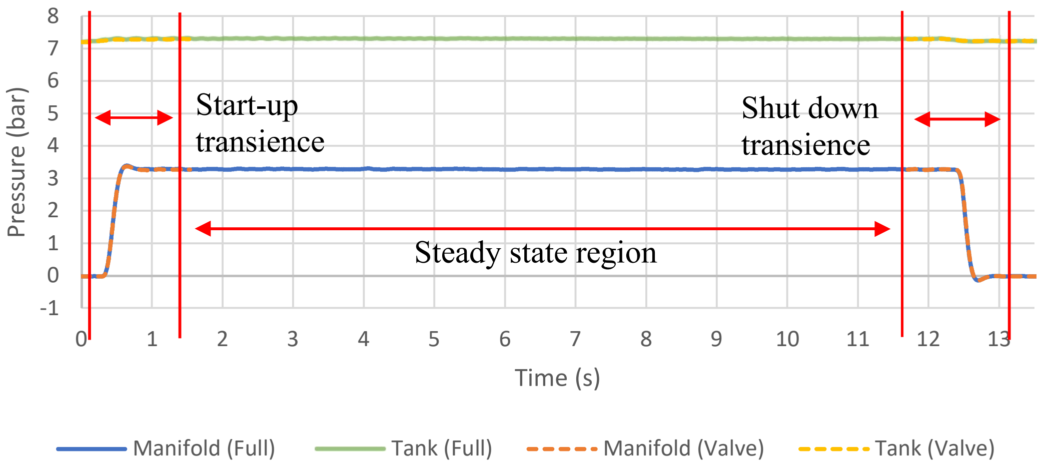


Figure 5-2: Resulting steady state pressures

The two-test method assumed that the two tests were conducted at identical mass flow rates. However, this was not practical as a slight variation in the tank ullage pressure would affect the mass flow rate. Therefore, an estimate of the uncertainty for the mass flow rate was calculated, resulting from slight mass flow rate variations between tests. The mass flow rate through a device for a given change in pressure across the device can be expressed using a flow function. An uncertainty was calculated using the flow function and the difference in manifold pressures between tests. In general, the difference in manifold pressures between tests was within 0.5% and 1% for the fuel and oxidiser tests respectively. These low-pressure differences resulted in a low uncertainty for the mass flow rate calculation. The estimated uncertainty calculations are covered in Chapter 6.

The steady state region in Figure 5-2 is composed of a full duration twelve-second test (Figure 5-3) and a truncated two-second transient valve test (Figure 5-4). Figure 5-2 depicts how the two-test approach was applied to remove the transient periods during start-up and shutdown. The result is a constant manifold pressure for the calculated mass flow rate. This data was required for a direct comparison to the simulation results. The manifold pressure in the steady state region was averaged and the standard deviation calculated to account for the uncertainty in this mean value.

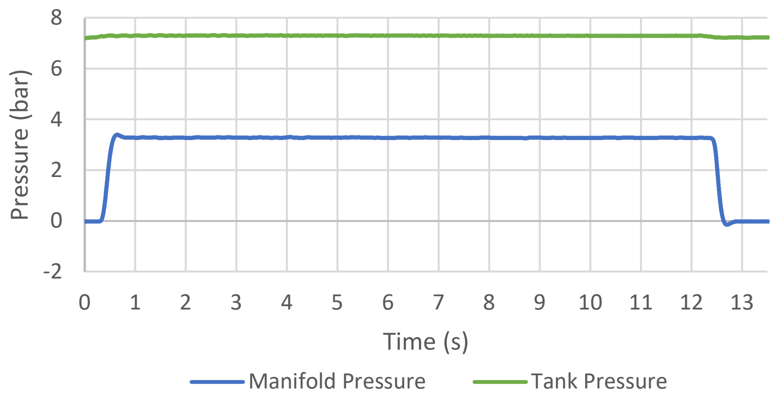


Figure 5-3: 12-second full duration test pressures

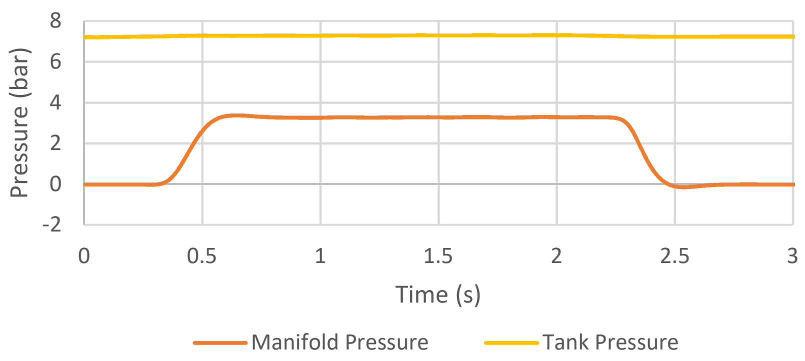


Figure 5-4: 2-second valve test pressures

### 5.4 Mathematical Model

Validating the simulation performance parameters involved determining the required injector pressure drop for a mass flow rate of 2.28 kg/s and 5.66 kg/s for the fuel and oxidiser domains respectively. The actual injector pressure drops for the design mass flow rates were evaluated based on two sets of ten repeated experiments. Obtaining an identical mass flow rate per test was impractical due to the sensitivity of the initial tank ullage pressure and the resulting flow rate. Therefore, as per the recommendations made by the AIAA (1998), a mathematical model was used that describes the relationship between the pressure drop and the mass flow rate. The flow factor $K_v$ is given by Equation 2-5. It describes the mass flow rate as a function of the pressure drop across a device and the specific gravity of the fluid.

### 5.5 Test Articles

The SAFFIRE injector contains two independent flow domains. These domains are integrated, as shown in Figure 5-5. They can be separated into the fuel and oxidiser domains. These domains contain independent inlets and outlets. Therefore, the performance of each flow domain was tested independently by machining two test articles, namely the fuel test article and the LOX test article. Since the domains were separated, propellant mixing tests were not conducted.

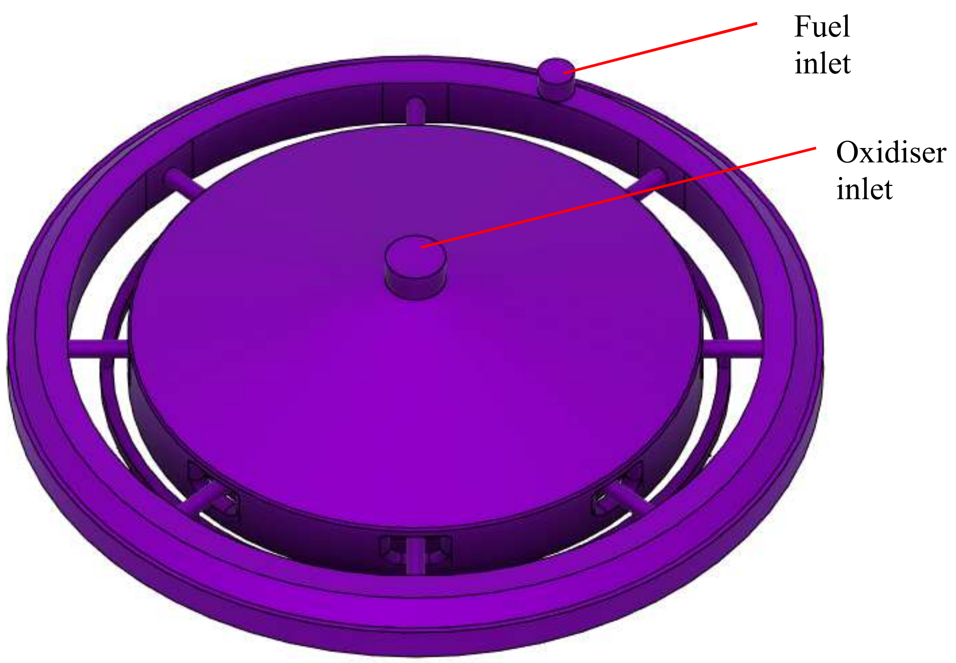


Figure 5-5: Integrated flow domains

## 5.6 Fuel Test Piece

The fuel test article contains the fuel domain, shown in Figure 5-6. It was designed to deliver kerosene to the combustion chamber at a flow rate of 2.28 kg/s. The fuel inlet supplies an annular manifold where a total of 8 radial passages with various diameter down-coming holes supply five concentric rings. The four inner rings supply the main orifices with 2.07 kg/s of propellant. The outer ring supplies the film cooling orifices with 0.207 kg/s of propellant.

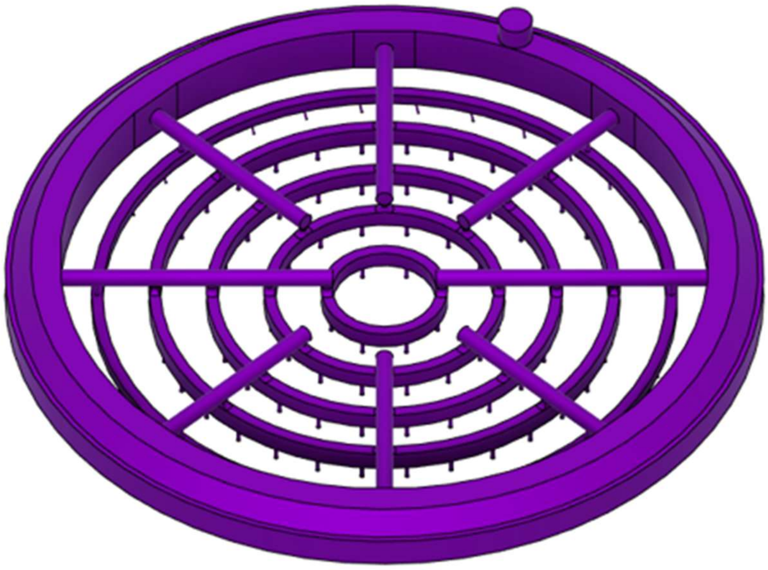

Figure 5-6: Fuel domain

The nominal design mass flow rates for the fuel and film cooling orifices are listed in Table 5-1. These are the required flow rates for each orifice to deliver the same amount of propellant for the correct O/F ratio across the chamber cross-section. This O/F ratio excluded the film cooling. These nominal mass flow rates were used to calculate the percentage deviation per orifice for the distribution results.

Table 5-1: Nominal mass flow rates per orifice

| **Parameter** | **Main orifice (kg/s)** | **Film cooling (kg/s)** |
|---|---|---|
| Mass flow rate | 0.0259 | 0.0069 |

The fuel domain was manufactured from three bodies, namely the top manifold, injector body and faceplate. These bodies were secured together using annular bolts. The faceplate was secured to the injector body using countersunk cap screws to ensure the O-ring seals did not leak from the deflection of the faceplate during operation. The manufacturing drawings can be found in Appendix E.

The fuel test article was machined from aluminium 6082 T6 plate. This aluminium was selected because of its corrosion resistance, high yield strength and availability. Finite Element Analysis (FEA) simulations and analytical calculations verified the bolted connections and O-ring sealing design. A total of 12 O-rings sealed the fuel test article, including 10 O-rings to seal the concentric rings and 2 O-rings to seal the injector body and top manifold interface. An integrated injector with both flow domains requires the faceplate to be brazed to the injector body. This is due to geometric constraints and sealing material compatibility issues (Martin, 2019). The faceplate was machined from stainless steel 316 plate. All the orifices were Wire Electrical Discharge Machining (WEDM) cut based on the high tolerance requirements on the 1.5 mm and 0.8 mm holes. An exploded view of the fuel test piece is shown in Figure 5-7.

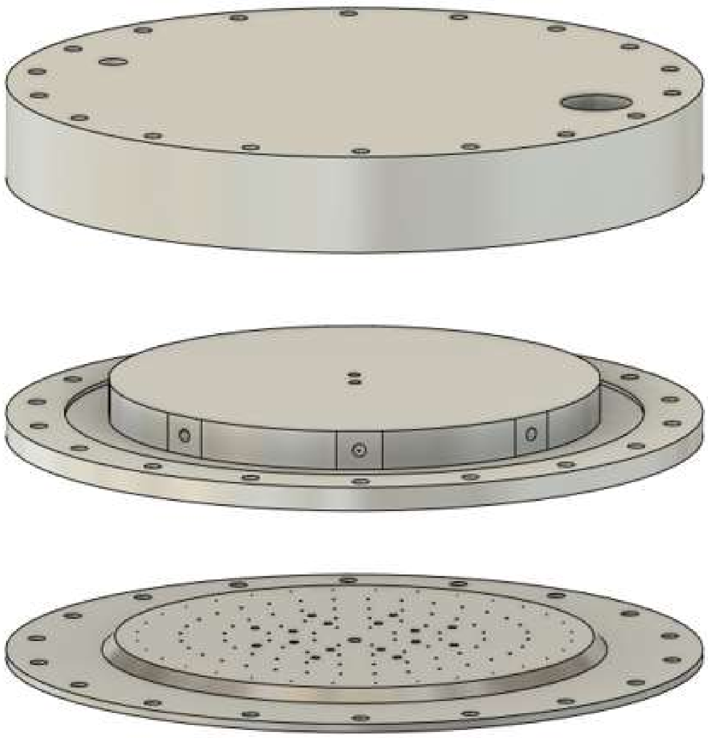

Figure 5-7: Fuel test piece exploded view

The simulations performed by Martin (2019) for the fuel domain included a 12 mm inlet due to geometrical constraints. The fuel test article was machined with a 12 mm inlet to resemble the modelled geometry. The test stand used 1” tubing with an internal diameter of 19.8 mm. Figure 5-8 depicts the step at the inlet. This step had a large analytical pressure drop of roughly 0.72 bar and was added to the simulation CAD model (Fox *et al.*, 2012).

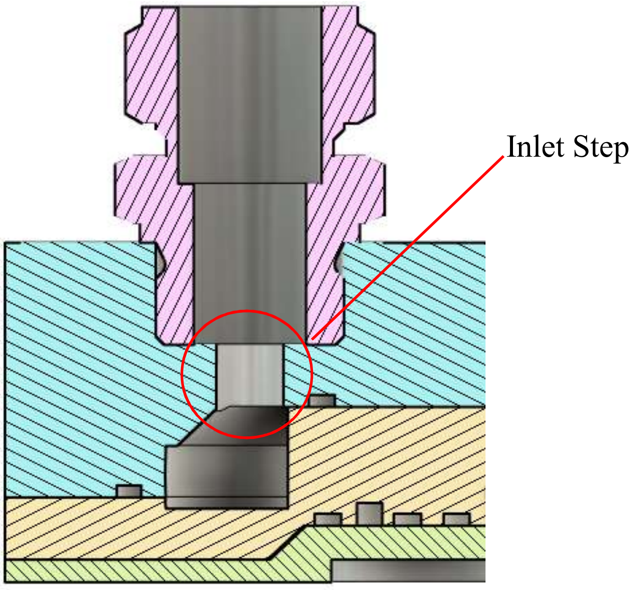


Figure 5-8: Fuel inlet step

The fuel simulation calculated the average pressure at the inlet surface of the injector domain. This was impractical from a testing perspective due to the mounting position of the injector. Instead, a pressure tap was inserted into the test article for a pressure transducer, as shown in Figure 5-9. This monitored the pressure at an accessible reference point within the annular manifold. An identical pressure tap was inserted into the simulation CAD model for a direct comparison.

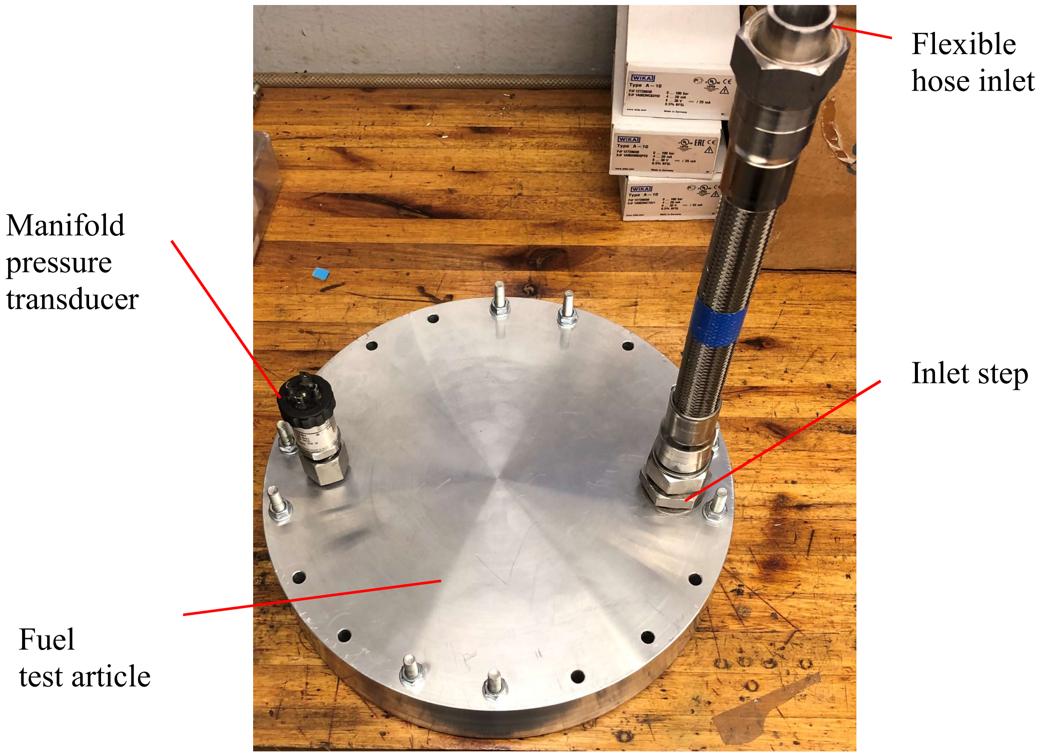


Figure 5-9: Fuel test article pressure tap

### 5.6.1. Geometric Simplifications

Geometric simplifications reduced the machining complexity of the fuel test article. The fillet on the down-coming holes from the radial passages into each concentric ring were removed. This is shown in Figure 5-10 and Figure 5-11. This simplification was included in the simulation CAD model. The CAD model update ensured the model being simulated and the physical model being tested contained the same geometry.

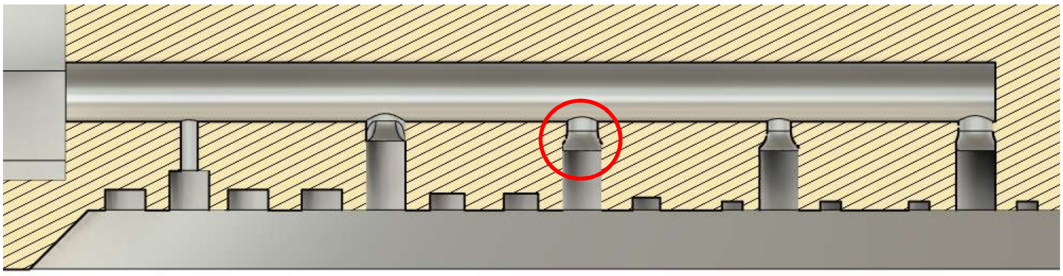

Figure 5-10: Fillets on down-coming holes

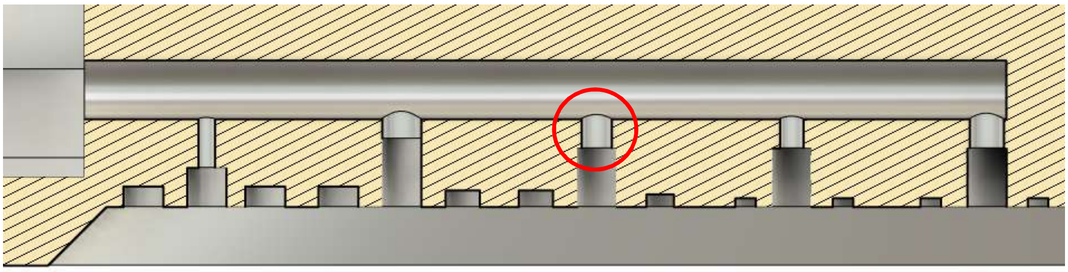

Figure 5-11: Updated geometry with no fillets

### 5.6.2. Machining Imperfections

The down-coming hole diameters for rings two and five were increased due to poor tolerances during machining. To save both time and money the simulation CAD model was updated to include these geometric deviations. The new dimensions are tabulated below in Table 5-2.

Table 5-2: Fuel test piece dimension updates

| Down-coming hole location | Design diameter (mm) | Machined diameter (mm) |
|---|---|---|
| Ring 2 | 2.4 | 2.6 |
| Ring 5 | 1.7 | 1.75 |

The fuel faceplate comprises 112 small diameter orifices, including 80 main orifices and 32 film cooling orifices with diameters of 1.5 mm and 0.8 mm respectively. These orifices were WEDM cut perpendicular to the faceplate. Therefore, the profile projector ( Figure 5-12), with a ×50 magnification, was used to measure the diameters with a high degree of accuracy.

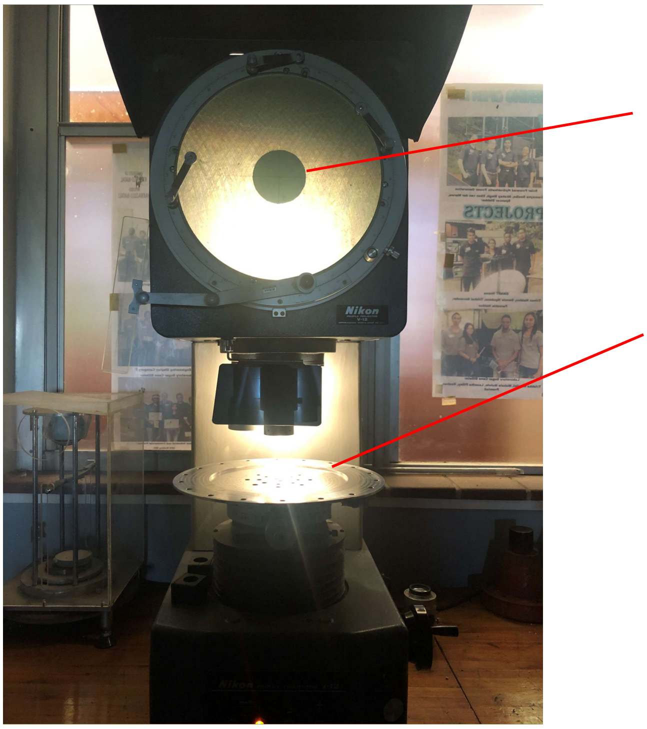


Figure 5-12: Profile projector with fuel faceplate

The profile projector bed contained a vertical and horizontal adjustment resolution of one micron. Each orifice diameter was calculated based on two bed measurements taken using reference markers and the magnified image on the display. The main orifices on average had a +30 microns tolerance, although the shape was adequately circular as shown in Figure 5-13.

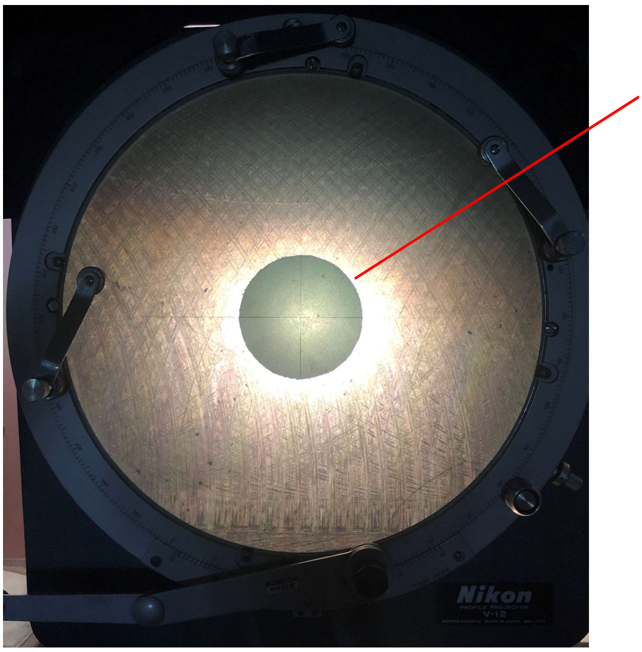


Figure 5-13: Fuel faceplate main orifice

The process of WEDM cutting a hole involves spark eroding a 0.5 mm hole for the wire to be fed through before the specific hole diameter is cut. This process is often done on a separate machine which increases the possibility of an error resulting from an offset in the second setup. When inspecting the film cooling orifices, Figure 5-14, it was evident that a setup error had occurred as majority of the orifices were oval. An average edge to edge distance was taken and the simulation CAD model was updated to include both the main and film cooling orifices tolerances. This process involved measuring all 112 orifices on the test article faceplate and updating the diameters of the simulation CAD model to ensure the simulation results did not contain errors due to geometric deviations.

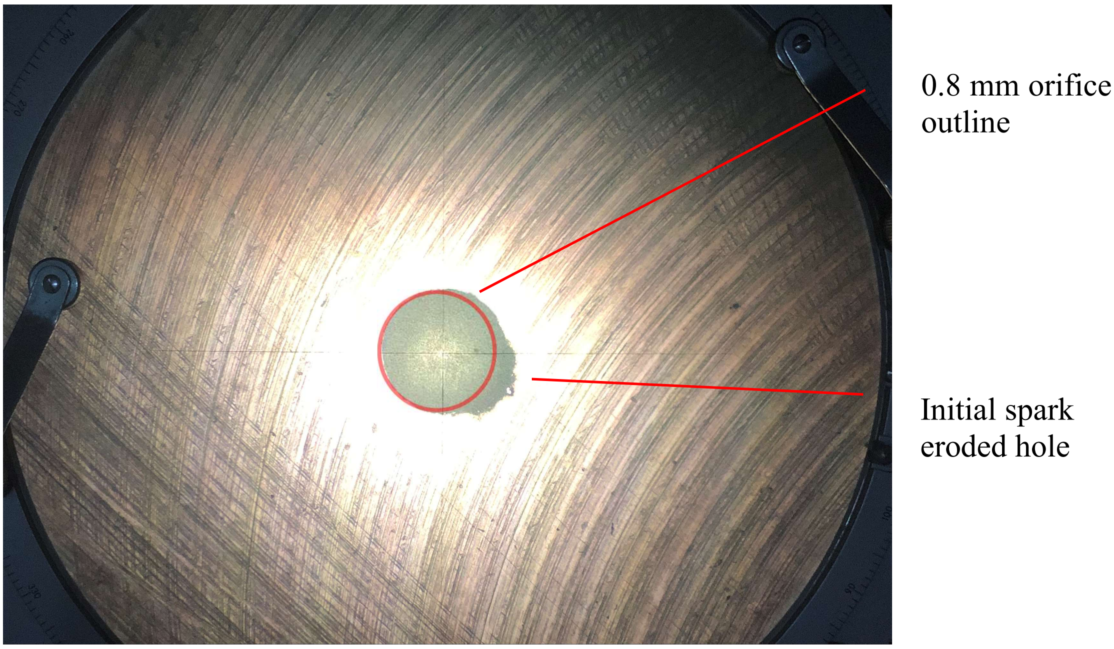


Figure 5-14: Film cooling defective orifice

### 5.6.3. Structural Design & FEA Analysis

The structural integrity and sealing design were verified using a Finite Element Analysis on ANSYS Mechanical. The bolted connections and material strength were verified to ensure the components did not yield under the operating pressures. A maximum of 5 bar (static pressure) was experienced during steady state conditions. A safety factor of two was used on this static pressure. Therefore, the static pressure was set to 10 bar for the simulation and analytical calculations. The main force separating the three components resulted from the internal pressure. This force was resisted by the annular bolts and countersunk screws.

The simulation treated the bolts and threaded interfaces as bonded sections. This resulted in unrealistic stresses at cell vertices within these bonded regions, as shown in Figure 5-15. These stresses were disregarded based on the analytical thread calculation shown in Appendix B.

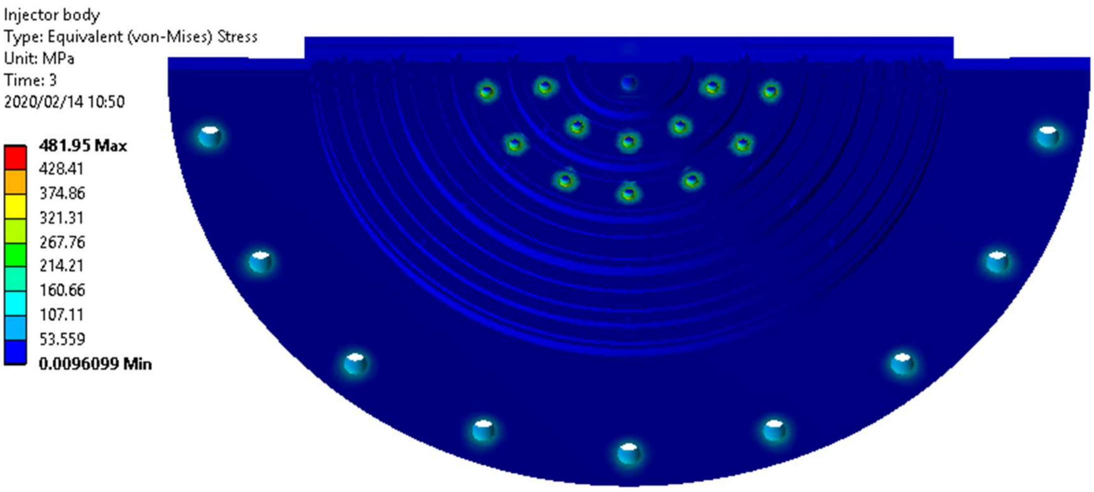


Figure 5-15: Injector body maximum Von-Mises stress

The deformation of the faceplate was modelled to ensure leak paths were not introduced during operating conditions. Countersunk cap screws torqued to 5 Nm secured the faceplate to the injector body and ensured minimal separation occurred due to internal pressures. The maximum deformation of the faceplate near the concentric rings was 0.0351 mm, as depicted in Figure 5-16. The resulting decrease in compression of the O-rings in this maximum deflection region was 1.167%. The initial compression of the O-rings was roughly 40% with a lower recommended limit of 30%. Therefore, this deflection was deemed acceptable.

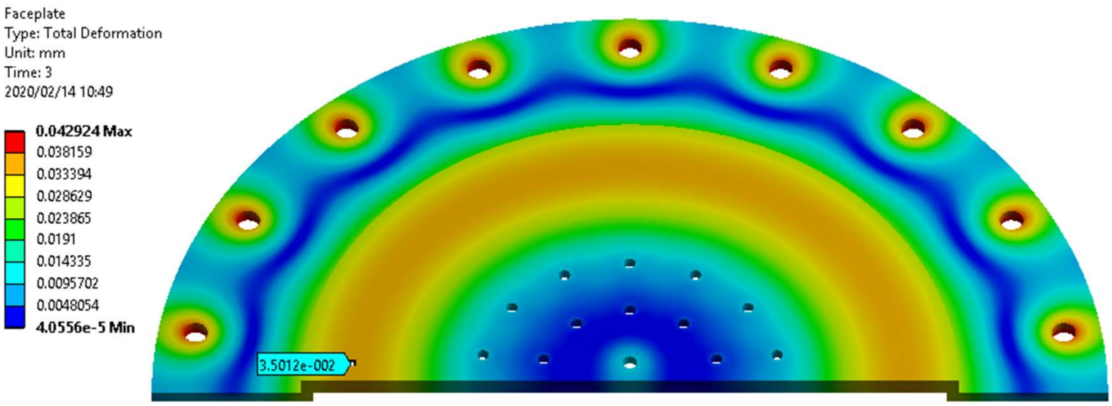


Figure 5-16: Faceplate deformation

## 5.7 LOX Test Piece

The LOX test article was constructed from three parts, namely the LOX dome, injector body and faceplate. The oxidiser inlet was positioned at the centre of the LOX dome. The injector body contained down-coming slots that supplied five concentric rings with propellant. These rings were located above the orifice inlets on the faceplate.

Figure 5-17: LOX domain top and bottom view

The LOX dome and injector body were manufactured from aluminium 6082 T6 due to its high yield strength and corrosive resistance. The faceplate was machined from a half-hardened copper plate to evaluate the expected tolerances of the 1.4 mm holes drilled at an angle of 30°. This was the original material selected for the SAFFIRE injector faceplate (Martin, 2019). O-rings were used to seal the concentric rings and the injector body/LOX dome interface. The LOX test article exploded view is shown in Figure 5-18. The manufacturing drawings can be found in Appendix E.

Figure 5-18: Lox test piece exploded view

A pressure tap was machined into the LOX test article for a practical reference pressure. The pressure tap was also inserted into the simulation CAD model at the same position for a comparative pressure reading. The LOX test article is shown in Figure 5-19.

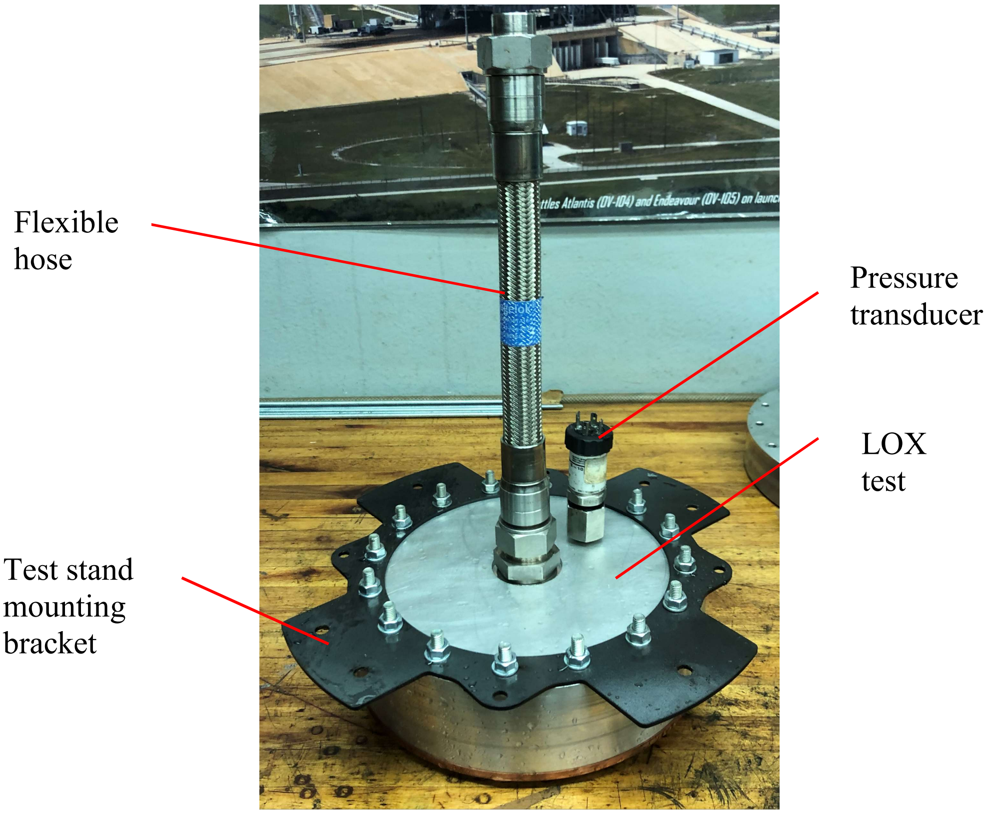


Figure 5-19: LOX test article

### 5.7.1. Machining Tolerances

The LOX faceplate included 160 orifices with a design diameter of 1.4 mm. The orifices were drilled then reamed at a 30° angle. Therefore, using the profile projector was not a viable method of measuring the orifices tolerances. A sample part containing five perpendicular holes made with the same manufacturing techniques, shown in Figure 5-20, was supplied with the faceplate. The tolerances of the sample holes were measured using the profile projector. It was concluded that the average tolerance for the perpendicular orifices was ±4 microns.

Figure 5-20: 1.4 mm hole sample

The manufacturer ensured all the angled orifices were within ±10 microns of 1.40 mm. This was determined using a pin gauge set with a 10-micron resolution. The quality of the entry and exit holes of each orifice was inspected using a profile projector. Figure 5-21 shows the exit quality of an orifice in ring 1. This exit quality was similar for rings 2 and 3. Rings 4 and 5 contained orifices with poor quality exits, as depicted in Figure 5-22. The possible effects of the poor-quality exits are discussed in Chapter 7.

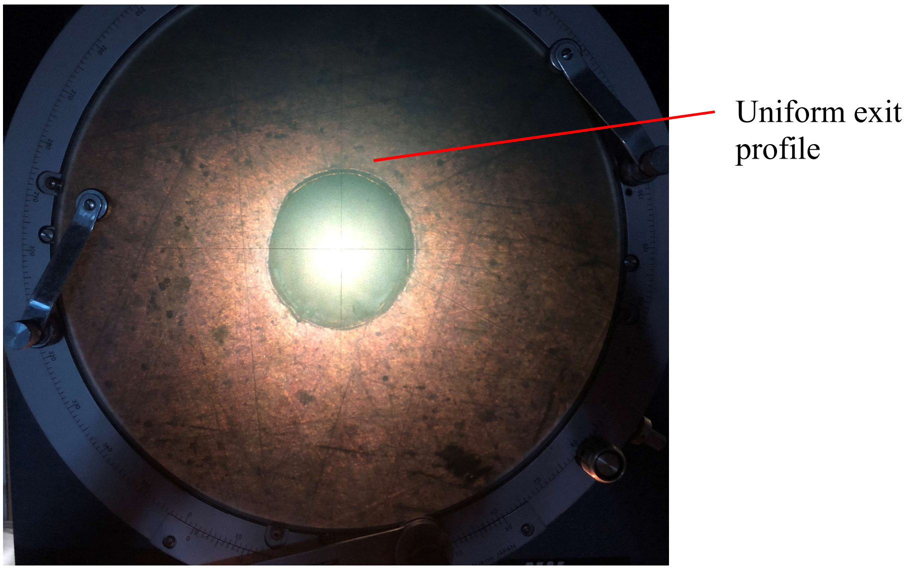


Figure 5-21: Ring 1 orifice exit (good quality)

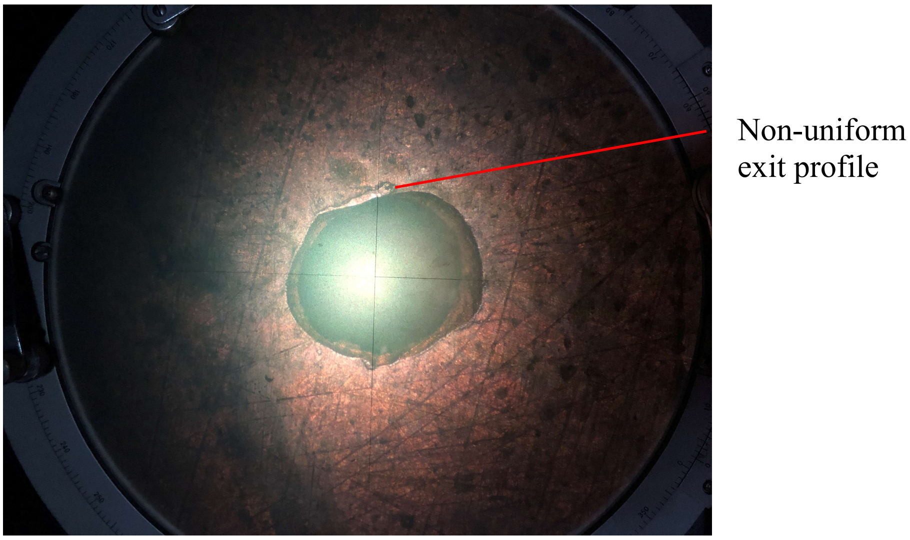


Figure 5-22: Ring 4 orifice exit (poor quality)

### 5.7.2. Structural Design & FEA Analysis

The LOX test article structural and sealing designs were verified using an FEA simulation on ANSYS Mechanical. This simulation used symmetry to reduce the computational requirements. The main objective of this simulation was to ensure the deflection between the injector body and faceplate did not cause an O-ring seal to leak due to inadequate compression. Secondly, the bolt and screw connections were verified. The resulting Von-Mises stresses on the injector were below the yield stress of 240 MPa for aluminium 6082 T6, as shown in Figure 5-23. Further analytical calculations shown in Appendix B verified the bolted connections were adequate for an internal pressure of 10 bar.

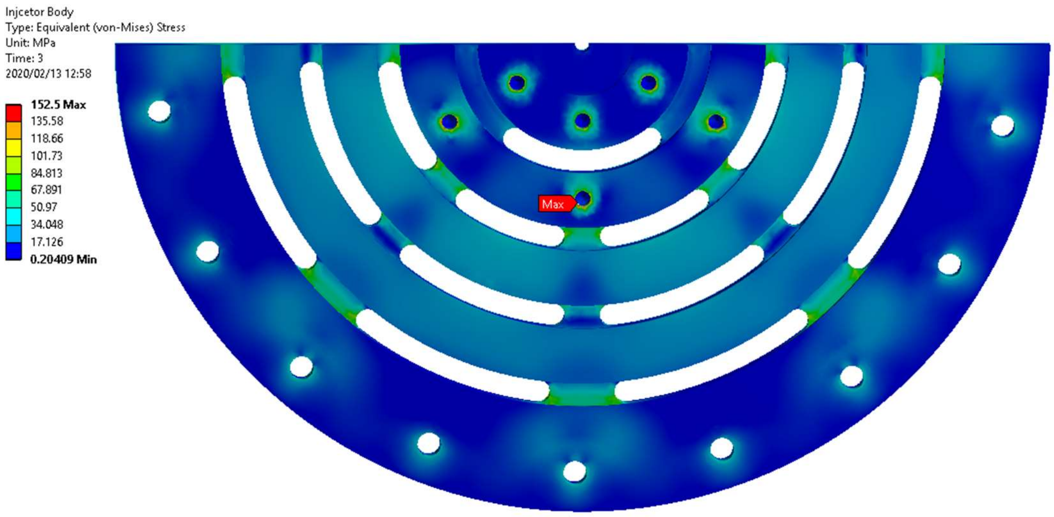


Figure 5-23: Injector body Von-Mises stress

Proper sealing of the concentric rings was required to obtain accurate distribution results. The sealing design of the faceplate and injector body was verified by inspecting the gap of the faceplate and injector body at a maximum operating pressure of 10 bar. Figure 5-24 depicts the gap between the faceplate and injector body due to internal pressures. The scale assumes a gap below the injector body surface is negative. Therefore, the maximum gap is near the outer ring, with a magnitude of 0.0206 mm. This results in a maximum decrease in compression of 0.69 %. Therefore, based on the minimum required compression of 30% and the initial compression of 40 %, the O-ring sealing design was accepted.

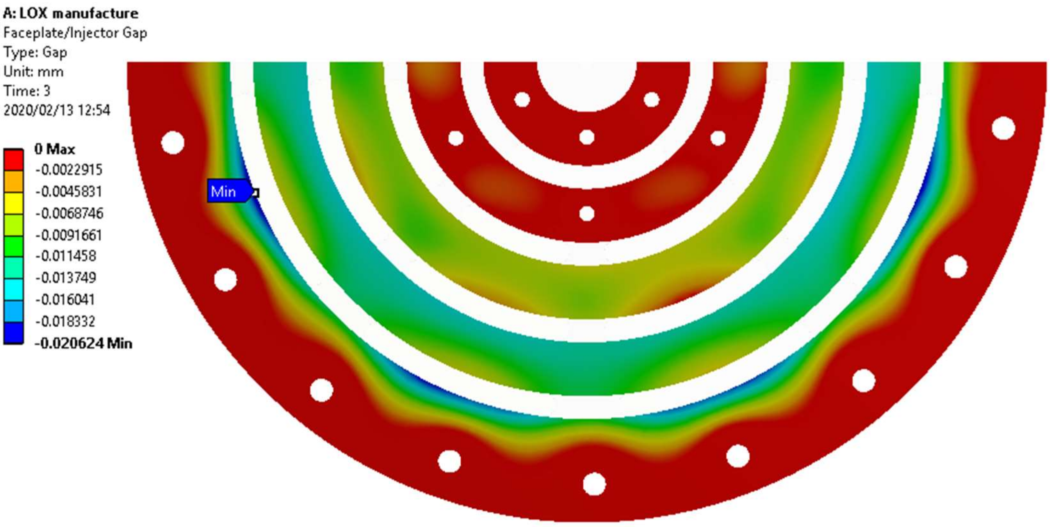


Figure 5-24: LOX faceplate and injector body gap

## 5.8 CFD Simulations Updates

The SAFFIRE injector design was modelled using commercial CFD software, Star CCM+. The objective was to aid the design phase and to determine the predicted performance of the injector. The full methods and results of the simulations have been published by Martin (2019) who conducted separate simulations to determine the performance of the fuel and oxidiser domains with the fluids set as kerosene and liquid oxygen respectively. A user-defined velocity inlet boundary set the mass flow rate at the injector inlet. The orifices exits were set as separate pressure outlet boundaries with the pressures defined as the chamber pressure. The performance outputs for each domain were the pressure drop across the injector, for the user-defined mass flow rate, and the propellant distribution. The propellant distribution was calculated using the modelled mass flow rate and the nominal mass flow rate per orifice. The nominal mass flow rate per orifice was calculated using the total mass flow rate and the total number of orifices. The simulations conducted by Martin concluded the orifices mass flow rates varied by a maximum of 2 % from the nominal mass flow rates per orifice. The simulation boundary conditions and solver settings for the oxidiser and fuel simulations are given in *Table 5-3*.

Table 5-3: LOX and fuel simulation boundary conditions and solver settings

| | **LOX** | **Fuel** |
|---|---|---|
| Inlet boundary | Velocity inlet | Velocity inlet |
| Outlet boundary | Pressure outlet | Pressure outlet |
| Physics continuum | Steady<br>Three dimensional<br>Turbulent<br>K-Epsilon Turbulent<br>Liquid<br>Two-layer all y+ wall treatment<br>Reynolds averaged Navier Stokes<br>Realizable K-Epsilon two-layer<br>Constant density<br>Segregated flow | Steady<br>Three dimensional<br>Turbulent<br>K-Epsilon Turbulent<br>Liquid<br>Two-layer all y+ wall treatment<br>Reynolds averaged Navier Stokes<br>Realizable K-Epsilon two-layer<br>Constant density<br>Segregated flow |
| Mesh count (cells) | $7.79 \times 10^7$ | $9.78 \times 10^7$ |

The fuel and oxidiser simulations used the K-Epsilon turbulence model with different wall treatments to model the flow through the respective domains. The fuel simulation used an all wall $y^+$ model with the first cell near the wall having a $y^+$ value $\leq 1$. Therefore, the viscous sublayer was modelled rather than approximated with a wall function. The oxidiser simulation used an all wall $y^+$ model with the first cell near the wall having a $y^+$ range of $30 \leq y^+ \geq 300$. Therefore, the viscous sublayer and transient region were approximated with a wall function.

A mesh refinement was required since the working fluids for both simulations were changed to water. The update to the fluid properties, namely density and viscosity, resulted in different flow characteristics through the respective domains. Therefore, the prism layer mesh thicknesses for both simulation domains were updated, such that the $y^+$ values still fell within the acceptable ranges. Acceptable $y^+$ ranges for both simulations were achieved, as shown in Figure 5-25 and Figure 5-26. The mesh refinement process involved an iterative approach of refining the prism layer thickness and observing the $y^+$ values.

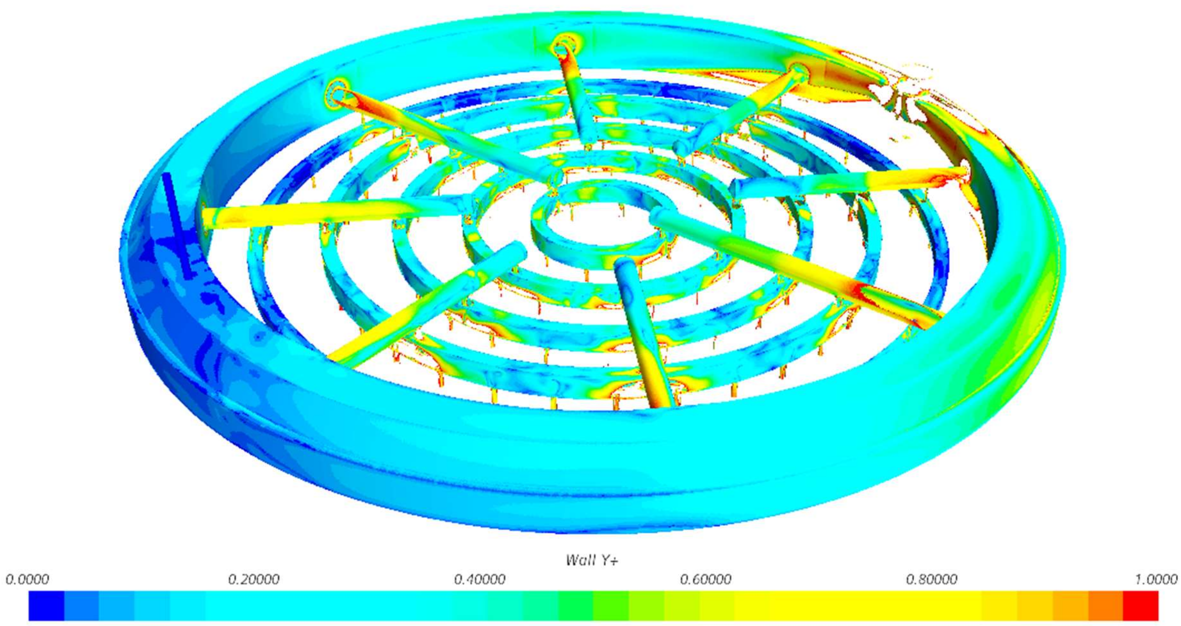


Figure 5-25: Fuel domain wall $y^+$ scene

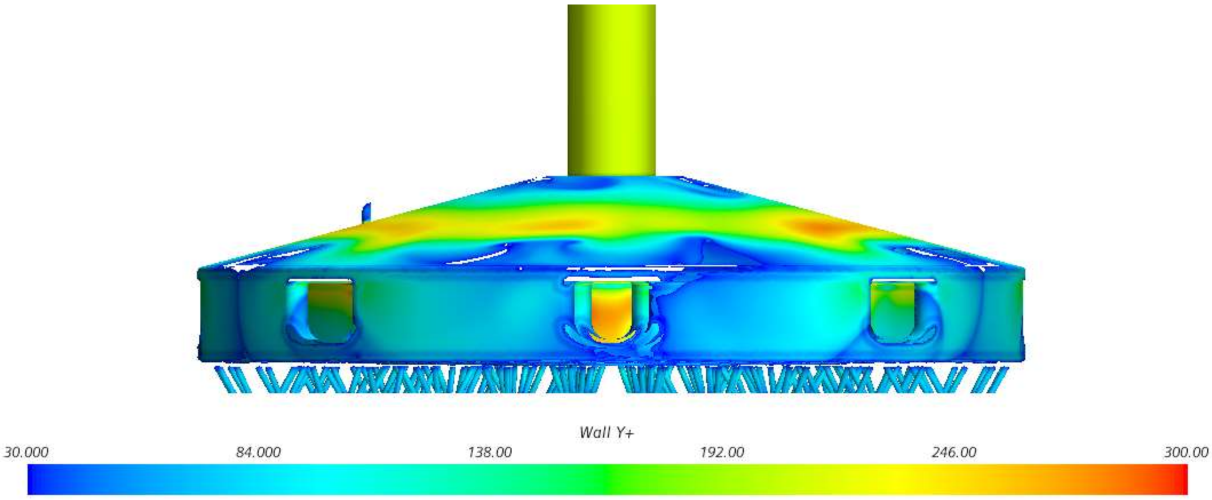


Figure 5-26: LOX domain wall $y^+$ scene

## 5.9 Fuel Propellant Distribution

The fuel domain was designed to deliver 2.28 kg/s of kerosene through 80 main orifices and 32 film cooling orifices with diameters of 1.5 mm and 0.8 mm respectively. The main orifices are the centres of each triplet element and were drilled perpendicular to the faceplate to produce axial liquid jets, as shown in Figure 5-27. The injector design stated that the individual orifices mass flow rates were within 2 % of the nominal mass flow rate. The nominal value was based off even distribution across the injector.

Figure 5-27: Fuel faceplate during cold flow testing

To determine the mass flow rate per orifice, a collection attachment was designed and manufactured specifically for the fuel element layout. It was attached below the faceplate, as shown in Figure 5-28. The attachment separated the axial streams into separate 500 ml containers. Each container was supplied by a short length of PVC flexible tubing.

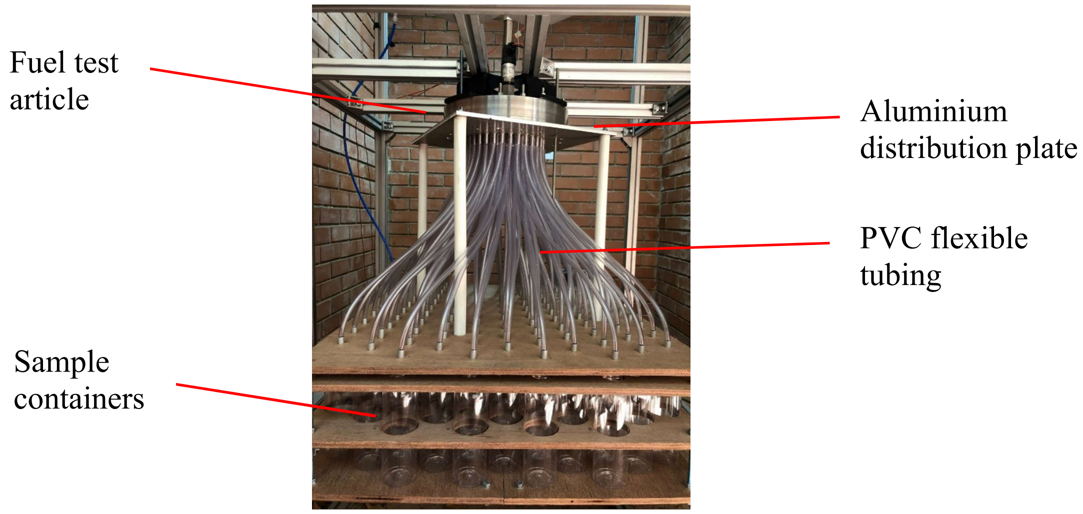


Figure 5-28: Fuel collection attachment

The PVC tubes were connected to an aluminium distribution plate with aluminium tubes positioned below each orifice, as shown in Figure 5-29.

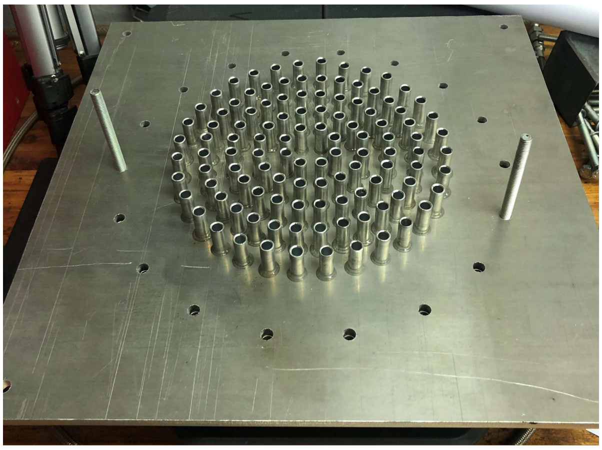

Figure 5-29: Distribution plate construction

The aluminium distribution plate was secured to the injector faceplate with a clearance of 1 mm to ensure the water in the tubes remained under free-flow conditions. The collection apparatus could be separated into two parts, namely the upper portion, containing the tubes and aluminium distribution plate, and the lower removable section housing the containers. The lower section was removed after each test to determine the weight collected in each 500 ml container. The individual bottle masses were measured using a laboratory scale.

A large sample mass per container was required due to the low mass flow rates for the orifices. This ensured the residual water contained in each flexible tube did not account for a large percentage of the collected samples.

Figure 5-30 shows the percentage error of a main and film cooling orifice sample, based on an excess of 1 g of water for varying test durations. The estimation of 1 g of water was a conservative value as on average the total residual water in each tube was 0.75 g. This value was calculated using the weight of the upper section of the collection attachment before and after numerous cold flow tests. It was assumed the increase in weight due to residual water was evenly distributed between the 112 tubes. The film cooling samples included the most significant error percentage due to the lower flow rate. A test duration of twelve seconds was deemed acceptable with an error percentage of ±0.32 % and ±1.29 % for the main and film cooling orifices respectively. The uncertainty due to the residual water in the tubes was a component of the combined standard uncertainty associated with the distribution results calculated in Chapter 6.

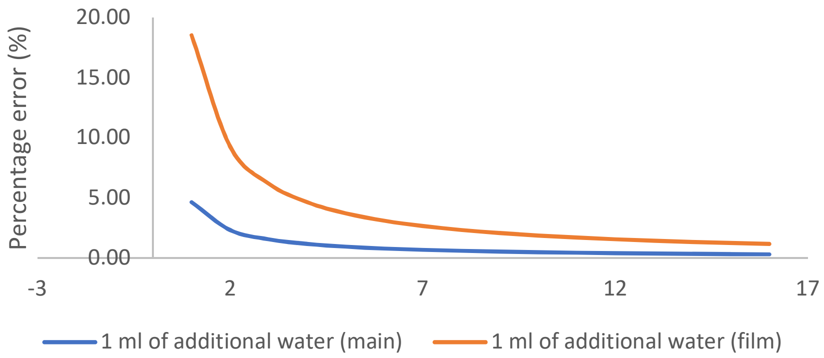


Figure 5-30: Percentage error of an additional 1g of water vs test duration

Comparing the propellant distribution results for the simulation orifices and test article orifices required the respective orifices to be matched according to their x and y coordinates. The simulation data containing the orifices mass flow rates and coordinates were exported as a .txt file from Star CCM+. A mapping chart was made during the construction of the propellant distribution apparatus. This linked each orifice to a numbered container. Each container was given an x and y coordinate based on the orifice supplying the container. A Python script sorted and matched the mass flow rates based on the corresponding x and y coordinates. This process ensured the correct orifices were being compared for the simulation and experimental distribution results.

### 5.10 LOX Propellant Distribution

The oxidiser domain was designed to deliver 5.66 kg/s of LOX to the combustion chamber through 160 orifices with diameters of 1.4 mm. The unlike triplet element configuration selected results in 80 orifice pairs that impinge below the faceplate as depicted in Figure 5-31. The orifice pairs are supplied from 5 concentric rings. Each ring only supplies propellant to one of the two orifices within an element pair. It is critical for each orifice in a pair to contain the same momentum to ensure no skewed spray fans cause hot spots of the chamber walls (Huzel & Huang, 1992).

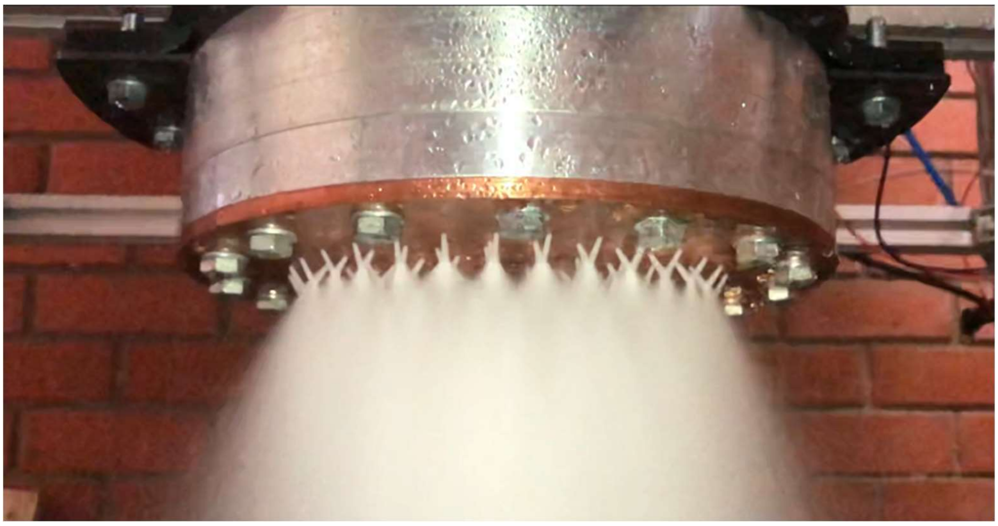

Figure 5-31: LOX domain impinging streams

To measure the flow rate through each orifice, a collection attachment was designed and manufactured specifically for the LOX element layout. This device incorporated a 3D printed distribution part shown in Figure 5-32. PVC tubing pressed into this part which included a slight angle at the inlet to reduce the effects of blowback caused by the high-velocity jets impinging on the tubing walls.

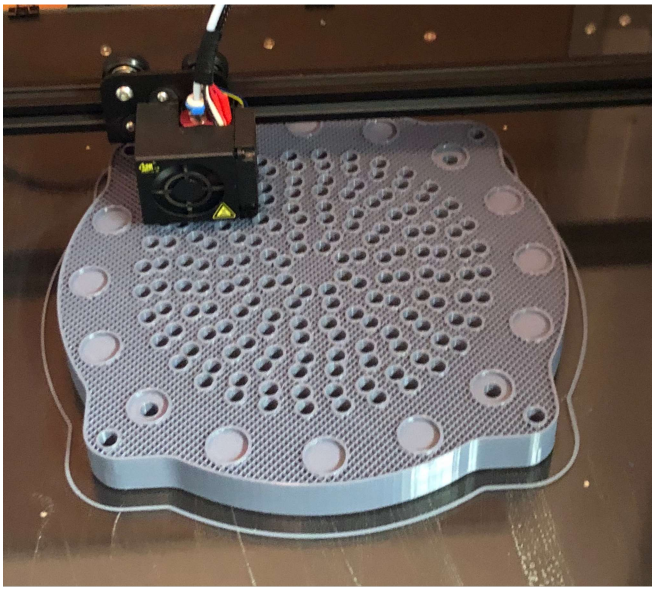

Figure 5-32: 3D printed distribution attachment

The part was positioned below the injector faceplate to separate the liquid jets before impingement occurred, as shown in Figure 5-33. The PVC tubes directed the liquid jets into individual 500 ml containers. The mass flow rates of each orifice were calculated based on the mass collected using the two-test method. This collection apparatus was positioned 0.5 mm below the faceplate to ensure the water in the tubes remained in a free-flowing state.

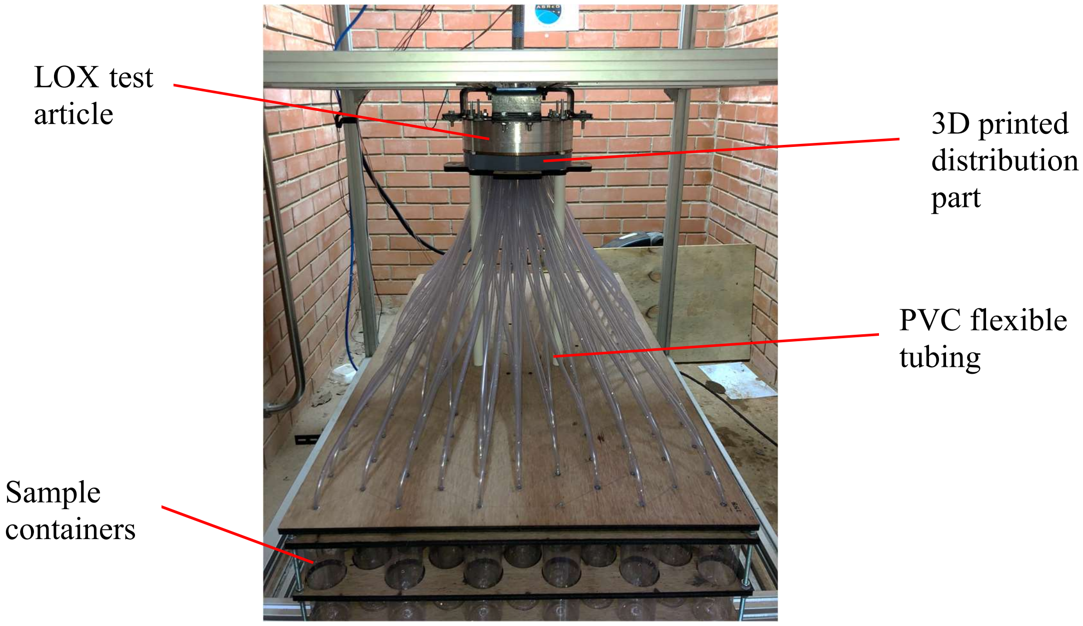


Figure 5-33: LOX distribution attachment

The error percentage of the mass flow rate for a full duration test of 12 seconds with a variance of 1 g of water was ± 0.28 % for the LOX tests. This conservative 1 g estimate was based on the difference in weight of the upper portion of the distribution attachment before and after cold flow testing. It was assumed that each tube contained an equal amount of water after testing. By dividing the total additional weight by 160 tubes, it was estimated that each tube on average contained 0.9 g of residual water.

## 5.11 Summary

The validation methodology described in this chapter was developed to gather steady state experimental data to determine the accuracy of CFD simulation results conducted on the fuel and oxidiser domains of the injector. This was accomplished by utilising a two-test method. This method yielded an accurate measurement of the experimental mass flow rate without the use of a flow meter. The flow rate was determined using this method for the pressure drop and distribution tests. The design and manufacturing procedures for each test article were covered. This included the required updates applied to the simulation CAD model due to machining tolerances and errors. Lastly, the procedures and devices developed to determine the propellant distributions were discussed. The actual test results obtained from the cold flow rig are described in Chapter 7.

# 6. Uncertainty Calculations and Measurement Limitations

## 6.1 Introduction

This chapter summarises the uncertainties of each performance parameter measured or calculated using the experimental methods described in Chapter 5. This includes uncertainties relating to pressure readings, calculated mass flow rates, flow factor calculations and propellant distribution results for both the fuel and oxidiser domains. The uncertainty components of the measured or calculated parameters are shown in Figure 6-1.

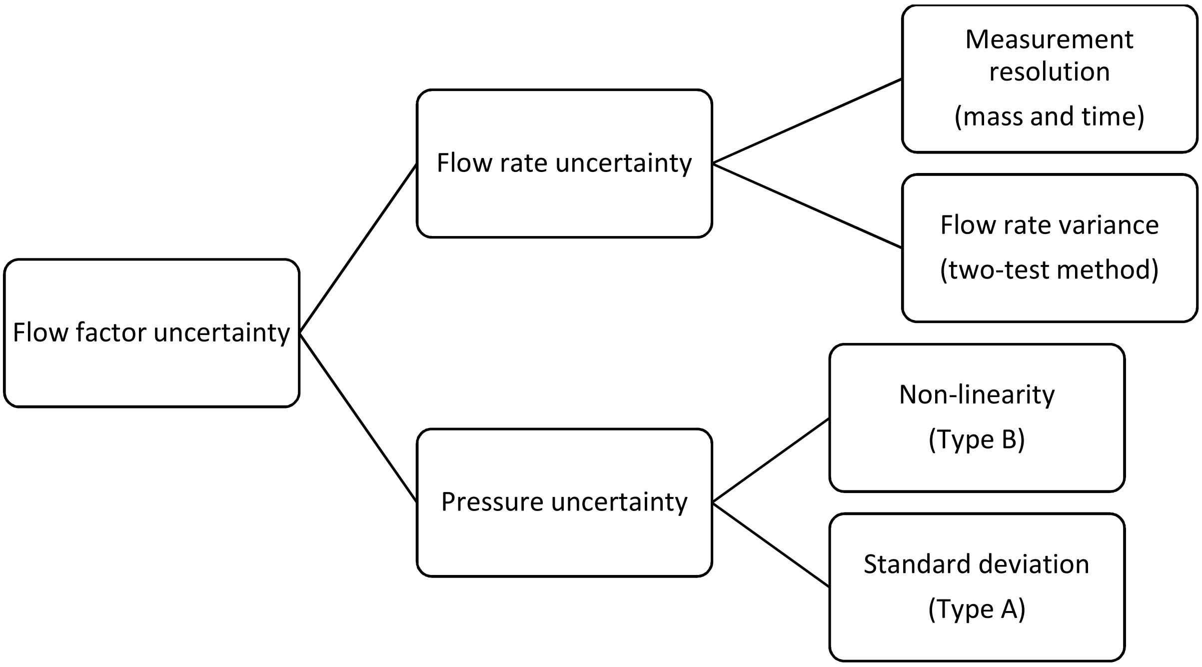


Figure 6-1: Uncertainty components

## 6.2 Pressure Drop and Flow Factor Uncertainties

Based on the literature reviewed in Chapter 2, the most appropriate method of expressing uncertainty for a term that combines multiple measurement uncertainties is a combined standard uncertainty. To calculate the flow factor ($K_v$) the manifold pressure was required which was measured by an S-10 pressure transducer. The flow rate was also required. The calculated flow rate was based on mass and test duration differences referred to as the 'two-test method' in Chapter 5. The calculated mass flow rate contained uncertainties due to the practical considerations that the mass flow rates were not identical for the two tests. Therefore, to state the calculated flow factor with confidence, it should encompass all the combined uncertainties that contribute to its evaluation. Additionally, a coverage factor and confidence interval must be calculated based on repeated measurements. This process was applied to the distribution results where applicable.

The calculated uncertainty of the flow factor is based on the manifold pressure ($P_{\mathrm{M}}$) and flow rate ($Q$) which is based on a test weight $m$, the test duration $t$, and the fluid density ($\rho$). All these terms have an uncertainty ($u(\,)$) based on the measurement method used. These values are given in Table 6-1.

Table 6-1: Absolute uncertainties of flow factor variables

| Parameter | Measurement device uncertainty | Resolution uncertainty (rectangular probability distribution) | Total uncertainty |
|---|---|---|---|
| $u(P_{\mathrm{M}})$ | ± 0.0625 bar (Non-linearity) | $2.89\ \times 10^{-7} bar$ (Negligible) | ± 0.063 bar |
| $u(m)$ | Resolution limited | 0.0029 kg | ± 0.0029 kg |
| $u(t)$ | Resolution limited | 0.00058 s | ± 0.00058 s |

A relative uncertainty method was adopted to calculate the uncertainties of $Q$ and $K_{\mathrm{v}}$ due to the process involving multiplication or division of uncertainties with unlike units. The relative uncertainty for the flow rate, $u(Q\ )$, was calculated using Equation 6-1:

$$\frac{u(Q\ )}{Q} = \sqrt{\left(\frac{u(m)}{m}\right)^2 + \left(\frac{u(t)}{t}\right)^2} \tag{6-1}$$

The relative uncertainty for the flow factor, $u(K_{\mathrm{v}}\ )$, was calculated using Equation 6-2:

$$\frac{u(K_{\mathrm{v}}\ )}{K_{\mathrm{v}}} = \sqrt{\left(\frac{u(Q)}{Q}\right)^2 + \left(\frac{u(P_{\mathrm{M}})}{2 \times P_{\mathrm{M}}}\right)^2} \tag{6-2}$$

The measurement method for obtaining the manifold pressure was an average based on the steady-state portion of a test as described in Chapter 5. This mean pressure included the standard deviation. This will be a component of the combined standard uncertainty attached to the data points in Chapter 7 as it varied between tests.

## 6.3 Propellant Distribution Uncertainties

The propellant distribution results were obtained using the apparatus described in Chapter 5. The sample measurements ($m_{\mathrm{D}}$) contained uncertainties based on 1) the residual water in the PVC tubes, 2) the measurement resolution of the scale used to measure the weight of each sample and 3) the two-test method flow rate uncertainty. The value of each uncertainty is tabulated in Table 6-2. The combined uncertainty for the sample measurements is given by $u(m_{\mathrm{D}})$. Furthermore, the type A uncertainties determined from multiple distribution tests were calculated as the standard deviation from the mean. This value was added to the type B uncertainties and summarised as a combined standard uncertainty in Chapter 7.

Table 6-2: Distribution test uncertainty

| **Parameter** | **Measurement device uncertainty** | **Resolution uncertainty (rectangular probability distribution)** | **Combined standard uncertainty** |
|---|---|---|---|
| $u(m_D)$ | ± 1.14 g (Method) | $0.029\ g$ | ± 1.14g |

For the distribution tests, the deviation ($D_{\mathrm{orifice}}$) is based on the collected mass of water per orifice ($m_{\mathrm{orifice}}$) relative to the nominal mass of water ($\overline{\mathrm{m}}_{\mathrm{orifice}}$). The nominal mass is given by Equation 6-3:

$$\overline{\mathrm{m}}_{\mathrm{orifice}} = \frac{m_{\mathrm{total}}}{N_{\mathrm{orifices}}} \tag{6-3}$$

Here, $N_{\mathrm{orifices}}$ is the number of orifices and $m_{\mathrm{total}}$ is the total mass collected from all the samples. The percentage deviation per orifice ($\mathrm{D}_{\mathrm{orifice}}$) from the nominal mass is given by Equation 6-4:

$$D_{\mathrm{orifice}} = \frac{m_{\mathrm{orifice}} - \overline{m}_{\mathrm{orifice}}}{\overline{m}_{\mathrm{orifice}}} \times 100 \tag{6-4}$$

Table 6-3 summarises the uncertainties for the fuel and LOX distribution tests. This total uncertainty is based on the measurement uncertainty of ±1.143 g of water per orifice sample. These values indicate the type B uncertainties and will be added to the type A uncertainties obtained through repeated experiments in Chapter 7.

Table 6-3: Distribution test deviation uncertainty

| **Parameter** | **Fuel distribution** | **Film cooling** | **LOX distribution** |
|---|---|---|---|
| Orifice deviation uncertainty (%) | 0.44 | 1.77 | 0.36 |

## 6.4 Two-Test Method Justification

The uncertainty for the two-test method is determined below. This uncertainty is the result of a slight manifold pressure variation between tests, leading to a minor difference in flow rates for the full duration and transient tests. The two-test method calculates the mass flow rate corresponding to a manifold pressure using two tests, assuming the flow rates are identical. Therefore, the uncertainty for this calculated mass flow rate was required. A detailed description of the two-test method can be found in Chapter 5.

A flow factor was calculated using the total mass collected during a full duration test and the steady state time period. This flow factor was used to calculate the predicted change in mass flow rate due to a manifold pressure variation between tests. The variation of the calculated mass flow rate, using the flow factor, was used to determine an uncertainty for the mass flow rate measurement, determined using the two-test method.

A maximum allowable pressure deviation of 0.5 % was set for the pressure deviations between tests. A sample calculation for the maximum pressure deviation and the resulting change in the mass flow rate can be found in Appendix C.

Based on the method described above, using a maximum pressure difference of 0.5 %, the difference in mass flow rates between the full duration test and the transient test was 0.0057 kg/s. Over the transient test duration, a maximum of 15.35 g of additional water could be collected due to this mass flow rate difference. This yields a relative uncertainty of 0.066 % for the calculated mass flow rate using the two-test method.

The uncertainties for the actual pressure deviations between tests are shown in Table 6-4. These tests all had a relative mass flow rate uncertainty of less than 0.066 %. Therefore, it was concluded that the two-test method is an appropriate method for calculating the mass flow rates accurately.

Table 6-4: Fuel manifold pressure variation between tests

| Full duration manifold pressure (bar) | Transient test manifold pressure (bar) | Average manifold pressure (bar) | Percentage difference (%) | Flow rate uncertainty (g/s) |
|---|---|---|---|---|
| 3.394 | 3.390 | 3.392 | -0.11 | 0.17 |
| 3.392 | 3.390 | 3.391 | -0.06 | 0.09 |
| 3.334 | 3.338 | 3.337 | 0.01 | 0.03 |
| 3.338 | 3.338 | 3.337 | 0.01 | 0.01 |
| 3.323 | 3.312 | 3.318 | -0.32 | 0.44 |

The above method was applied for the LOX test results. The uncertainties based on the difference in manifold pressure between full duration and transient tests are shown in Table 6-5.

Table 6-5: LOX manifold pressure for start-up and full duration tests

| Full duration manifold pressure (bar) | Start-up and shut down test manifold pressure (bar) | Average manifold pressure (bar) | Percentage difference (%) | Flow rate uncertainties (g/s) |
|---|---|---|---|---|
| 5.000 | 4.994 | 4.997 | -0.13 | 3.9 |
| 5.035 | 5.003 | 5.019 | -0.64 | 19.0 |
| 5.123 | 5.121 | 5.122 | -0.04 | 1.1 |
| 5.121 | 5.088 | 5.104 | -0.65 | 18.9 |
| 5.059 | 5.035 | 5.047 | -0.45 | 13.1 |

## 6.5 CFD Verification

This study builds on the design and analysis of the SAFFIRE injector conducted by Martin (2019). The previous study included a verification based on the GCI for the steady state simulations and a basic validation based on benchmarks with similar fluids and complex geometries. This validation technique was used due to the lack of published experimental data regarding liquid rocket engine injectors.

This study updated the simulations performed by Martin (2019). The required changes included updating the working fluid to water and editing the meshes to capture the boundary layer flow adequately. The complex geometry and fine mesh size used by Martin required the resources of the Centre for High-Performance Computing (CHPC). This facility allows users to login, via a Linux terminal, to upload simulation and script files to run computations on a cluster with up to 240 cores. CHPC was employed for all the simulations conducted in this study due to the intensive resource requirement set by the recommended mesh sizes.

The GCI was calculated to ensure grid convergence was achieved for the updated simulations. The continuum value monitored was the static inlet pressure for the fuel and LOX domains. The values for each mesh refinement are shown in Table 6-6 and Table 6-7.

Table 6-6: Fuel simulation GCI inputs

| Mesh | Base size (mm) | Inlet static pressure (bar) |
|---|---|---|
| Coarse | 1.2 | 3.56 |
| Medium | 0.6 | 3.34 |
| Fine | 0.3 | 3.33 |

Table 6-7: LOX simulation GCI inputs

| Mesh | Base size (mm) | Inlet static pressure (bar) |
|---|---|---|
| Coarse | 1.2 | 5.598 |
| Medium | 0.6 | 4.680 |
| Fine | 0.3 | 4.676 |

The static inlet pressure for a zero-grid spacing and a GCI comparison were obtained using the output data and Equations 2-6 and 2-7 respectively. An estimated error, listed in Table 6-8, was calculated based on the fine mesh continuum value and the zero-grid spacing output for both domains.

Table 6-8: Grid convergence outputs

| Simulation | $P_{\text{inlet,zero-grid}}$ (bar) | GCI comparison | Estimated error (%) |
|---|---|---|---|
| Fuel | 3.33 | 0.996 | -0.024 |
| LOX | 4.68 | 0.999 | -0.00037 |

The GCI comparisons and estimated errors show the simulations are mesh independent using a base size of 0.3 mm. The residual graphs for each simulation can be found in Appendix A. These graphs show the two simulations converge with mesh refinement occurring for base sizes of 1.2, 0.6 and 0.3 mm.

The validation of the CFD simulations performed by Martin (2019) for the K-Epsilon turbulence model was based on previous experimental studies conducted on systems resembling the geometry and flow features of a liquid rocket engine injector. These included the flow around a butterfly valve (Song & Park, 2007), a diesel intake manifold (Emara, Ahmed & Razek, 2015), flow through an orifice (Shah, Joshi, Kalsi, *et al.*, 2012) and two co-axial injector elements (Farmer, *et al.*, 2016). Before this study, the two most appropriate models were the K-Epsilon and K-Omega SST turbulence models (Martin, 2019). Therefore, using the acquired experimental data, the accuracy of the K-Epsilon and K-Omega SST models were investigated. The results of these investigations are discussed in Chapter 7.

### 6.6 Summary

This chapter addressed the evaluation of specific uncertainties of measured and calculated quantities expected to be encountered in the testing of the injector, namely the flow factor, mass flow rate and distribution deviation. The uncertainty arising from the two-test method is determined using the relationship of the pressure drop through a device and the mass flow rate, given by the flow factor.

The chapter also addresses the verification of CFD simulations based on the residuals of a mesh independence study.

# 7. Experimental and Simulation Results

## 7.1 Introduction

The data obtained through the experimental methods described in Chapter 5 were compared to the updated simulation results. A direct comparison between the results could be made due to the simulation updates and calibration procedures described in Chapters 4 and 5. The data collected contained the uncertainties documented in Chapter 6. The modelled pressure drop, flow factor and propellant distribution accuracies were determined based on the deviation from the experimental results.

## 7.2 Simulation Data

The performance parameters from the updated simulations, described in Chapter 5, were used as an initial testing point to validate the pressure drop through the SAFFIRE injector. The performance parameters for both flow domains are summarised in Table 7-1.

Table 7-1: Fuel Simulation and Experimental starting point

| **Variable** | **Fuel domain** | **LOX domain** |
|---|---|---|
| Fluid | Water | Water |
| Gauge outlet pressure (bar) | 0 | 0 |
| Manifold pressure (bar) | 3.39 | 4.36 |
| Mass flow rate (kg/s) | 2.28 | 5.66 |
| Flow factor ($m^3/h.bar^{0.5}$) | 4.46 | 9.76 |

## 7.3 Pressure Drop Calibration

Numerous inlet pressures were tested to calibrate the mass flow rate through the fuel test article. Figure 7-1 shows the simulation manifold pressures and the testing manifold pressures for 20 experiments resulting in various mass flow rates. From the initial calibration tests, it was concluded that on average, the simulation was under predicting the manifold pressure by approximately 9%. At this stage, the injector was inspected for possible geometric imperfections. The down-coming holes were cleaned to ensure no burs were influencing the flow. During this process a small amount of material was removed from the down-coming holes of ring two. Figure 7-1 shows the mass flow rates and manifold pressures for the simulation and experimental results. Error bars are not shown as uncertainties were not considered during calibration tests.

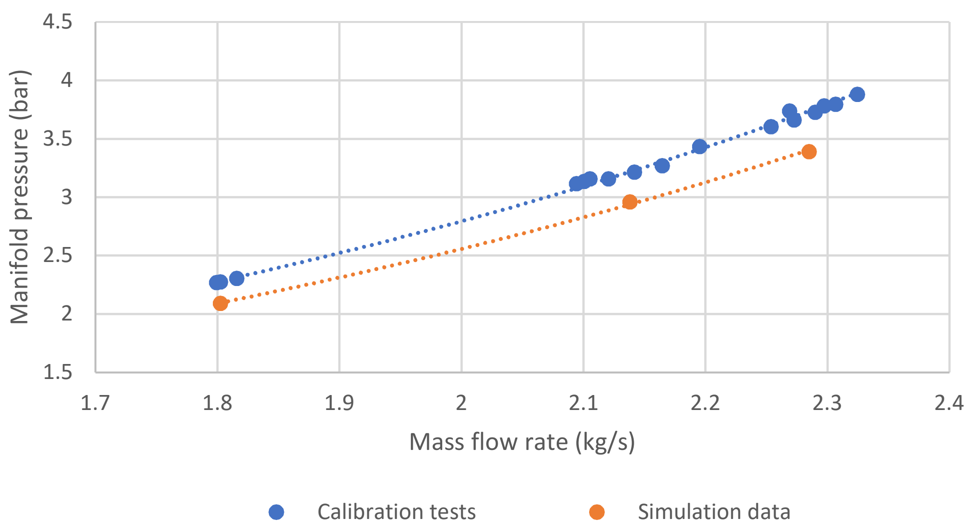


Figure 7-1: Mass flow rate vs manifold pressure

## 7.4 Fuel Domain Results

The fuel test article was cold flow tested to determine the pressure drop and propellant distribution using water as an analogue fluid. The simulation results were gathered using the updated simulations discussed in Chapter 5 with the same fluid and geometry.

### 7.4.1. Pressure Drop

Based on the required tank pressure determined during the calibration tests, a final set of tests was conducted within 1% of the design mass flow rate of 2.28 kg/s. A total of ten tests were conducted, including five full duration and five transient tests. The tank pressure and manifold pressure varied slightly between tests due to practical limitation. The two-test method was used to calculate the mass flow rate corresponding to the manifold pressure of each test. The mass flow rate uncertainties were based on the two-test method uncertainties and the measurement limitations described in Chapter 6.

In Figure 7-2, the mass flow rates and manifold pressures are shown for the simulation and experimental results. The error bars show the range of a 99 % confidence interval for the experimental data points. The minimum deviation obtainable was ±2.48 %. This was mainly attributed to a 0.0625 bar uncertainty for the manifold pressure transducer. Each simulation data point fell within the uncertainty limits of the experimental results as shown in Figure 7-2. It was concluded that the simulation accurately predicted the required pressure drop across the injector for the design mass flow rate of the fuel domain. The simulation deviated by a maximum of 2.48 % from the experimental results.

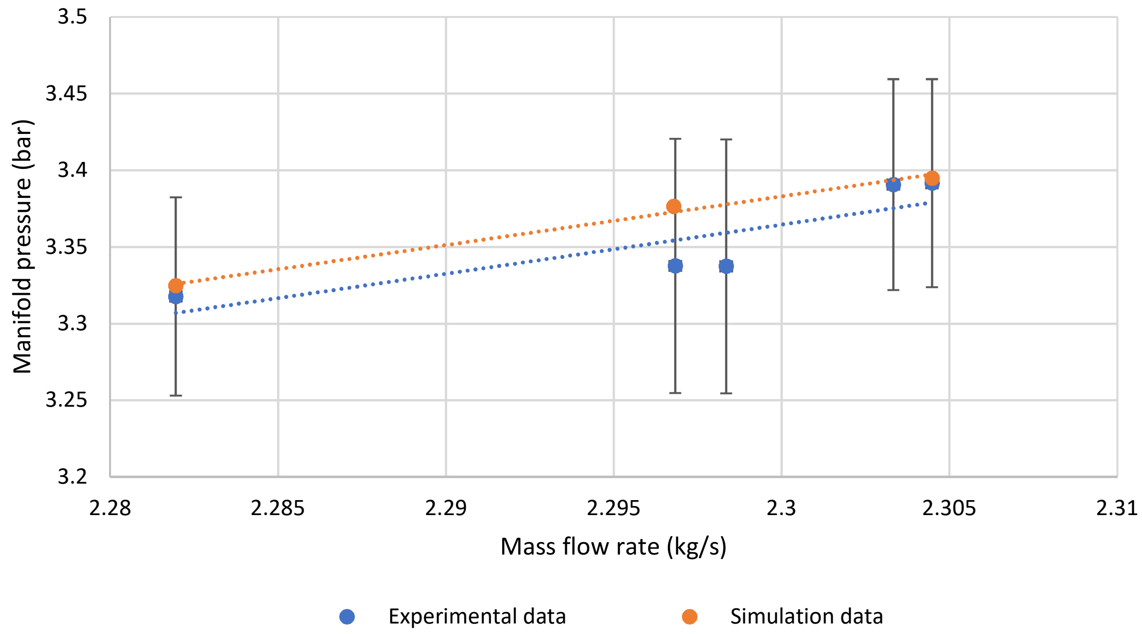


Figure 7-2: Manifold pressure vs mass flow rate

Another method of comparing the simulation and experimental data points was to calculate the flow factor ($K_v$) as described in Chapter 2. Table 7-2 shows the calculated flow factor, including the combined uncertainty for the experimental results. When comparing the simulation and experimental data, it was evident that on average, the simulation underestimated the flow factor by 2.22 % with an uncertainty of 1.1 %. This method shows that the pressure drop was accurately modelled by the simulation with a high degree of accuracy.

Table 7-2: Flow factor for experimental and simulation data

| **Parameter** | **Experimental Data** | **Simulation** | **Percentage Difference** |
|---|---|---|---|
| Flow factor ($m^3/h.bar^{0.5}$) | 4.5 ± 0.049 | 4.46 | 2.22 % ± 1.1% |

### 7.4.2. Turbulence Model Sensitivity

An investigation was conducted using different wall treatments and turbulence models for the simulation to determine the accuracy of the modelled pressure drop and propellant distribution. The pressure drop accuracy was based on the percentage difference between the experimental manifold pressure and the simulation manifold pressure for the same mass flow rate. The distribution accuracy was based on the total average deviation from the experimental distribution results. This data was normalised based on the contribution to the total mass flow rate.

Switching to a high $y^+$ wall treatment decreased the accuracy based on the experimental data. The high $y^+$ wall treatment model required five custom volume meshes to achieve an adequate $y^+$ range for the entire domain. The decrease in accuracy and lengthy setup time leads to the conclusion that the most appropriate wall treatment was the low $y^+$ model.

The K-Omega turbulence model did not increase the accuracy, although a good correlation between the experimental results was still present. The percentage deviation of each model is shown in Figure 7-3. It is evident that the original turbulence model selected (K-Epsilon low $y^+$) yielded the lowest deviation from the experimental results for the pressure drop and propellant distribution.

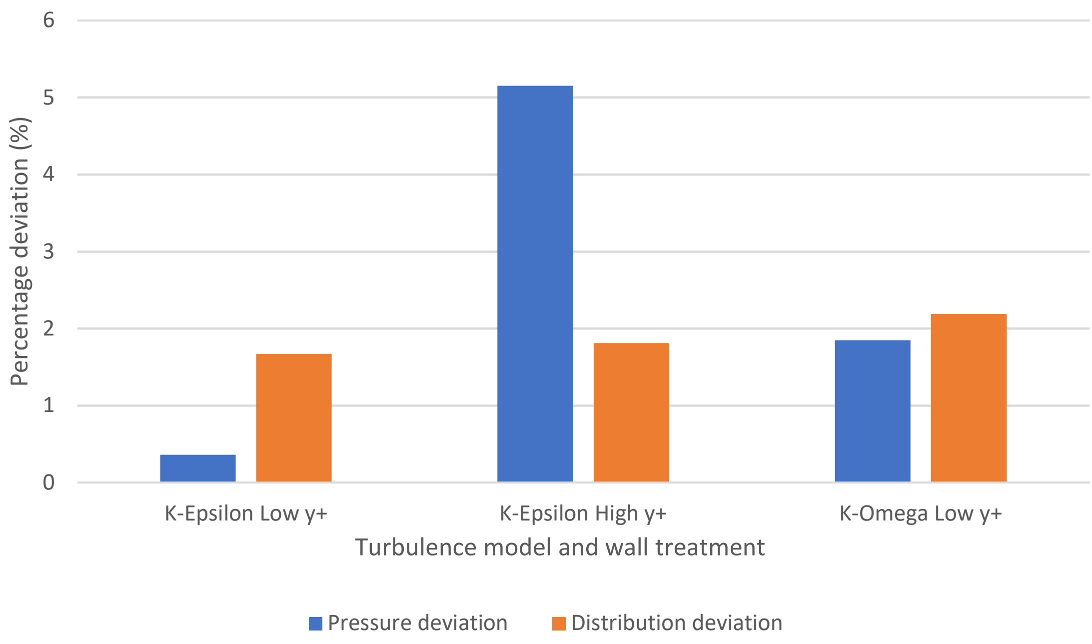


Figure 7-3: Turbulence model and wall treatment sensitivity

### 7.4.3. Fuel Distribution Results

Each fuel distribution test resulted in a total of 112 mass flow rate data points. Each mass flow rate corresponded to an orifice on the fuel faceplate. The element layout contained five concentric rings, as shown in the chamber section of the fuel faceplate (Figure 7-4). Rings one to five contained 8, 16, 24, 32 and 32 orifices respectively. The fastener holes used to secure the faceplate to the injector body are also shown. The average distribution per ring was investigated followed by an in-depth study of each ring. The orifice indices used for the distribution results were based on a counterclockwise rotation about the faceplate centre of the orifice index line shown in Figure 7-4.

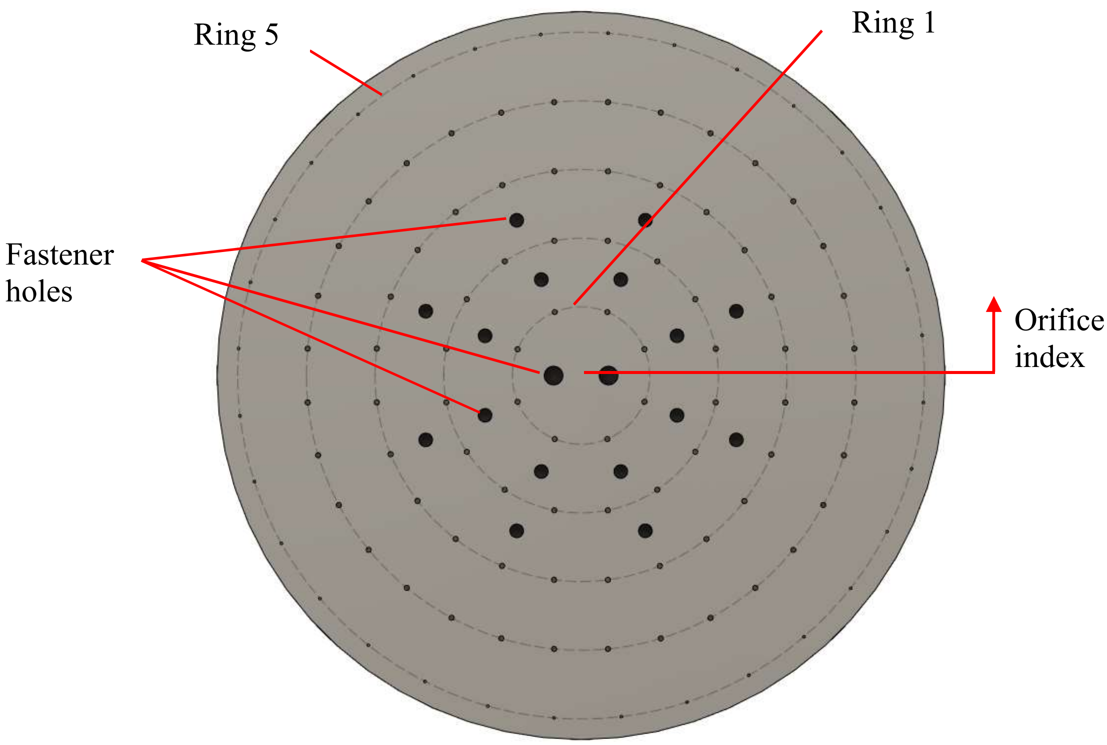


Figure 7-4: Fuel faceplate chamber section

The experimental distribution results were compared to those of the updated simulation. The deviation from the nominal mass flow rate per orifice was calculated for the experimental and simulation results. The deviations were based on the design criteria of even distribution. This assumes that each orifice injects an equal amount of propellant into the combustion chamber. The deviations of the orifices flow rates from the nominal flow rate were calculated using Equation 6-3 and 6-4.

The average deviation of each ring was calculated using the nominal flow rate per orifice and the number of orifices per ring. An important note is that the deviations from the nominal mass flow rates are not completely representative of the original injector design due to the minor geometric changes discussed in Chapter 5 and the change in fluid properties. The output from the distribution tests was the accuracy of the simulation when evaluating the propellant distribution, compared to the experimental data. Table 7-3 shows how the simulation distribution results deviated from the experimental results for each concentric ring. The contribution per ring to the total mass flow rate is also given. This represents the significance of the accuracy. The percentages are based on the mass flow rate for the correct O/F ratio. Film cooling was considered as additional fuel. Therefore, the four inner rings make up 100 % of the 2.07 kg/s designated for the main orifices and ring 5 makes up 100 % of the 0.207 kg/s designated for the film cooling.

Table 7-3: Simulation percentage deviation from experimental results

| Ring | Simulation deviation from experimental data (%) | Total specific flow rate percentage (%) |
|---|---|---|
| 1 | 3.26 | 10 (Main orifices) |
| 2 | -0.76 | 20 (Main orifices) |
| 3 | -2.09 | 30 (Main orifices) |
| 4 | 1.11 | 40 (Main orifices) |
| 5 | 0.11 | 100 (Film cooling) |

Figure 7-5 shows the percentage deviation from the nominal mass flow rate for each ring. A 0 % deviation represents the nominal mass flow rate. The flow rate deviations are shown for the experimental data (blue) and the simulation data (orange). The CFD simulation on average predicts the propellant distribution with a relatively high accuracy based on the experimental results. On average rings two, three, four and five deviated by less than 3 % from the experimental results with ring one having the highest deviation of 3.26 %. Ring one contains the least significant contribution to the total mass flow rate. The film cooling ring has the largest uncertainty due to the lower flow rate.

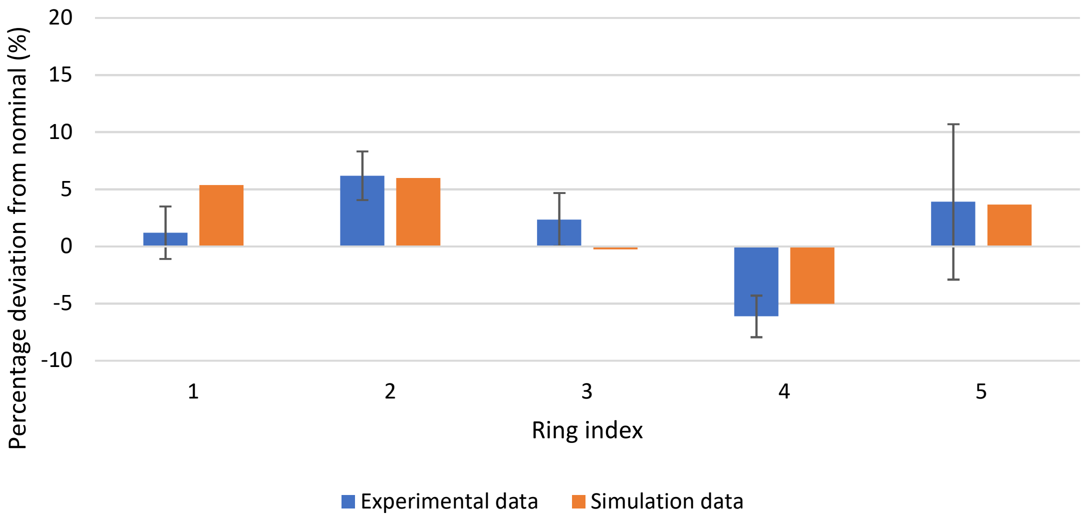


Figure 7-5: Average propellant distribution deviation per ring

When analysing the individual rings, it was evident that the maximum deviation from the experimental results was larger than 3.26 %. However, the CFD simulation did accurately predict the change in flow rate between each orifice. These trends are visualised using a line graph with each orifice as a data point. Each data point is an average taken from ten tests, including five full duration tests and five transient tests. The error bars for the experimental data include all the uncertainties discussed in Chapter 6. These uncertainties are multiplied by a coverage factor of 3.747 based on a t-distribution with four degrees of freedom to result in a level of confidence of 99 %.

Figure 7-6 shows the data for ring one, containing eight orifices. This graph shows the deviation from the nominal mass flow rate for the simulation and experimental data. The specific deviation from the simulation results can be found in Appendix A. The data was representing in this manner as it displays the ability of the simulation to capture the trends between orifices. Red indicators identify simulation data points that fall outside of the experimental data error bars. The down-coming holes for this ring are positioned between orifices 4-5 and 8-1. The data from ring one implies the orifices near the down-coming holes are captured more accurately by the simulation.

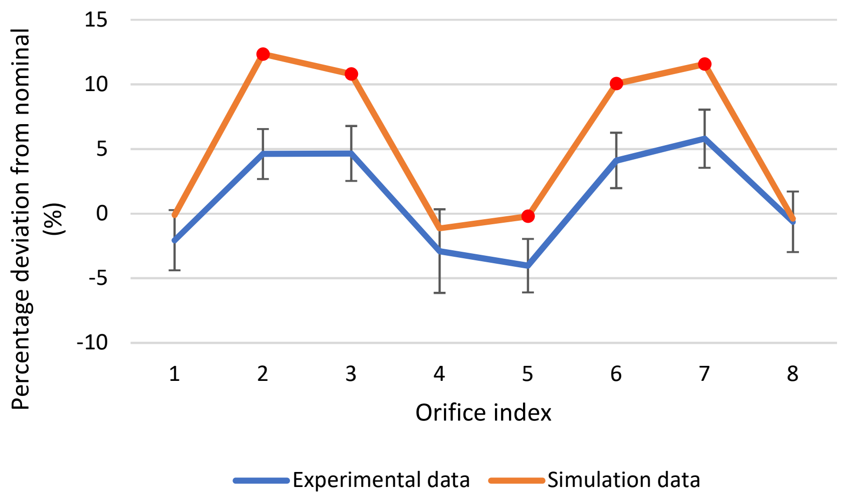


Figure 7-6: Ring one simulation vs experimental distribution data

Figure 7-7 displays the 16 orifices from ring two. A total of 20 % of the total flow rate is supplied from this ring. The simulation data shows good correlation to the experimental data as 14 of the 16 orifices are accuracy modelled by the simulation.

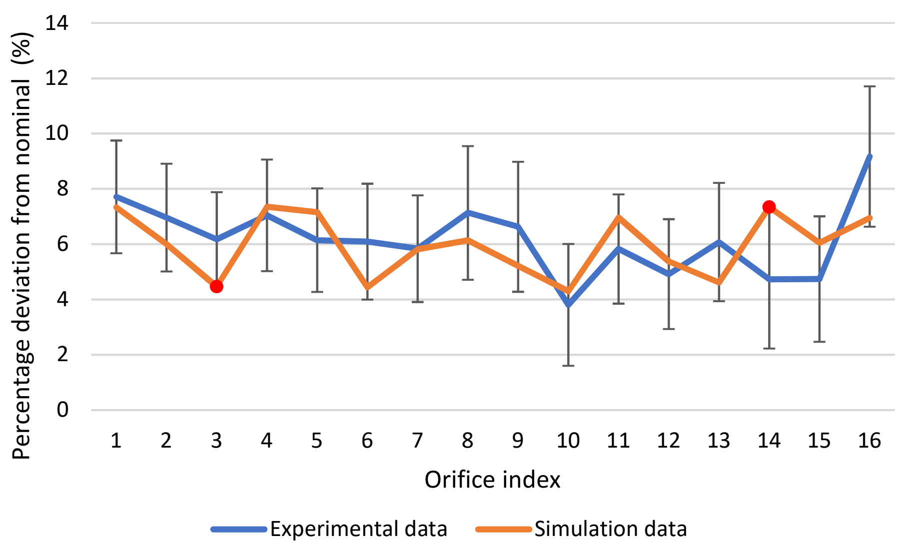


Figure 7-7: Ring two simulation vs experimental distribution data

Ring three supplies 30 % of the total flow rate to the combustion chamber through a total of 24 orifices. The simulation only modelled 58 % of the orifices accurately. However, the change in mass flow rate between orifices is accurately modelled.

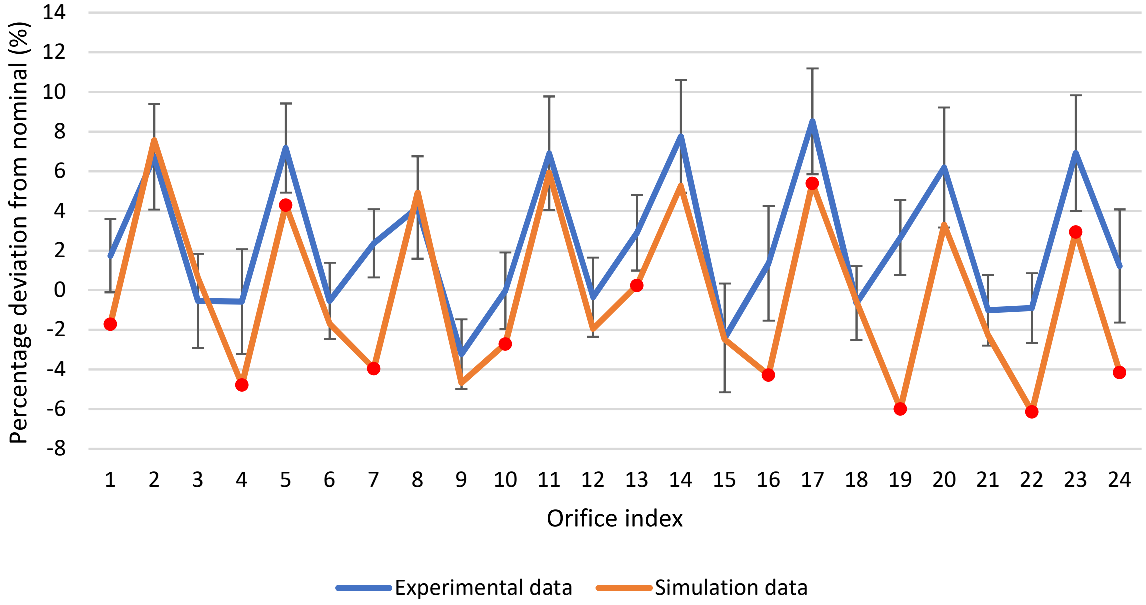


Figure 7-8: Ring three simulation and experimental data

Ring four supplies 40 % of the mass flow rate through 32 orifices. This is the largest contribution from a single ring. Figure 7-9 shows the major outliers for the simulation are located at orifices 12 and 13. The simulation was unable to accurately model the flow near these orifices. A custom volumetric mesh did not solve the problem of the simulation underpredicting the mass flow rate at these orifices. Ring four contains four orifices between each down-coming hole, identical to ring one, which suggested the simulation accuracy reduces as the distance of the orifice from the down-comer hole increases. This relationship was not reiterated by ring four.

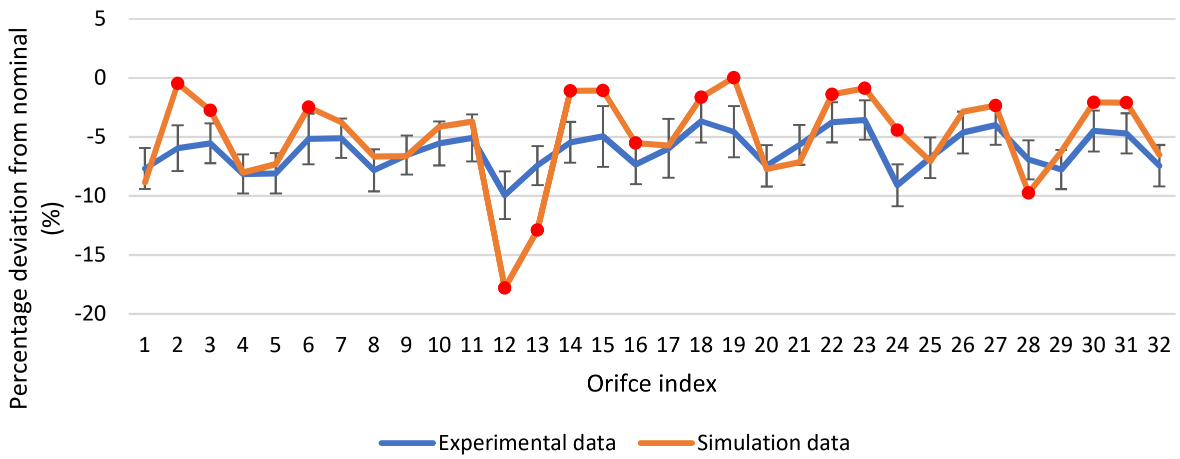


Figure 7-9: Ring four simulation and experimental data

The film cooling was designed to supply a total of 0.207 kg/s of propellant to the outer ring of the combustion chamber, containing 0.8 mm orifices. The small diameter orifices increased the flow rate sensitivity caused by slight diameter variations. The diameters of the film cooling orifices were increased on the simulation CAD model to match the test article due to machining defects discussed in Chapter 5. Orifices 14, 15 and 16 contained the largest diameter variations. A minor diameter variation increased the flow rate by roughly 25 %. However, the simulation was able to capture this increase in flow rate accurately, as shown in Figure 7-10. The uncertainties of the film cooling distribution tests were larger than the main orifices due to the lower flow rates. Therefore, additional water collected amounted to a larger propellant deviation than the main orifices. For the film cooling, 1.143 g of additional water equated to an increase in deviation of 1.77 %. The 1.143 g was used as a reference as this is the total uncertainty resulting from the measurement method used.

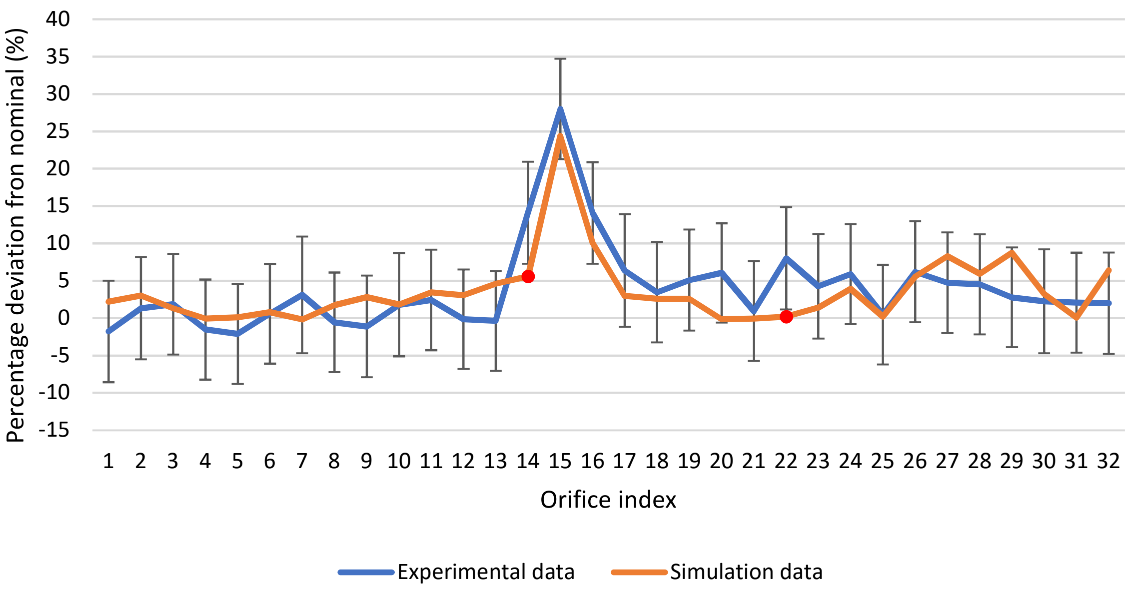


Figure 7-10: Ring five simulation and experimental distribution

## 7.5 LOX Domain Results

The LOX test article was tested to determine the pressure drop required to drive the analogue propellant through the SAFFIRE injector oxidiser domain at a mass flow rate of 5.66 kg/s. Secondly, the propellant distribution was determined using cold flow testing methods. The simulation results were compared to the experimental data. The experimental data shown below contains uncertainties based on the components described in Chapter 6. The experimental and simulation data consisted of water flowing through the same domain at the same mass flow rates, within practical limits. Therefore, the two data sets were compared to each other directly.

### 7.5.1. Pressure Drop

The pressure drop through the injector dictates the mass flow rate as described by the flow factor. The pressure tap positioned in the top manifold was used to compare the pressures from the simulation to the experimental results. Figure 7-11 shows the pressure readings and corresponding mass flow rates taken from a total of ten tests, including five full duration and five transient tests. These are compared to the updated simulations described in Chapter 5 at the same mass flow rates. Figure 7-11 shows that the simulation on average underpredicts the manifold pressure by 14.59 %. The turbulence model and wall treatment have a large impact on the modelled pressure in the flow domain. Therefore, a sensitivity study was conducted to evaluate the accuracy of different turbulence models and wall treatments.

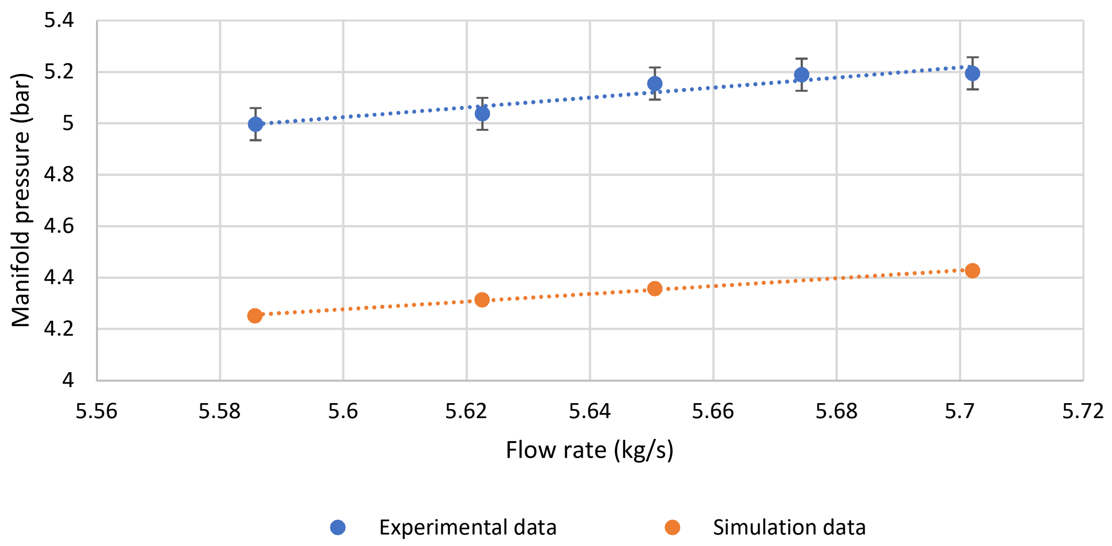


Figure 7-11: Manifold Pressure vs mass flow rate

The average flow factor calculated using Equation 2-5 for the simulation data was 9.76 $\mathrm{m}^3/\mathrm{h.bar}^{0.5}$. The experimental flow factor, including the uncertainty, was 8.99 $\pm 0.055\ \mathrm{m}^3/\mathrm{h.bar}^{0.5}$. The modelled flow factor deviated from the experimental data by 15.5 % ±0.74 %.

The required pressure drop across the orifices was 4.374 bar using the empirical model described by Equation 2-11. For the simulation, the average pressure of each ring supplying the orifices was 4.386 bar. The simulation and empirical model differ by 0.274%. Therefore, the empirical model was used to determine the increase in pressure due to a 10-micron decrease in orifice diameter. This quantity was used as the measurement resolution of the LOX orifices was 10 microns. The required pressure increased by 0.127 bar. The difference between the manifold pressures for the simulations and experiments was roughly 0.74 bar. This empirical comparison implies a small component of the difference in pressure drop could be a result of minor geometric deviations in the test article orifices.

### 7.5.2. Turbulence Model Sensitivity

Different wall treatments and turbulence models were investigated based on the recommendations made by Martin (2019). Additionally, a low $y^+$ wall treatment was used due to the high accuracy obtained from this model for the fuel domain validation. The pressure drop accuracy is based on the percentage difference of the simulation manifold pressure and the experimental manifold pressure. The distribution deviation is based on the average propellant deviation of each ring. This data was normalised based on the percentage of propellant supplied by each ring. The low $y^+$ model improved the accuracy of the modelled pressure drop. However, the accuracy of the propellant distribution was decreased. The K-Omega SST turbulence model increased the accuracy of the modelled pressure drop slightly while decreasing the accuracy of the normalised propellant distribution. Based on these data outputs, the recommended turbulence model and wall treatment is the K-Epsilon high $y^+$, because of the increased accuracy of the propellant distribution.

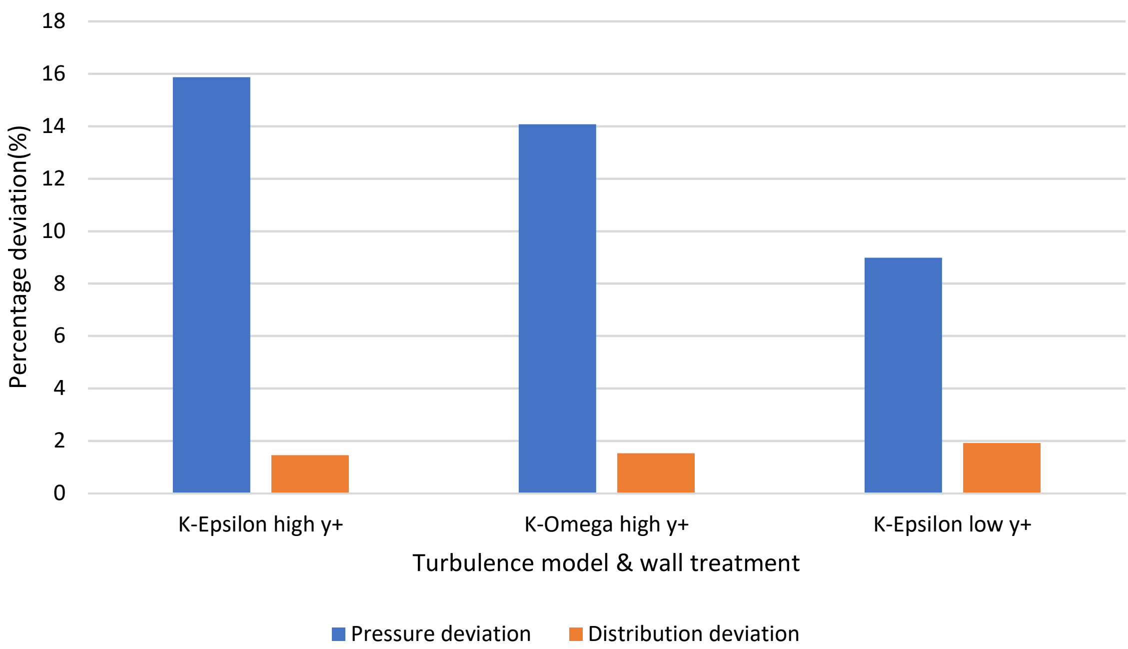


Figure 7-12: Turbulence model and wall treatment sensitivity study

### 7.5.3. LOX Distribution Results

A total of 160 data points were produced from each LOX distribution test. This data was used to determine the experimental deviation from the nominal orifice flow rate, using Equation 6-3 and 6-4. These results included the same measurement uncertainties described for the fuel distribution tests as the 'two-test' method and an identical measurement method was followed. The uncertainties are listed in Table 6-2. The combined uncertainties include type A uncertainties based on eight tests which include four full duration tests and four transient tests. The uncertainties were multiplied by a coverage factor of 4.541 based on a t-distribution with 3 degrees of freedom to yield a 99 % confidence interval.

The LOX faceplate contains five concentric rings. Ring one is positioned in the centre and ring five is the outer ring. The orifice indices used in the distribution results section are based on a counterclockwise rotation about the faceplate centre starting from the horizontal orifice index line shown in Figure 7-13.

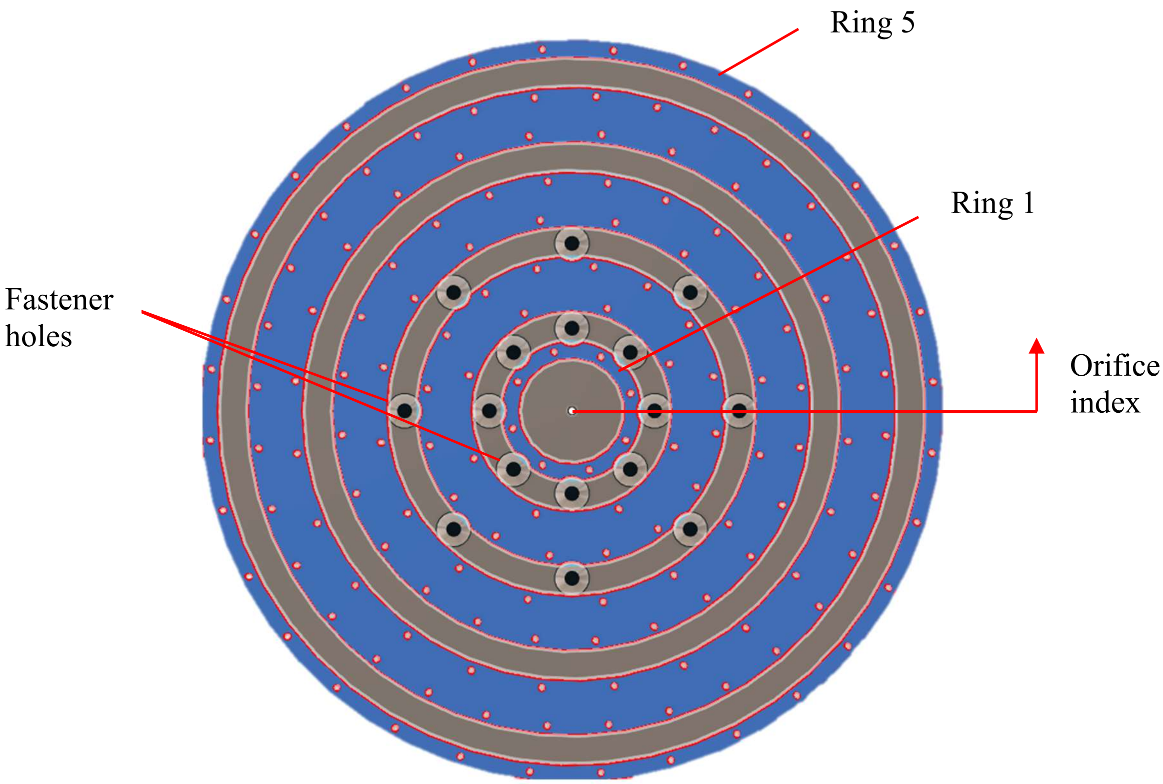


Figure 7-13: LOX faceplate

The average ring deviation from nominal is displayed in Figure 7-14. This graph shows that the simulation accurately predicts the average mass flow rate per orifice for rings two, three and four. This was concluded due to the average distribution for the simulation falling within the error bars of the actual distribution determined through cold flow testing. These rings account for 75 % of the total mass flow rate. The largest deviations between the simulation and experimental results are rings one and five with deviations of 2.82 % and 2.99 % respectively. Ring one accounts for 5 % of the total mass flow rate and contains only eight orifices. Ring five contains 32 orifices and is designed to supply 20 % of the propellant to the combustion chamber. The causes of these deviations are discussed when investigating the individual rings.

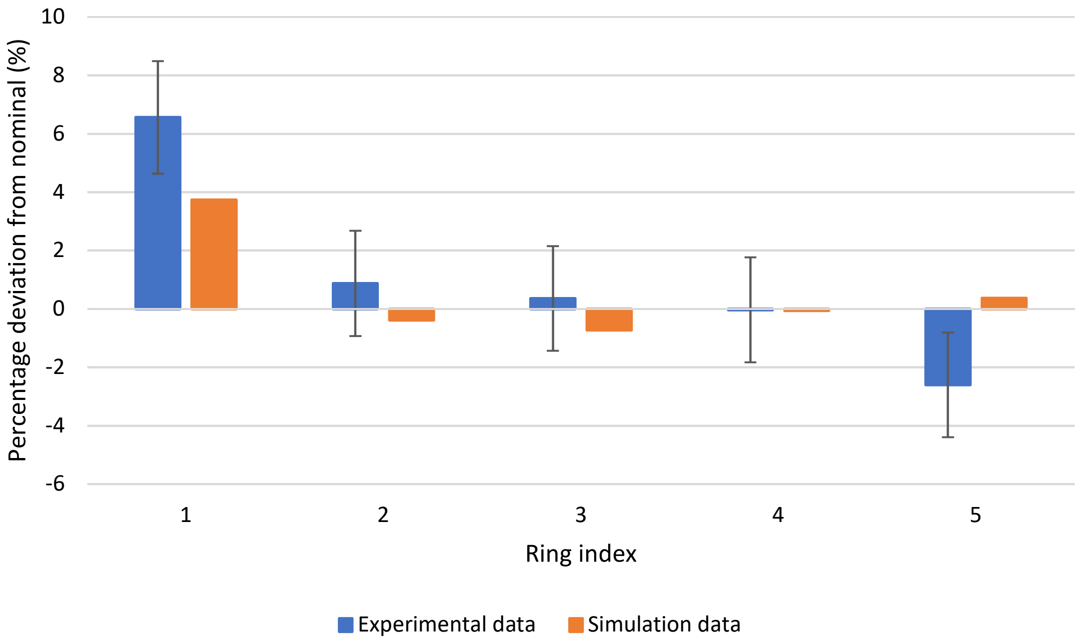


Figure 7-14: Ring average deviation for experimental and simulation data

The mass flow deviation from nominal, based on even distribution for each orifice, was calculated from the data collected during propellant distribution tests. The results showed that the original turbulence model and wall treatment accurately predict the mass flow rate for most of the orifices. Each ring was investigated separately and its significance is described as a percentage of the total flow rate based on even distribution. Red indicators identify simulation data points that fall outside of the experimental error bands. The specific deviations for the simulation data from the experimental data are shown for the LOX domain in Appendix A. The visual representation used below displays the ability of the simulation to model the change in flow rate between orifices.

Figure 7-15 shows the deviation for ring one which supplies 5 % of the total mass flow rate. The results indicate that the simulation on average underpredicts the mass flow rate by -2.82 %, although the general trends within the ring are accurately predicted. The centre down-comer geometry was deemed acceptable as the flow rate did not deviate by a large percentage between orifices. However, the down-coming centre hole diameter could be reduced to decrease the mass flow rate for the ring.

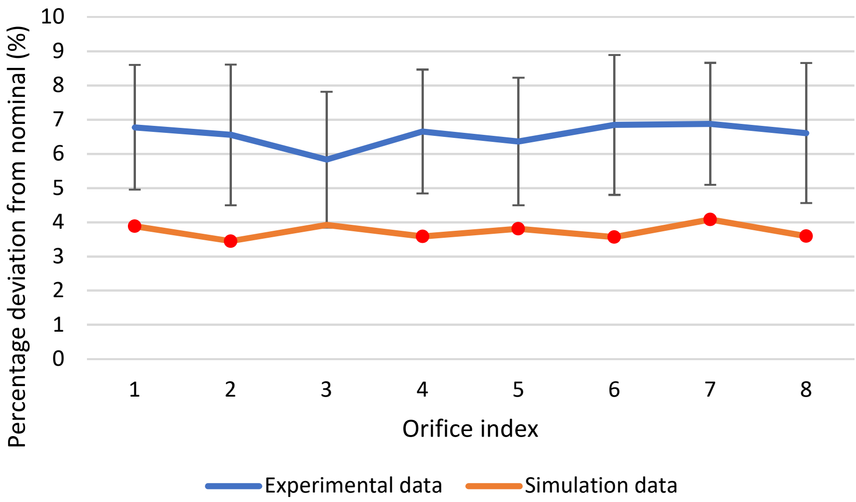


Figure 7-15: Ring one experimental and simulation deviation from nominal

Ring two supplies 15 % of the total flow rate through 24 orifices. Figure 7-16 shows the distribution results for the simulation and experimental data. Maximum and minimum valves are present at orifices 8, 9, 20 and 21. These were caused by the position of the down-coming slots in the internal geometry. This ring is accurately modelled. However, the overall trend is skewed due to an overestimation of the flow rate near orifices 5 and 17.

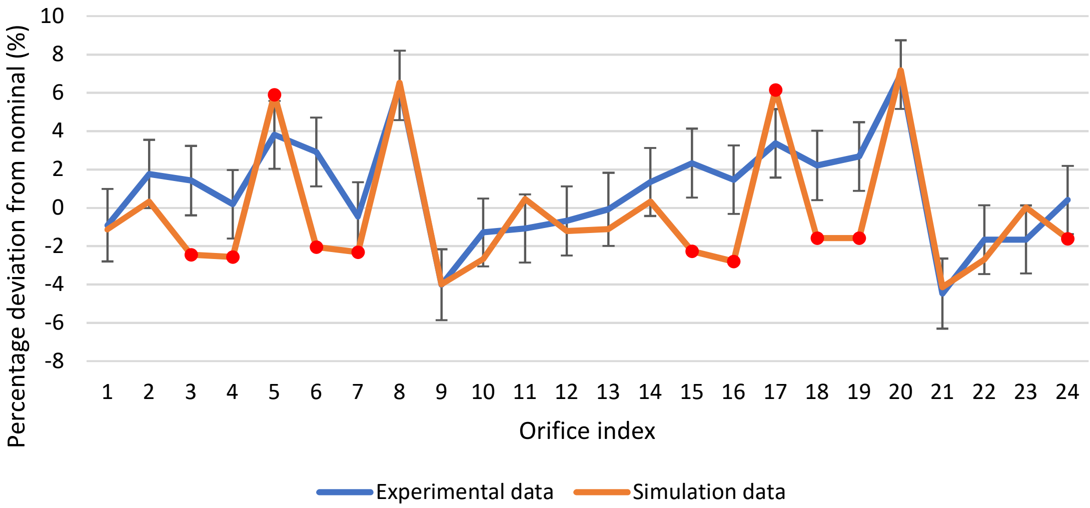


Figure 7-16: Ring two experimental and simulation deviation from nominal

Ring three supplies 25 % of the total flow rate through 40 orifices. The average deviation was accurately modelled by the simulation within -1.08 %. From Figure 7-17, the results show that the simulation predicts the trends between orifices with a high degree of accuracy. The variations are due to geometrical error or computational errors. It is likely that these variations are due to the tolerances on the orifices since some of the orifices are modelled accurately and other are not.

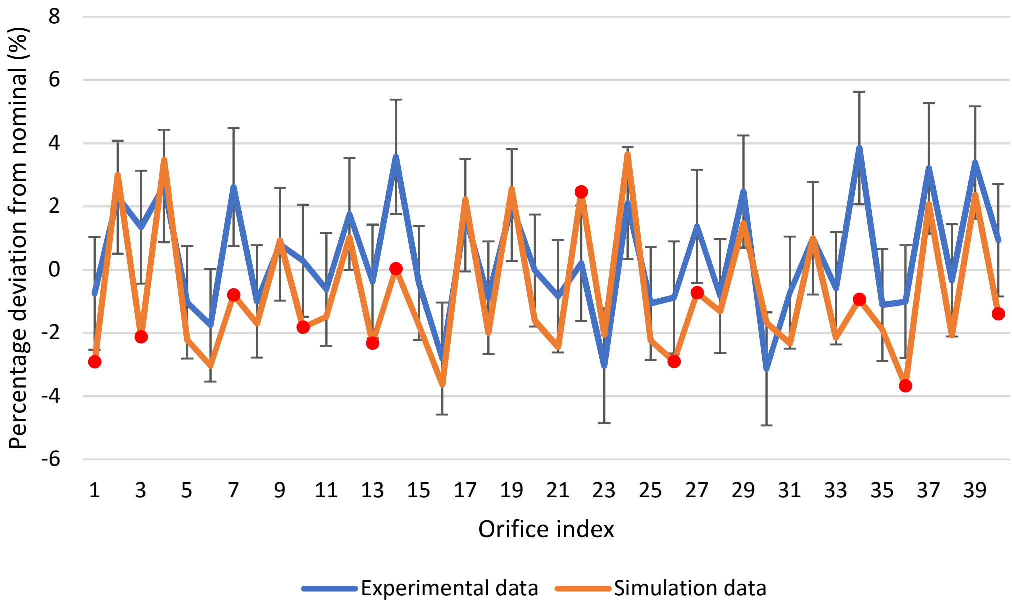


Figure 7-17: Ring three experimental and simulation deviation from nominal

Ring four has the largest contribution to the total mass flow rate of 35 % through 56 orifices. The simulation accurately predicted the propellant distribution for ring four with an average deviation from the experimental results of -0.03 %. However, there are orifices that deviate by as much as 7 %, as shown in Figure 7-18. These deviations in the experimental results were possibly caused by non-uniform orifices, as shown in Figure 5-22. The propellant distribution was expected to be symmetric due to the symmetrical nature of the injector flow domain.

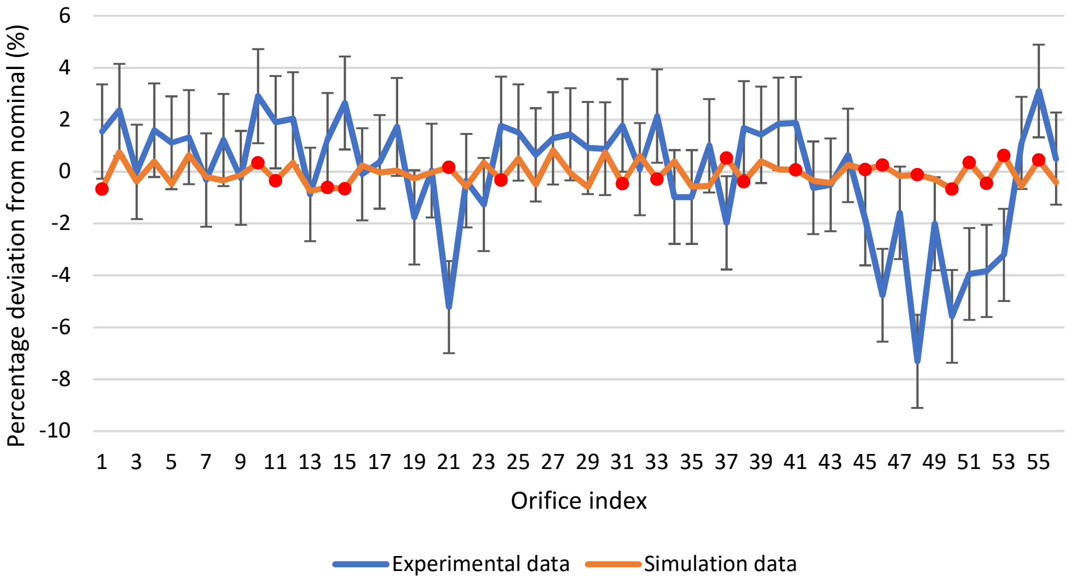


Figure 7-18: Ring four experimental and simulation deviation from nominal

Ring five supplies 20 % of the total oxidiser flow rate through 32 orifices. The asymmetric nature of the experimental data, shown in Figure 7-19, suggests a geometric or setup error was present during testing. This deviation was concluded as a mixture of orifice defects and a minor offset between the injector body and the top manifold due to a possible misalignment during assembly. Figure 7-19 shows roughly half of ring five (16 orifices) produced a below average mass flow rate.

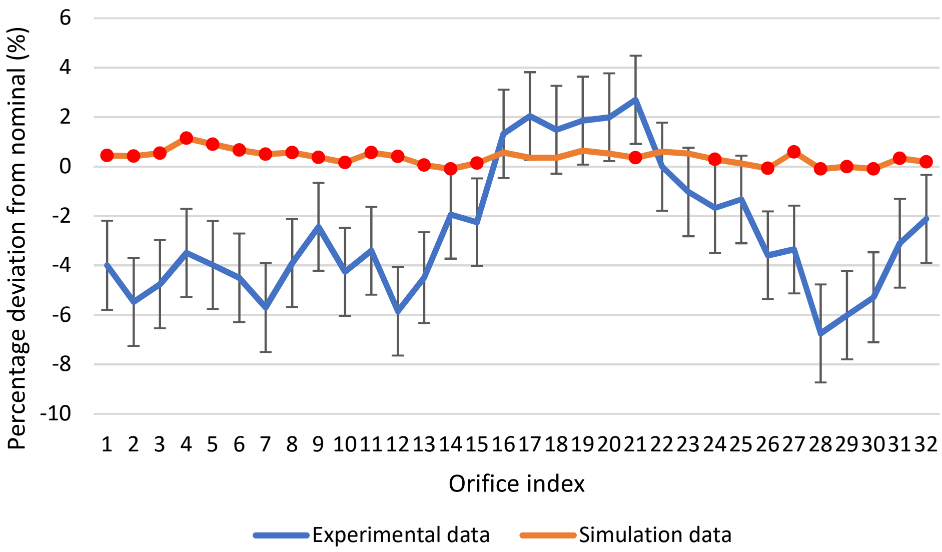


Figure 7-19: Ring five experimental and simulation deviation from nominal

Upon investigation, the total possible offset due to tolerances on the securing bolts was 0.5 mm. A simulation was run with a 0.5 mm step between the injector body and the top manifold. The results did not capture the same deviations as the experimental data, showed in Figure 7-19. An exaggerated offset of 1 mm resulted in the simulation diverging due to reverse flow within the LOX domain. The exact cause of the deviation was not identified. However, since roughly 50 % of the orifices produce a reduced flow rate, a geometric error was the most probable cause. Therefore, the interfaces of the injector bodies should not be positioned where a slight misalignment will cause an obstruction to the flow. Furthermore, the orifice entry, exit quality and diameters are critical for the correct performance of the SAFFIRE injector.

## 7.6 Summary

The results concluded that the simulations accurately predicted the average propellant distributions per ring for the fuel and LOX domains with a maximum average deviation per ring of 3.26 % and 2.97 % respectively. The individual orifice deviations showed the simulations accurately modelled the change in flow rate between orifices. This was achieved using the 112 and 160 data points collected when testing the fuel and LOX test articles respectively.

Analysing the simulation and experimental results for the fuel domain showed that the major geometrical factors influencing the distribution were the down coming holes and the orifice diameters. The simulation updates made in Chapter 5 ensured the simulation CAD model contained the machined orifice diameters. The down coming holes could not be measured with a high degree of accuracy due to their locations. Therefore, a geometric deviation could exist between the physical test article and the simulation model. This was concluded as the most likely cause for the difference between the simulation and experimental data. Alternatively, the deviation of the simulation data could be due to simplifications of the simulation model made to reduce the computational resource requirements, for example, using a steady state solver to model the turbulence within the injector domains.

The LOX test article contained high propellant distribution variations between the simulation and experimental data sets within rings four and five. The poor-quality orifice exits, shown in Figure 5-22, were concluded as the most probable cause for the propellant deviations in ring four. The deviations in ring five were concluded as the result of a geometric deviation caused by a misalignment between the injector body and top manifold since half the orifices performed poorly.

# 8. Conclusion and Recommendations

## 8.1 Overview

The objective of this study was to determine the accuracy of two CFD simulations used to evaluate the performance of the SAFFIRE liquid rocket engine injector. This involved developing a cold flow test stand and test articles to obtain experimental data.

The SAFFIRE cold flow test stand was developed and used to determine the performance of the fuel and LOX test articles. It was concluded that the fuel simulation accurately modelled the pressure drop and propellant distribution of the fuel domain with an average maximum deviation from the experimental results of 2.48 % and 3.26 % respectively. The LOX domain pressure drop and propellant distribution were modelled with a maximum average deviation from the experimental results of 15 % and 2.97 % respectively.

## 8.2 Conclusion

The research objectives discussed in Chapter 1 were concluded as follows:

*Objective 1: Develop an experimental testing methodology to determine the pressure drop across separate oxidiser and fuel injector domains, for nominal mass flow rates, and propellant distribution for the SAFFIRE injector.*

A two-test method was used to determine the mass flow rate through the injector corresponding to a manifold pressure. This included a full duration test and a transient test. The propellant distribution was determined by calculating the flow rate through each orifice for both the fuel and oxidiser domains using the same two-test method.

The testing method involved designing and manufacturing two separate test articles for the fuel and oxidiser domains. Each test article was tested independently with water as the analogue fluid to determine the actual performance. Since the injector domains were separated, injector mixing tests were not conducted. However, the distribution of each domain was evaluated based on the mass flow rate per orifice before impingement. The simulations were updated with water as the fluid to determine comparable results. These simulations were verified to reduce simulation errors.

The pressure drop tests involved collecting the total mass of water expelled through the injector during the test duration. The propellant distribution tests involved designing and manufacturing attachments which were mounted below the injector faceplate to determine the mass flow rate through each orifice.

*Objective 2: Develop and commission a SAFFIRE injector test stand incorporating an appropriate control system, flow control valves and data acquisition hardware to obtain the required experimental data.*

The test facility and test stand were designed to conduct multiple cold flow tests using a regulated blow down configuration with nitrogen as the pressurant gas. A relatively constant tank pressure was maintained during all tests by using a single-stage regulator and a pressurant inlet solenoid. The test stand used a LabVIEW control system with National Instruments hardware to control the valves and data acquisition for the entire system. A LabVIEW based auto sequence controlled the test procedure. The data acquisition was also managed by the control system via 4-20 mA signals received by the NI hardware.

*Objective 3: Obtain comparative data sets for the simulation and experimental results by conducting cold flow tests for the oxidiser and fuel domains, using water as an analogue fluid, and modelling the injector performance using the analogue fluid properties.*

Multiple cold flow tests were conducted to determine the operating conditions for the fuel and LOX test articles. Final tests were conducted on each test article within 1.1 % and 1.3 % of the design mass flow rates for the fuel and oxidiser domains respectively. These tests were used to determine the pressure drop and propellant distribution for each domain. Type A uncertainties were determined based on the repeated data points obtained from multiple tests. Type B uncertainties were included based on the equipment specifications and testing methods.

The simulation results were updated with water as the simulation fluid. The simulation fuel and LOX domains required remeshing due to the changes in density and viscosity of the fluids. New simulation performance parameters were obtained for the pressure drops and propellant distributions using water as the fluid. Therefore, these results could be compared to the experimental results.

*Objective 4: Determine the accuracy of the simulation results based on the deviation from the experimental data and validate the final performance of the injector.*

The actual pressure drop and propellant distribution for the design mass flow rate of 2.28 kg/s through the fuel domain were determined experimentally and compared to the simulation results. The simulation modelled the pressure drop and average propellant distribution per ring with an accuracy of 2.48 % and 3.26 % respectively. These accurate results validated the selection of the CFD assumptions, namely the turbulence model, wall treatment, mesh type and selected solving algorithm. Furthermore, based on the turbulence model and wall treatment investigation, the K-Epsilon all wall treatment with a wall $y^+ < 1$ was the most accurate model for the complex internal geometry of the fuel domain. For future injector designs where the flow is mainly distributed through internal passages with small diameter holes, a K-Epsilon turbulence model with a low $y^+$ wall treatment is recommended.

The experimental results for the LOX domain showed that the simulated pressure drop was modelled with a 15% deviation. The deviation was possibly caused by a simulation error, namely a model error caused by simplifications of the conceptual model. An alternative cause for the deviation was a geometrical error due to a misalignment or poor orifice exit quality in rings four and five. However, the average propellant distribution per ring was accurately modelled with a maximum deviation of 2.97 % from the experimental results.

This study concluded that the selected CFD models accurately captured the flow phenomena within each flow domain to yield an accurate result for the propellant distribution. This was achieved by comparing the simulation results, conducted with water as the simulation fluid, to the experimental data gathered using water as the testing fluid. The simulations used to obtain these results were performed with the same simulation assumptions and models as the SAFFIRE injector simulations, using kerosene and LOX as the fluids. Therefore, the simulated performance of the SAFFIRE injector, can be assumed to be as accurate as the results captured within this study. This assumption was made since the only major varying factors within the simulations were the fluid's densities and viscosities. This assumption limits the LOX simulation accuracy to the accuracy of the solver at higher Reynolds numbers. This method was adopted since using dynamic similarity was not practical.

## 8.3 Discussion of Problems Encountered

Manufacturing tolerances had a large influence on the propellant distribution results. This made it difficult to determine whether the deviations between the simulation and experimental results were due to simulation errors or geometrical errors. The main reason for this issue was the sensitivity introduced by the small diameter orifices.

The tank ullage pressure was difficult to set before testing due to the current pressurization system. The single-stage regulator and solenoid setup required the regulator to be calibrated to ensure a constant tank pressure was maintained for the required mass flow rate. A more sophisticated setup with a dome loaded pressure regulator would simplify the issue of maintaining a constant tank pressure during the duration of a test.

## 8.4 Recommendations and Future Work

During the calibration tests of the fuel domain, covered in section 7.4.1, the sensitivity of the distribution and pressure drop due to machining tolerance was made apparent. For the fuel domain, the down-coming holes had a minor effect on the pressure drop but a large effect on the propellant distribution. The orifices diameters had a large effect on the propellant distribution due to the small diameters of 1.5 mm and 0.8 mm for the main and film cooling orifices respectively. These holes were wire cut and had an average tolerance of +30 microns. For the LOX domain, the angled holes were drilled then reamed and resulted in a maximum tolerance of ± 10 microns for the angled orifices and ± 4 microns for a perpendicular orifice sample. Based on the sensitivity of the tolerances, it is recommended that the holes of the SAFFIRE injector are drilled then reamed for both the perpendicular fuel and angled LOX orifices.

The test articles were bolted together using an annular arrangement of bolts. This introduced the possibility of a 0.5 mm misalignment due to the tolerances of the bolt threads and holes. This misalignment created a step between the injector body and top manifold that possibly disturbed the flow within the LOX domain. This could lead to a misimpingement of LOX in ring five near the chamber wall, resulting in a possible hot spot. It is recommended that the final design of the SAFFIRE injector contains tapered alignment pins and slots to reduce the possibility of a misalignment. Furthermore, an in-depth quality control procedure must be performed on the highly sensitive manufactured parts, namely the faceplate orifices and injector body.

The deviations of the simulation propellant distribution results from the experimental data for the LOX domain were concluded as the result of possible geometric deviations between the orifices. To determine the relationship between the flow rate and orifice diameter/shape a high-resolution CFD study could be conducted, investigating the change in flow rate due to an increase or decrease in orifice diameter. Furthermore, to determine the simulation's accuracy of capturing each flow phenomena within the different injector features a more in-depth cold flow testing program could be conducted. This would involve separating the injector into unit problems to evaluate the simulation's accuracy for each flow feature.

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## Appendix A: Simulation Graphs

The specific deviations of the simulation results from the experimental data are shown for each orifice in the following figures. The orifices are separated into each concentric ring for the fuel and LOX domains. Error bars are shown to indicate the uncertainty with regards to the experimental results. A 0 % deviation indicates the simulation accurately modelled the orifice. Therefore, orifices, where the error bars include 0 %, are considered to be accurately modelled.

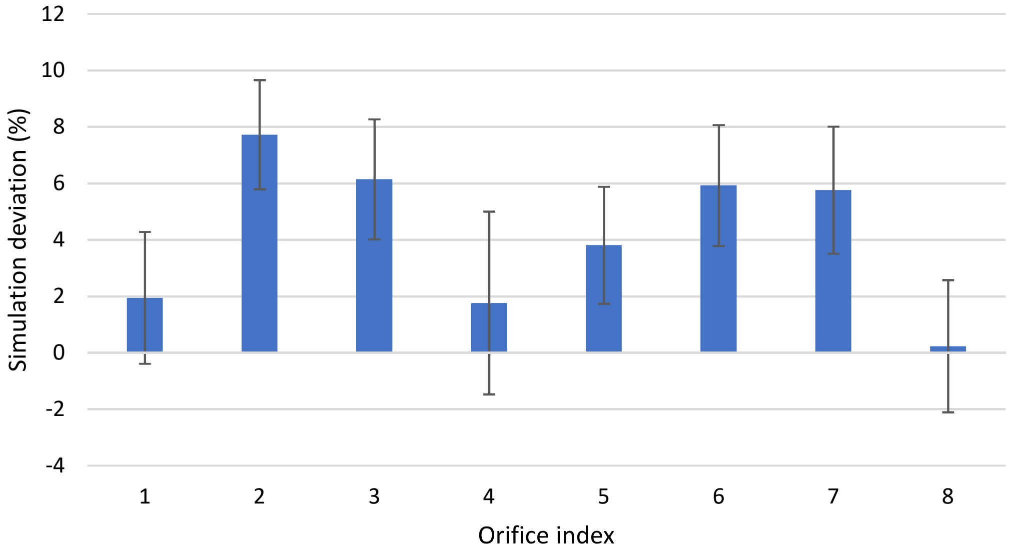


Figure A-1: F1 simulation deviation

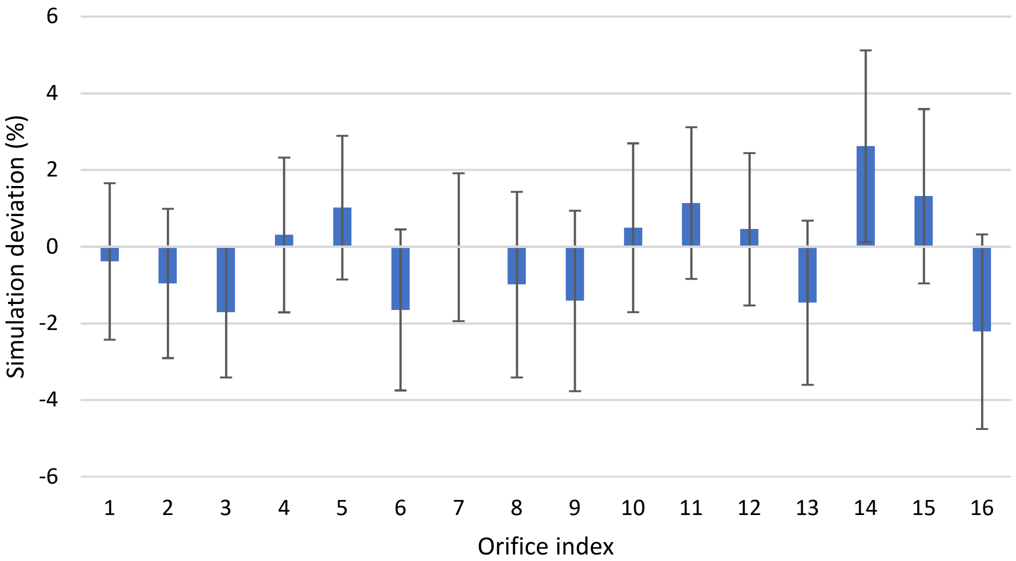


Figure A-2: F2 simulation deviation

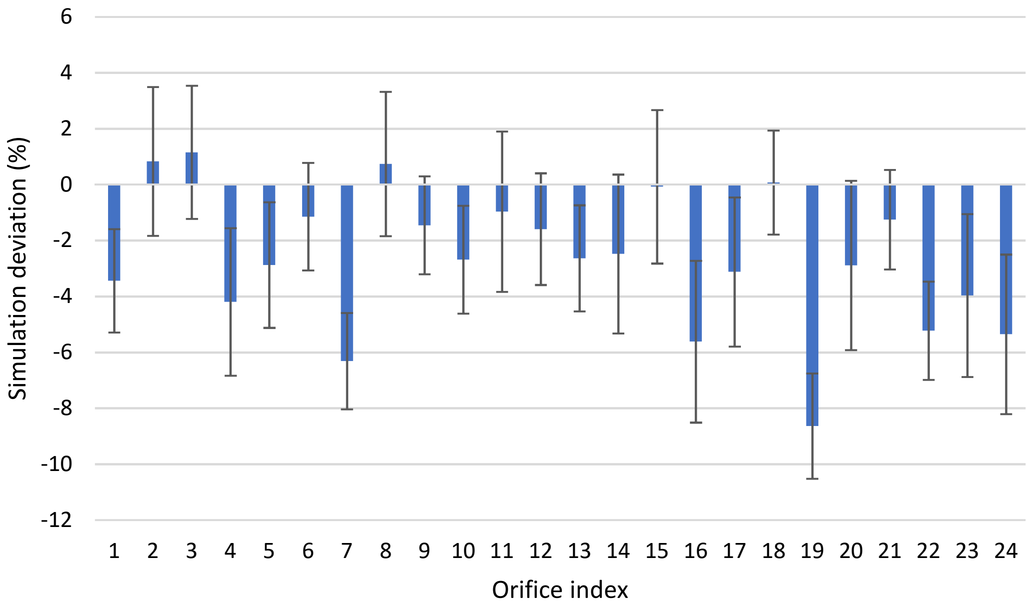


Figure A-3: F3 simulation deviation

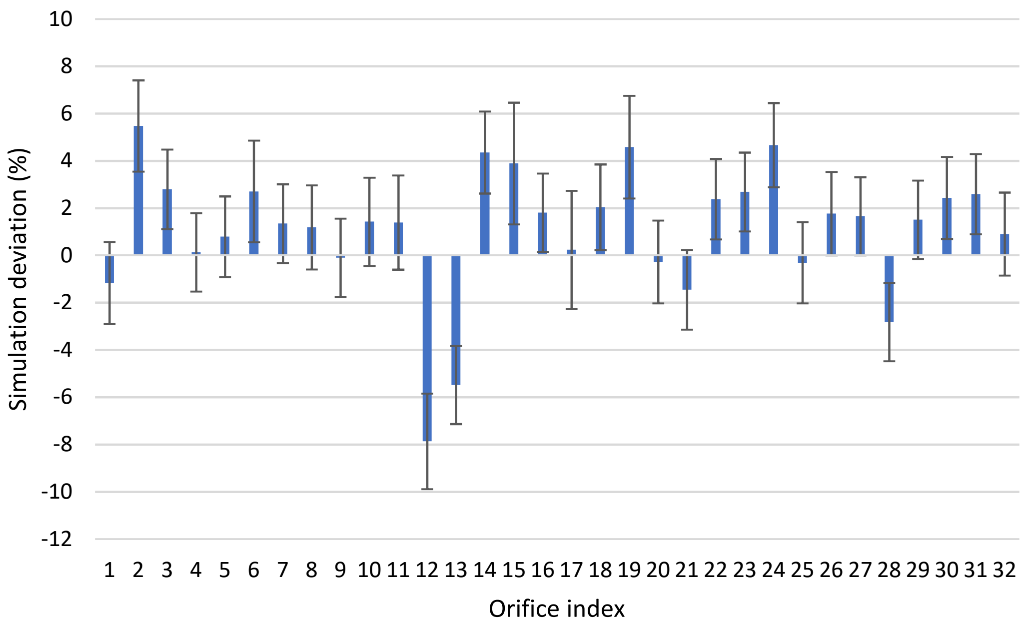


Figure A-4: F4 simulation deviation

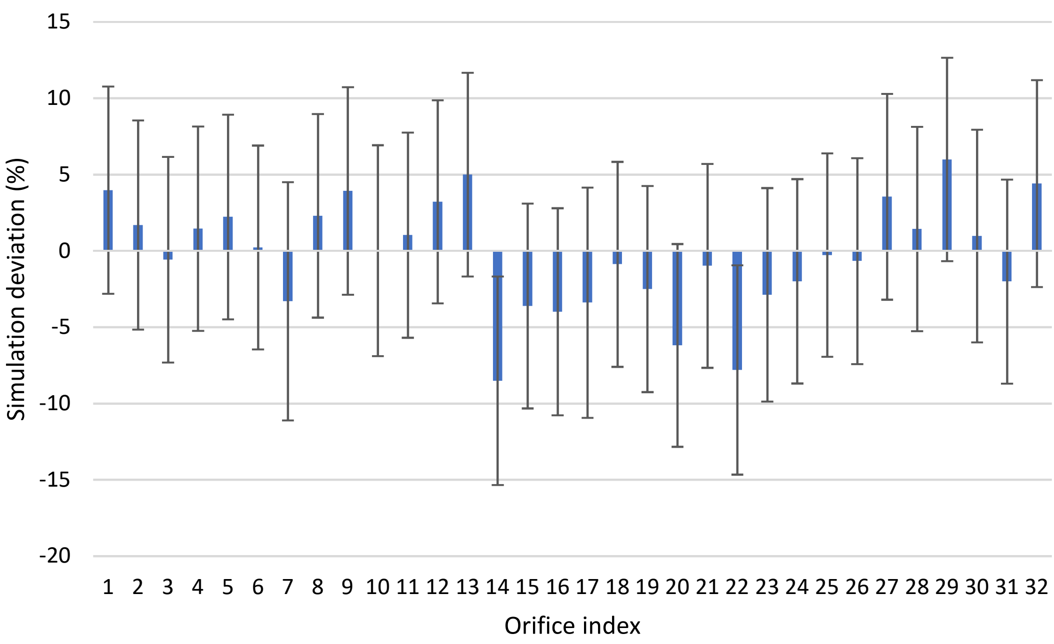


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Figure A-5: FF simulation deviation

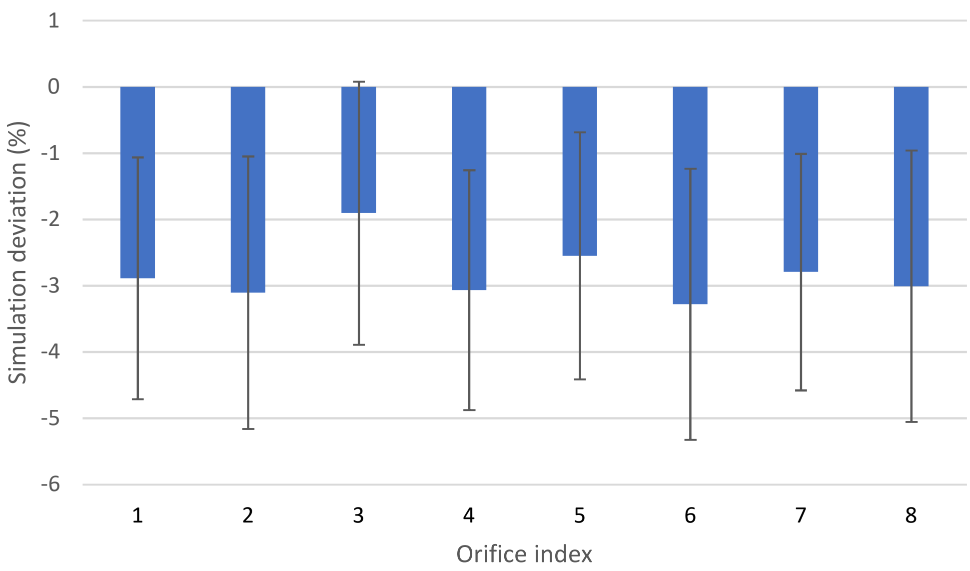


Figure A-6: L1 simulation deviation

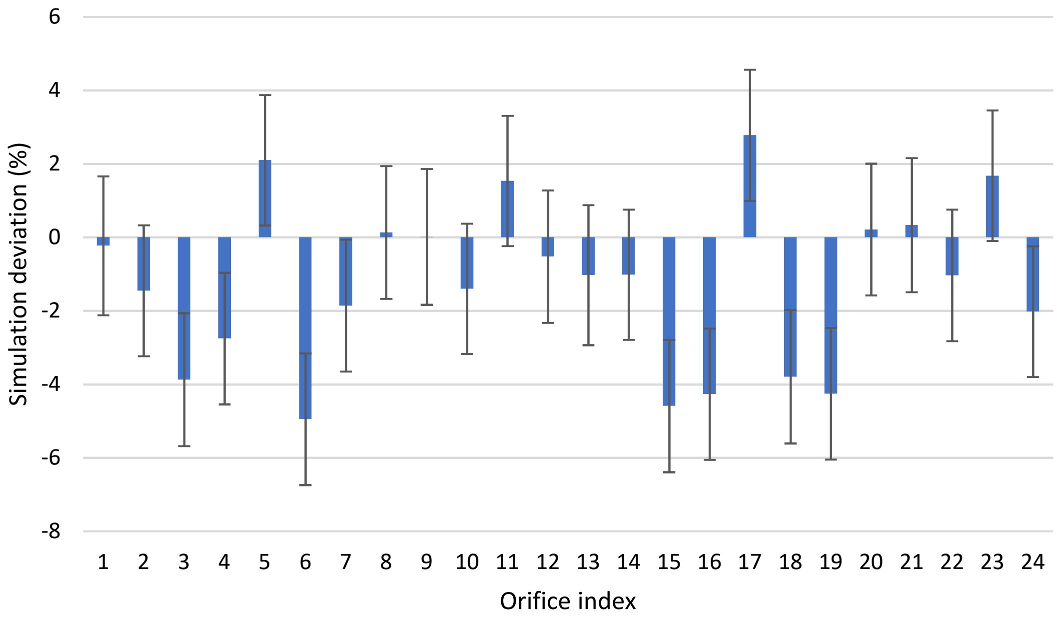


Figure A-7: L2 simulation deviation

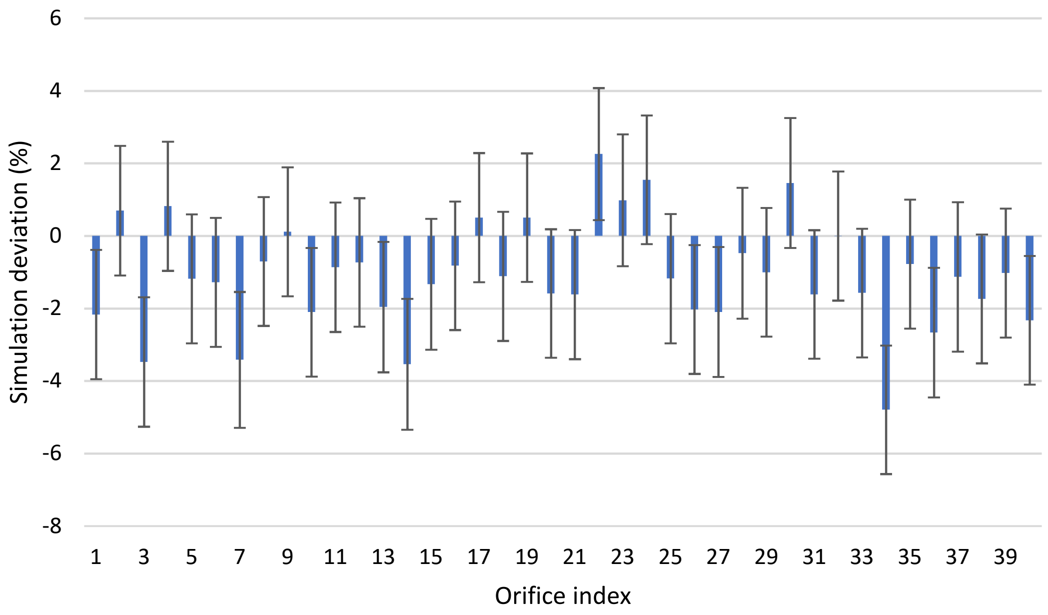


Figure A-8: L3 simulation deviation

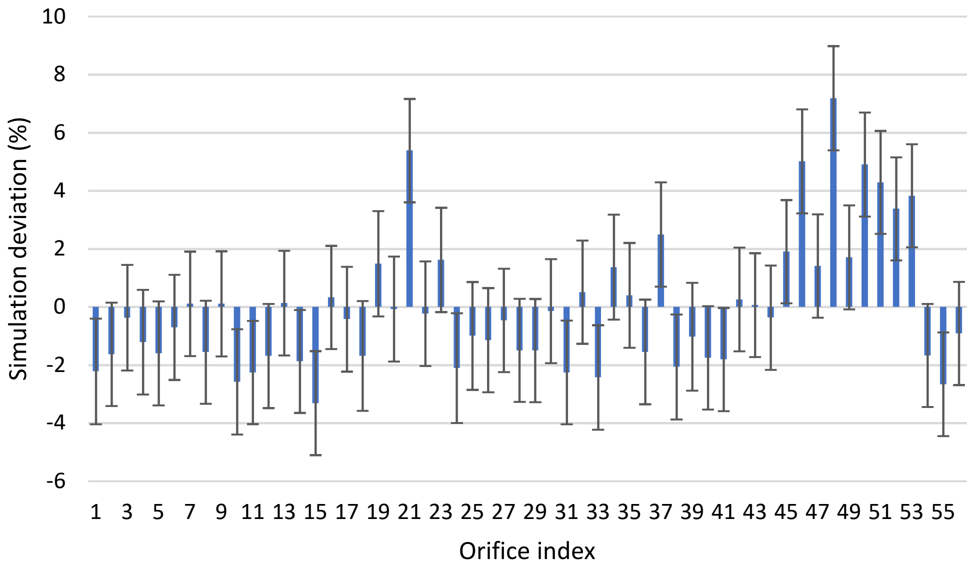


Figure A-9: L4 simulation deviation

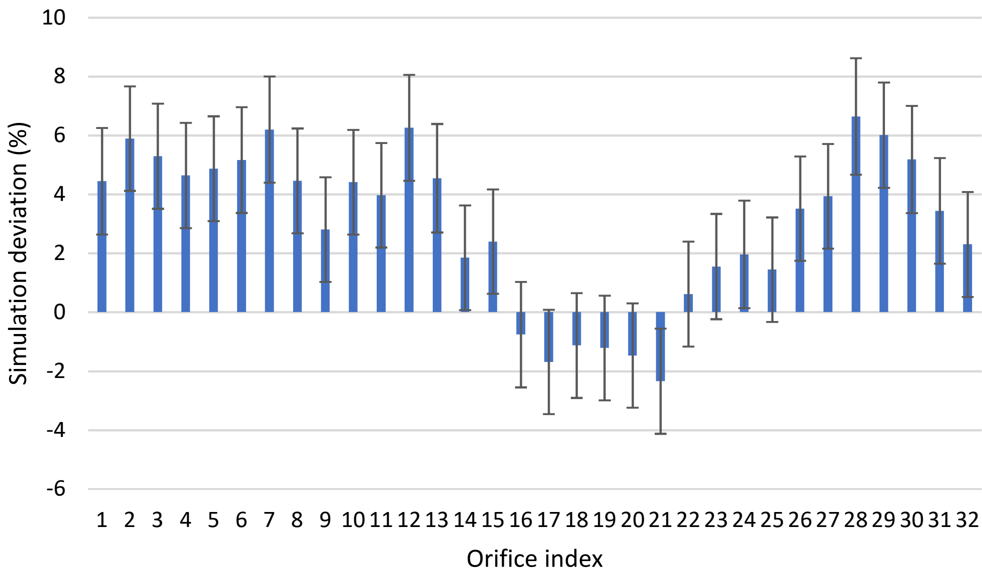


Figure A-10: L5 simulation deviation

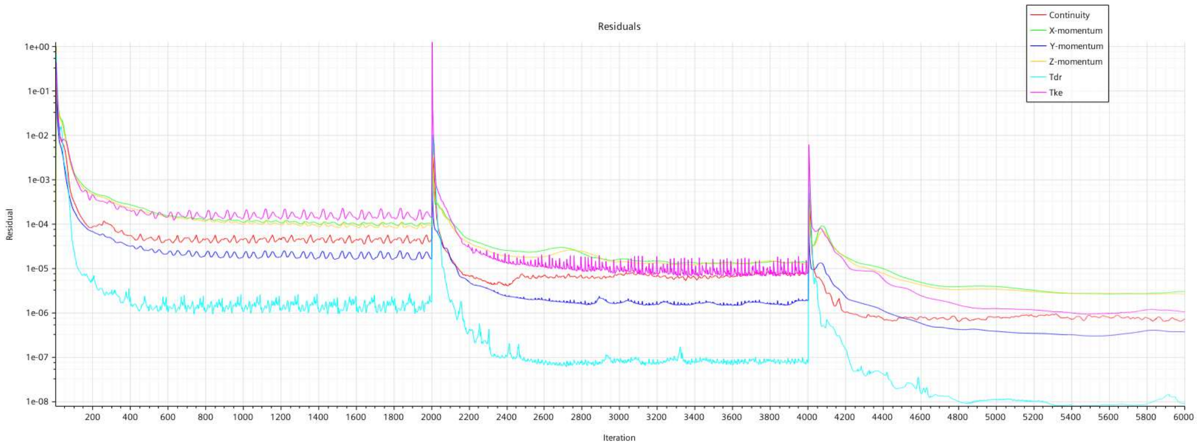


Figure A-11: Fuel simulation residuals

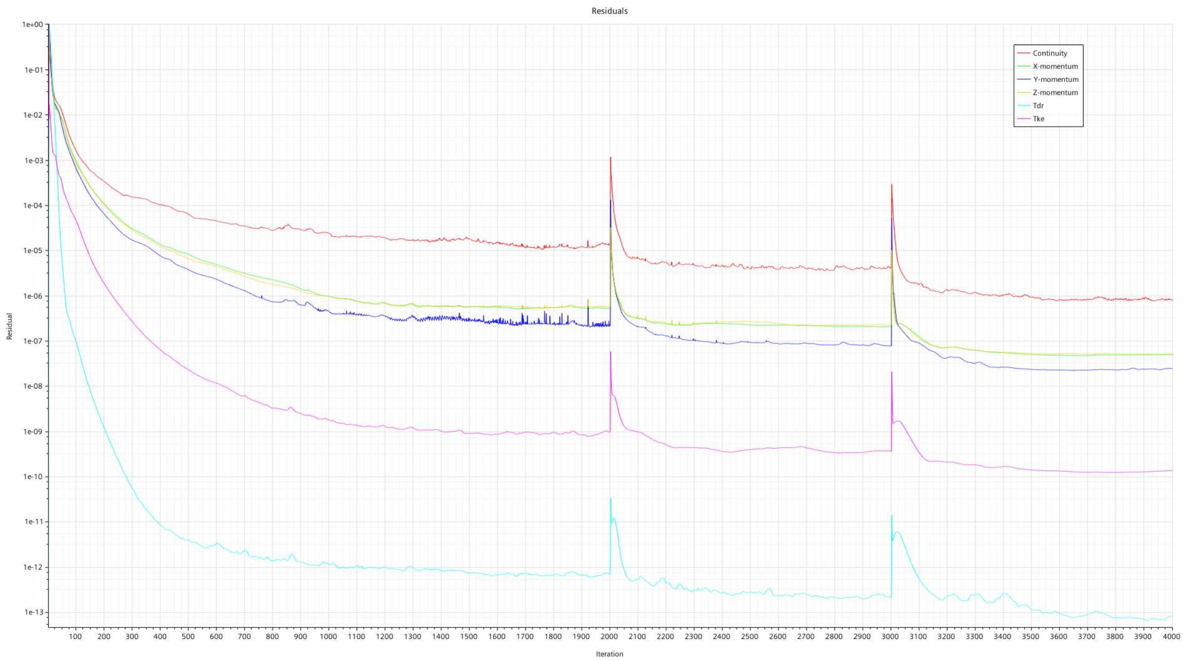


Figure A-12: LOX simulation residuals

# Appendix B: Bolted Connections

The fuel and LOX test articles included bolted connections. The theory and equations used to verify the design analytically are discussed below. This includes the thread requirements for both test articles due to an internal pressure maximum of 10 bar. The fuel test articles included 18 off M8 bolts, 2 off M5 screws, and 16 off M4 screws to secure the three main bodies together. The LOX test article included 16 off M8 bolts and 16 off M4 screws.

With an internal pressure of 10 bar, the external forces exerted on the bolts were 44.07 KN and 55.25 KN for the internal geometries of the fuel and LOX test articles respectively. The bolt force $F_{\mathrm{b}}$ and clamping force $F_{\mathrm{c}}$ were calculated using Equations B-1 and B-2:

$$F_{\mathrm{b}} = F_{\mathrm{i}} + \frac{k_{\mathrm{b}}}{k_{\mathrm{b}} + k_{\mathrm{c}}} F_{\mathrm{e}} \tag{B-1}$$

$$F_{\mathrm{c}} = F_{\mathrm{i}} - \frac{k_{\mathrm{c}}}{k_{\mathrm{b}} + k_{\mathrm{c}}} F_{\mathrm{e}} \tag{B-2}$$

Here, $F_{\mathrm{e}}$ is the external or internal force separating the components, $k_{\mathrm{b}}$ and $k_{\mathrm{c}}$ are the spring constants for the bolt and clamped materials respectively. For clamped components consisting of different materials, an equivalent spring constant is determined using Equation B-3:

$$\frac{1}{k_{\mathrm{c}}} = \frac{1}{k_{\mathrm{c1}}} + \frac{1}{k_{\mathrm{c2}}} + \frac{1}{k_{\mathrm{c3}}} + \cdots \tag{B-3}$$

$k_{\mathrm{b}}$ and $k_{\mathrm{c}}$ are calculated from Equations B-4 and B-5:

$$k_{\mathrm{b}} = \frac{A_{\mathrm{b}} E_{\mathrm{b}}}{L_{\mathrm{g}}} \tag{B-4}$$

$$k_c = \frac{A_c E_c}{L_g} \tag{B-5}$$

The initial bolt tension is given by $F_{\mathrm{i}}$ and is calculated with Equation B-6:

$$F_{\mathrm{i}} = K_{\mathrm{i}} A_{\mathrm{t}} S_{\mathrm{p}} \tag{B-6}$$

Here, $K_{\mathrm{i}}$ is the initial tightening factor set to 0.75 in this application, $A_{\mathrm{t}}$ is the tensile stress area of the thread and $S_{\mathrm{p}}$ is the proof strength of the bolt material.

The bolt and clamping forces are listed in Table B-1 for SAE Class 12.9 steel bolts and screws. The outer bolts and countersunk screws were designed to carry the full load of the internal pressure independently. This resulted in a larger safety factor for the test article. This maximum bolt force was used to calculate the required thread length to avoid thread stripping of the aluminium. The required thread length was 4.3 mm.

Table B-1: Fuel test article bolt calculation

| **Parameter** | **Symbol** | **Faceplate screws – M4** | **Annular bolts – M8** |
|---|---|---|---|
| Bolt force (kN) | $F_{\mathrm{b}}$ | 6.89 | 26.76 |
| External bolt force (kN) | $F_{\mathrm{e}}$ | 1.84 | 2.75 |
| Initial bolt force (kN) | $F_{\mathrm{i}}$ | 6.40 | 26.63 |
| Clamping bolt force (kN) | $F_{\mathrm{c}}$ | 5.16 | 24.07 |

The bolt forces for the LOX test article were calculated and are shown in Table B-2. The required thread length for the bolt forces on the screws was 4.2 mm to avoid thread stripping.

Table B-2: LOX test article bolt calculation

| **Parameter** | **Symbol** | **Faceplate screws – M4** | **Annular bolts – M8** |
|---|---|---|---|
| Bolt force (kN) | $F_{\mathrm{b}}$ | 6.62 | 27.06 |
| External bolt force (kN) | $F_{\mathrm{e}}$ | 0.99 | 3.45 |
| Initial bolt force (kN) | $F_{\mathrm{i}}$ | 6.40 | 26.63 |
| Clamping bolt force (kN) | $F_{\mathrm{c}}$ | 6.11 | 23.75 |

# Appendix C: Two-Test Method Sample Calculation

The mass flow rate corresponding to an injector manifold pressure was determined using the two-test method. A possible uncertainty was introduced because of the variation of the flow rate between the full duration and transient tests. The calculations below use the flow factor to quantify the uncertainty. The values used for the calculations were taken from a full duration and transient test for the fuel domain.

Table C-1: Example test results

| **Parameters** | **Transient test** | **Full duration test** |
|---|---|---|
| Total weight collected (kg) | 4.87 | 28.03 |
| Total test duration (s) | 2.69 | 12.41 |
| Steady state test duration (s) | 2.10 | 11.82 |
| Average manifold pressure (bar) | 3.41 | 3.39 |

The mass flow rate ($\dot{m}$) calculated below was based on the total weight ($m$) and steady state time period ($t$) of a full-duration test. Therefore, it results in a conservative flow factor as it includes the flow collected during the transient periods. Combined uncertainties are not considered during this process as the true value of the flow rate is not a concern. Only the difference in the flow rate due to the change in manifold pressure is being investigated. The flow rate ($Q$) is replaced with the more relevant mass flow rate term ($\dot{m}$ ) using Equation C-1:

$$Q = \frac{\dot{m}}{\rho} \tag{C-1}$$

Here, $\rho$ is the density of the fluid. The calculated mass flow rate is given by Equation C-2:

$$\dot{m} = \frac{m}{t} \tag{C-2}$$

Using the testing data from the above example, $\dot{m} = 2.371$ kg/s and $K_{\mathrm{v}} = 4.63\ \mathrm{m^3/h.\,bar^{0.5}}$.

Using this flow factor, a mass flow rate for a transient test with a manifold pressure increase of 0.5% was calculated. The mass flow rate due to a 0.5% manifold pressure was 2.3767 kg/s. This is an increase of 0.0057 kg/s. Over the transient test duration, a maximum of 15.35 g of additional water could be collected due to this mass flow rate difference. This yielded a relative uncertainty of 0.066% for the calculated mass flow rate using the two-test method.

# Appendix D: Control System Code

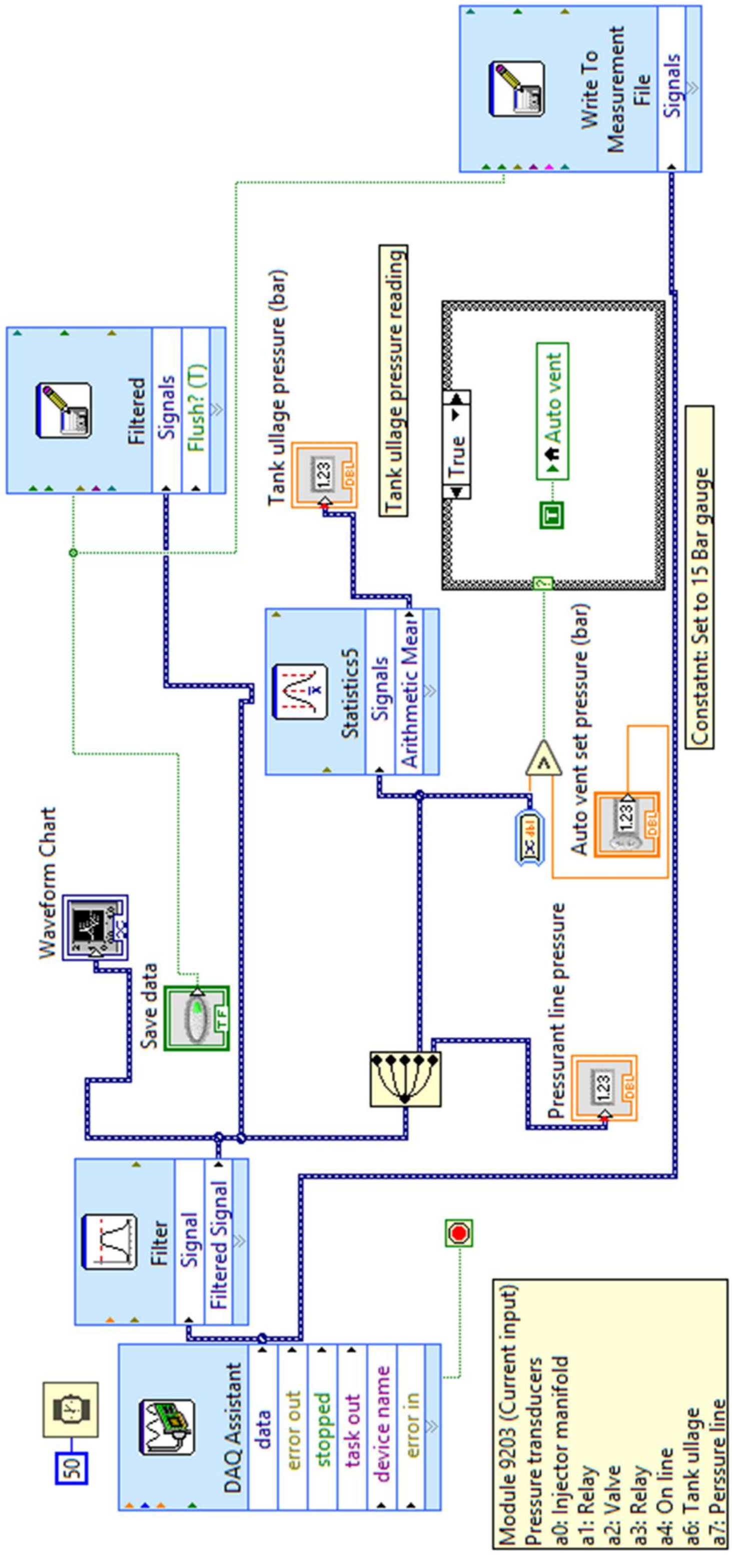


Figure D-1: LabVIEW primary control loop

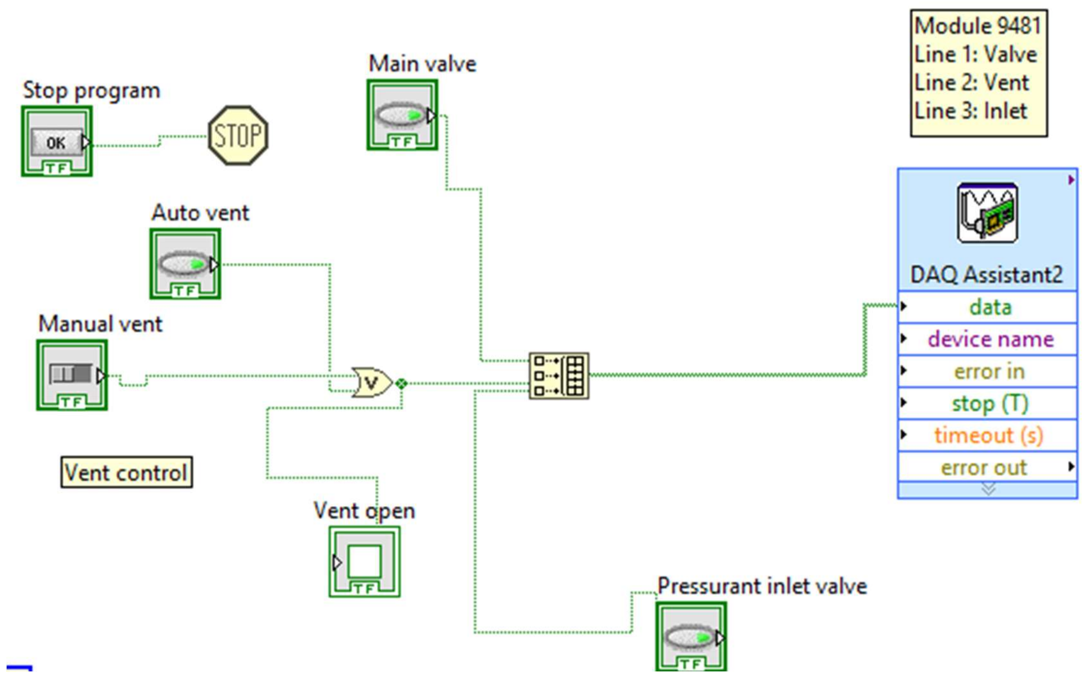


Figure D-2: LabVIEW primary valve control loop

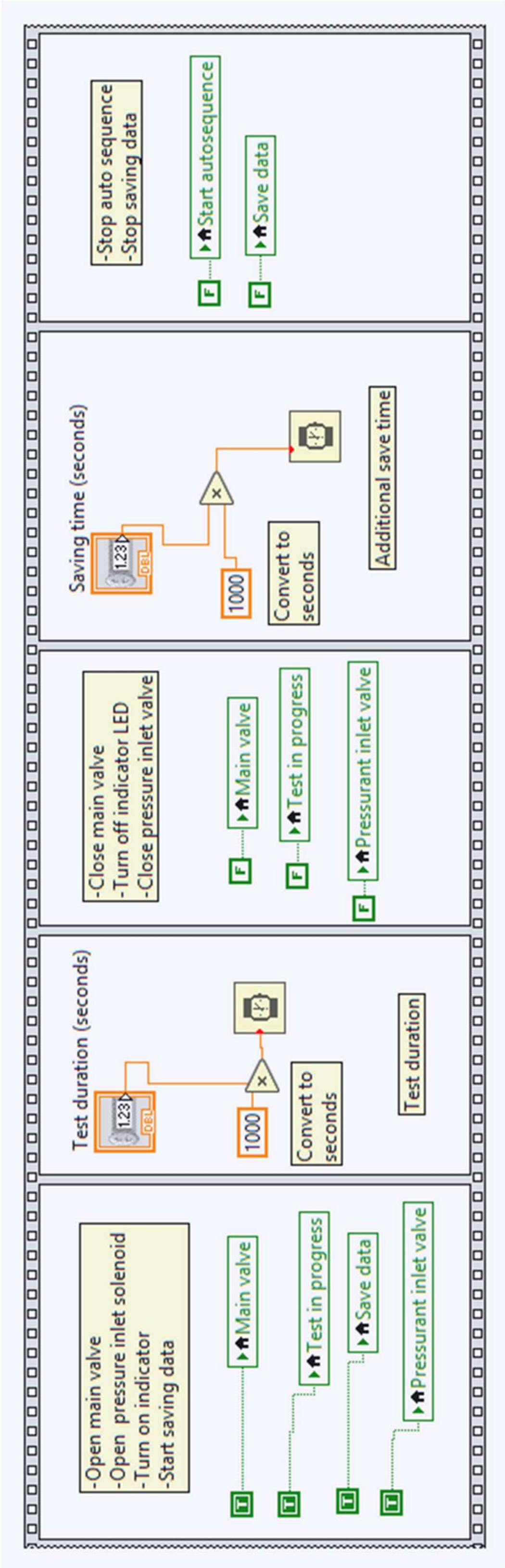


Figure D-3: LabVIEW auto sequence

# Appendix E: Machine Drawings

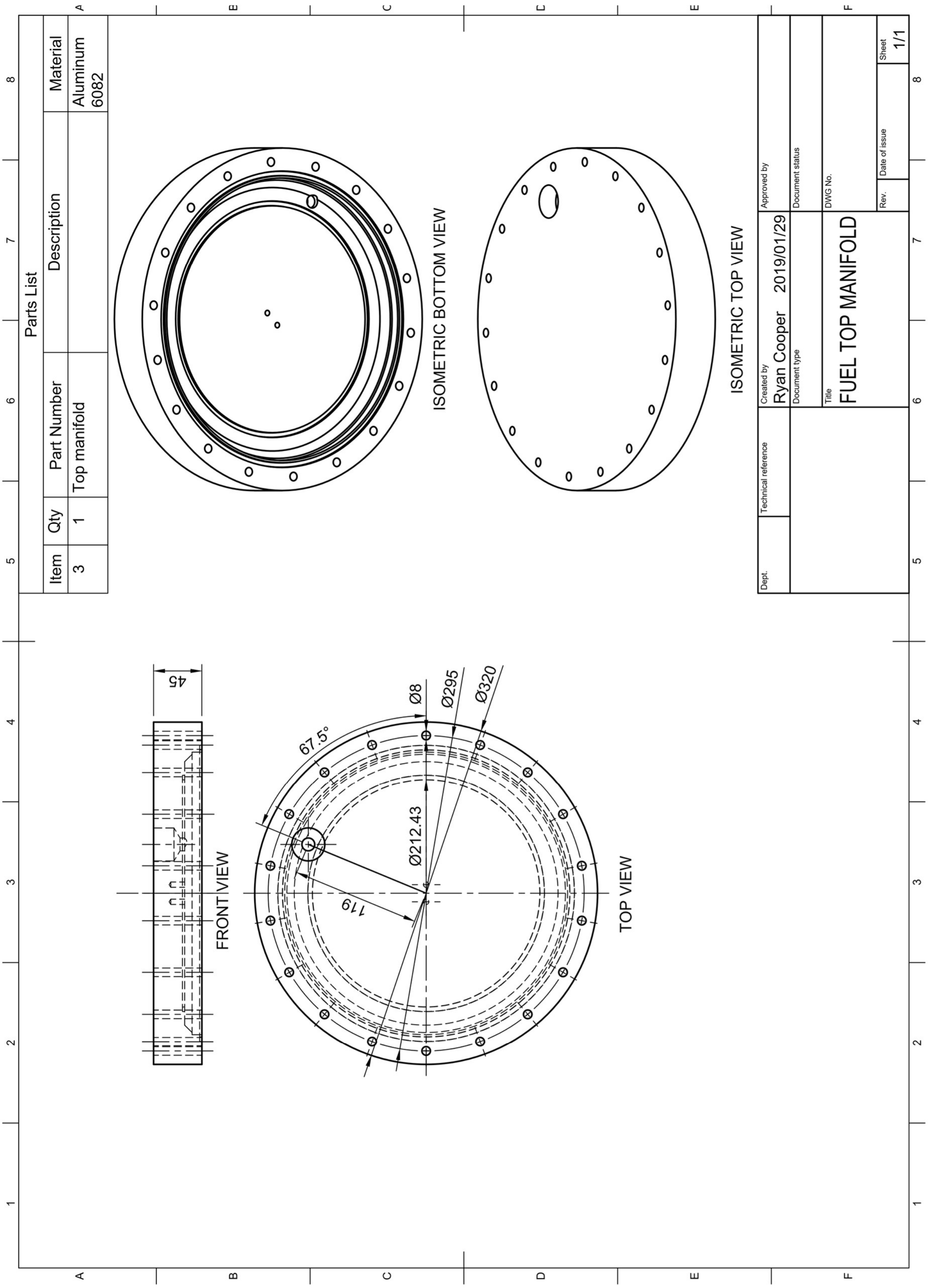


Figure E-1: Fuel top manifold

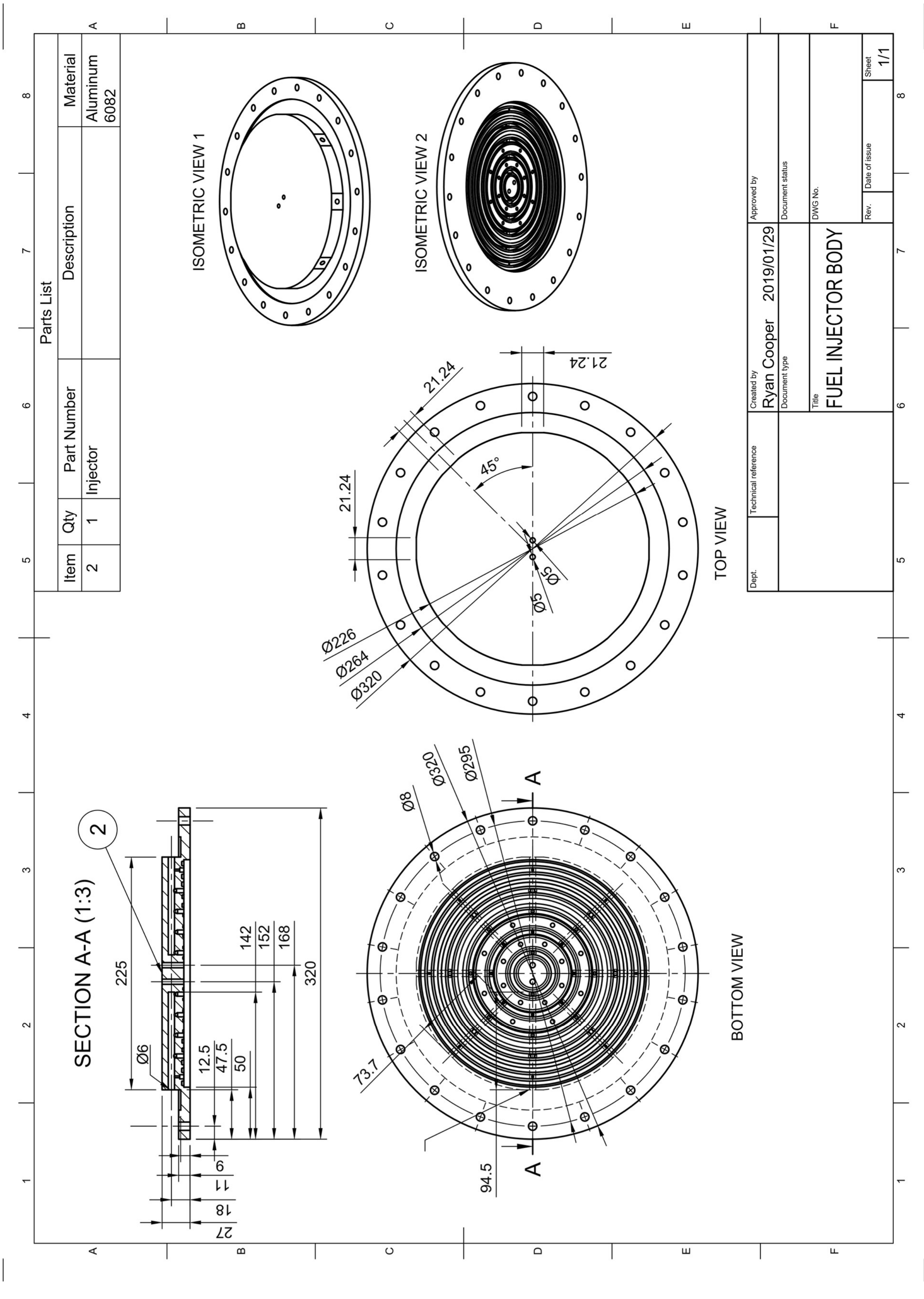


Figure E-2: Fuel injector body

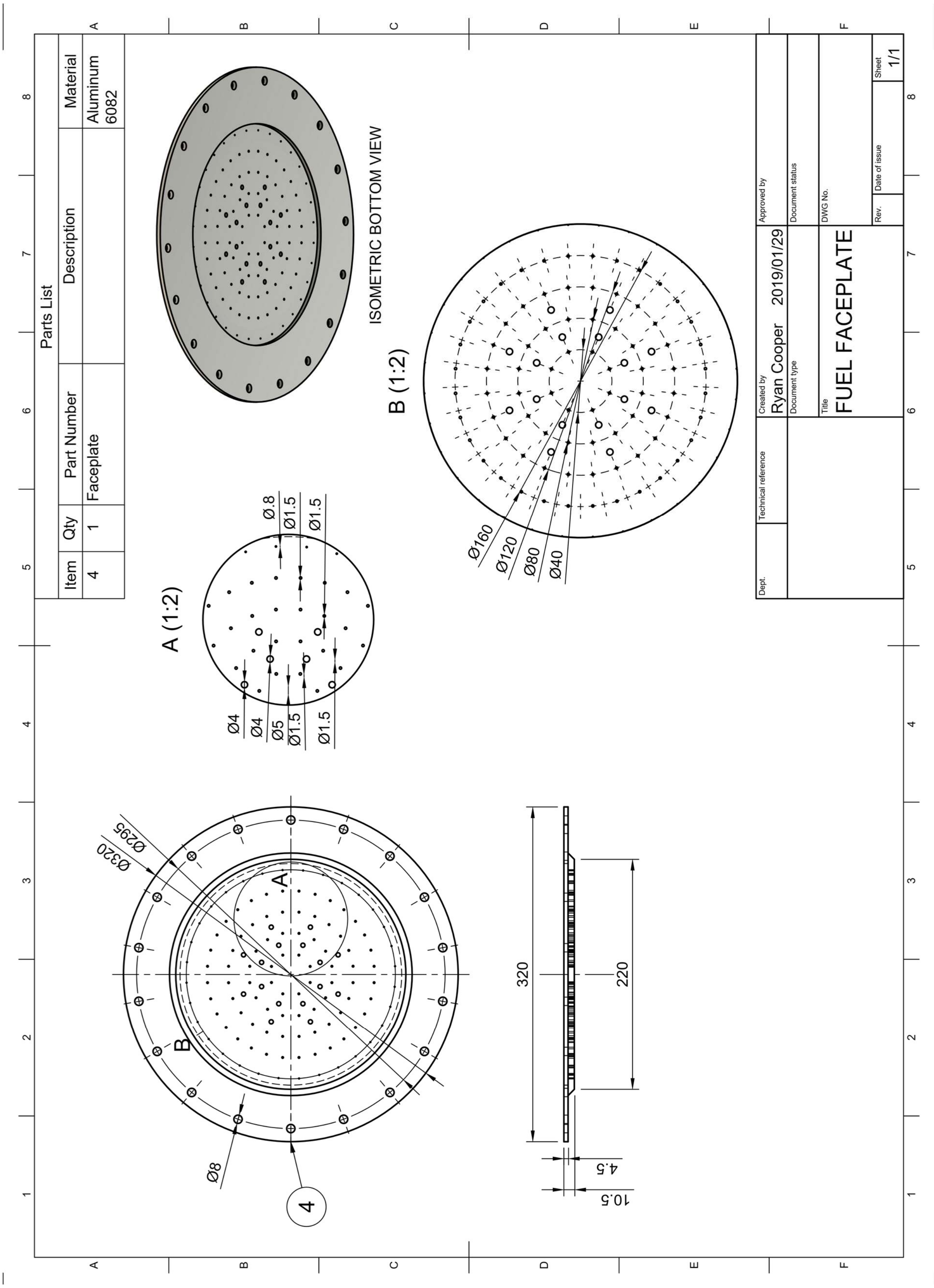


Figure E-3: Fuel faceplate

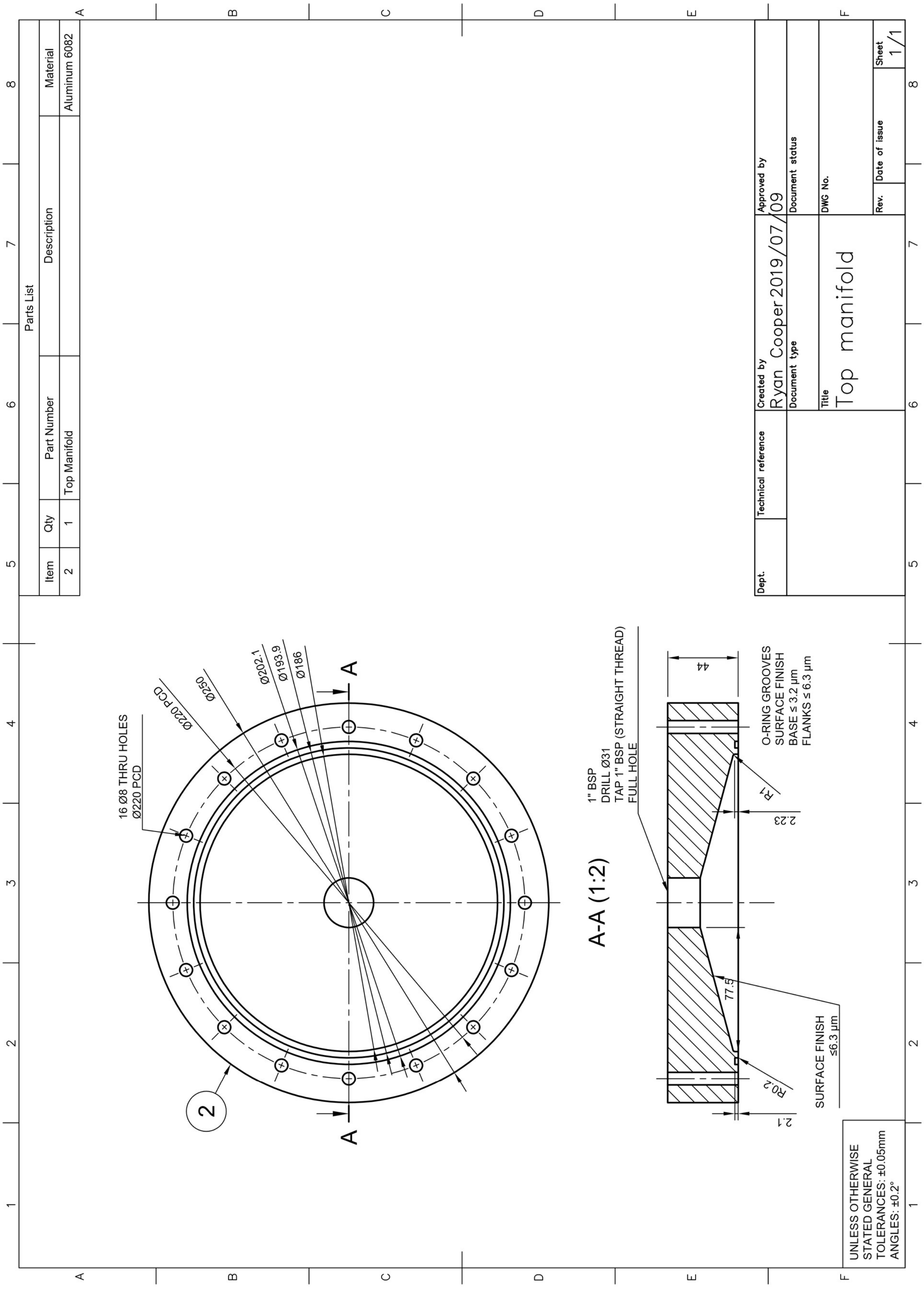


Figure E-4: LOX top manifold

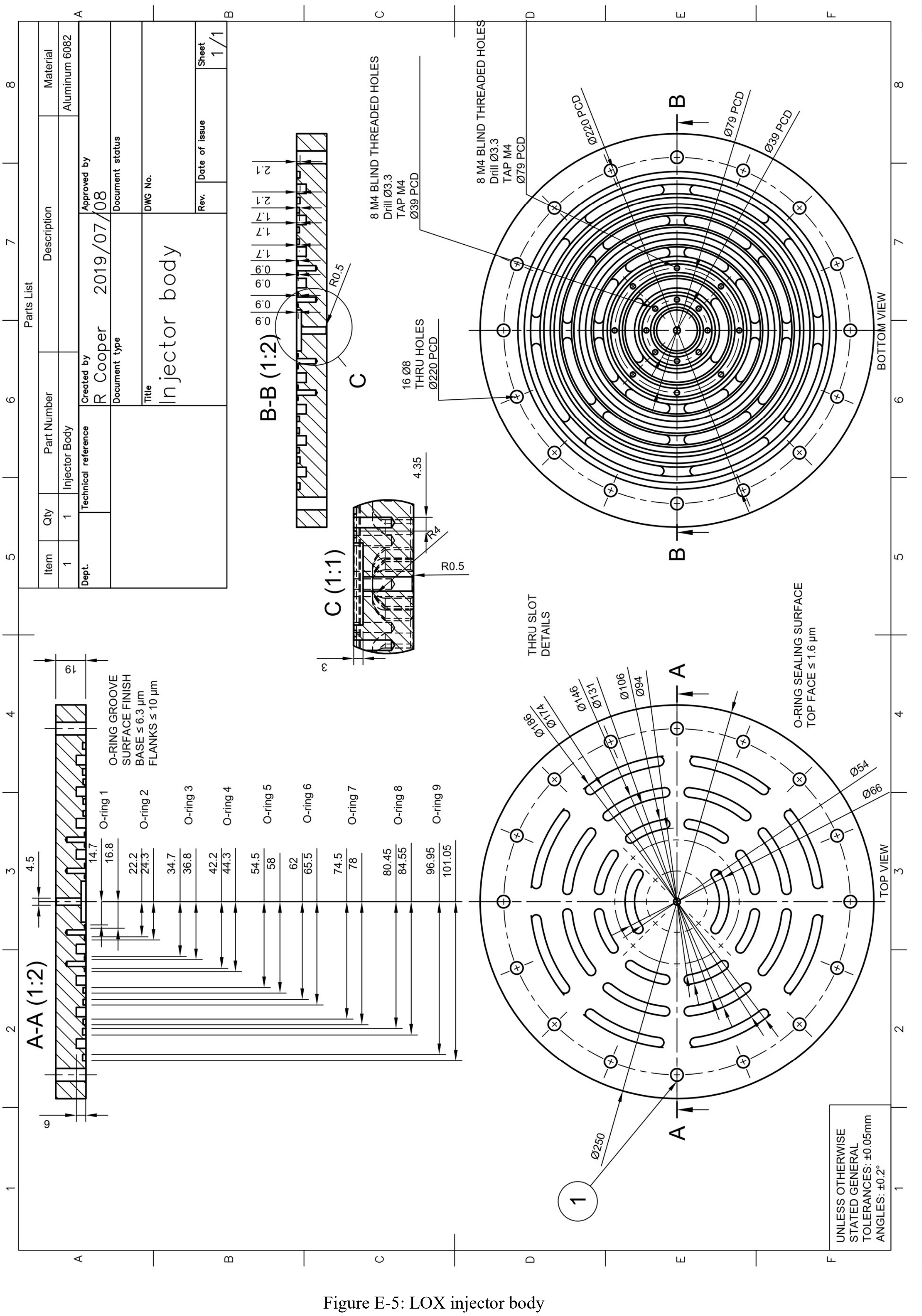


Figure E-5: LOX injector body

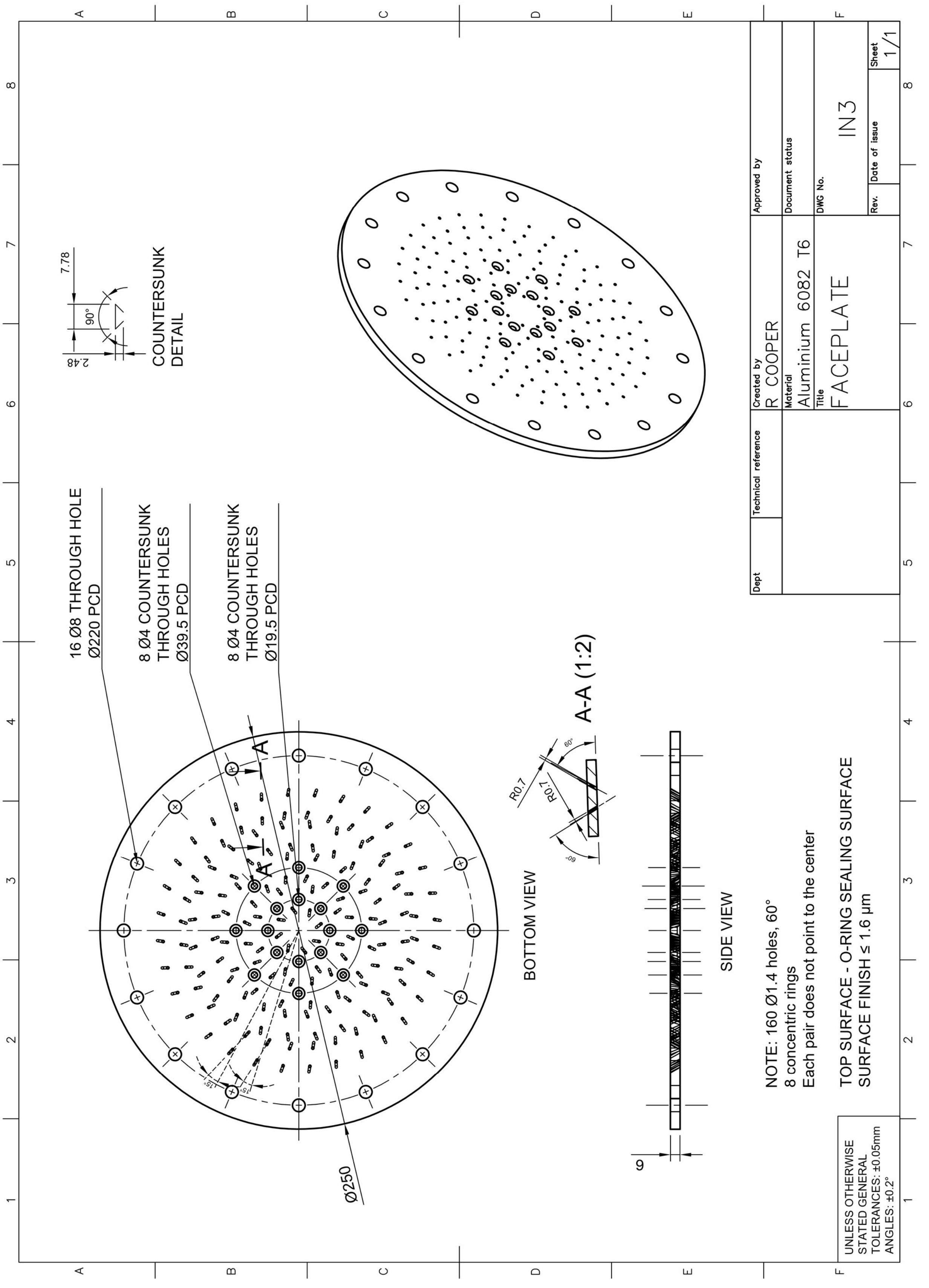


Figure E-6: LOX faceplate